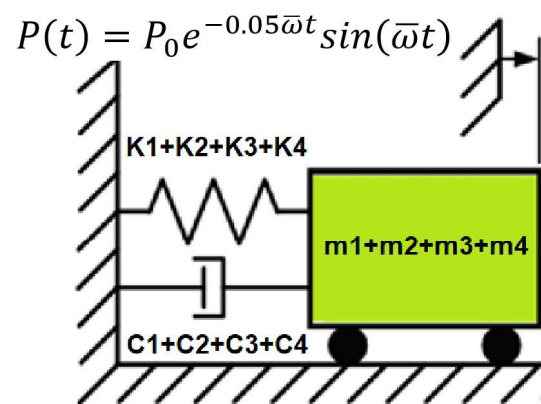


>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

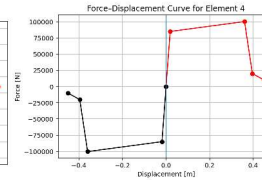
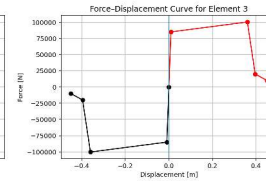
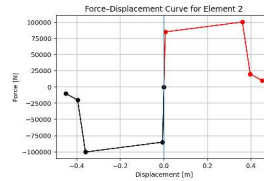
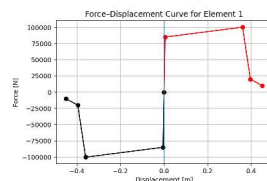
COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF A POST-FIRE SINGLE- DEGREE-FREEDOM WITH FOUR ELEMENTS STRUCTURE (SDOF) : AN OPENSEES FRAMEWORK FOR STATIC PUSHOVER, CYCLIC DEGRADATION, STATIC TIME- HISTORY AND DYNAMIC TIME-HISTORY ANALYSIS

THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)

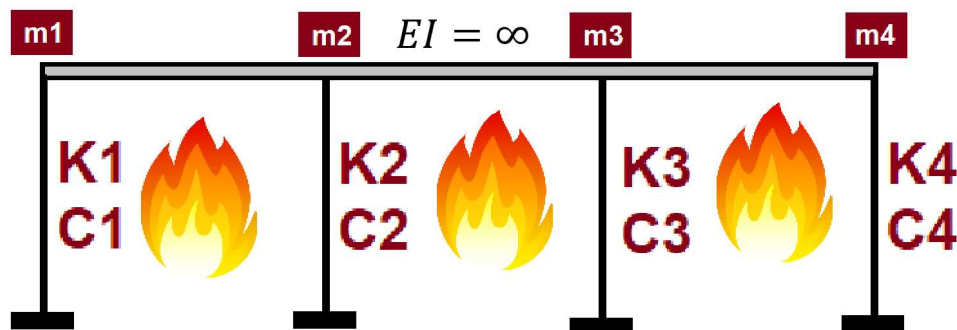
INCREMENTAL
DISPLACEMENT



Initial Displacement
Initial Velocity
Initial Acceleration



INCREMENTAL
DISPLACEMENT



$$\text{Structural Ductility Damage Index} = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

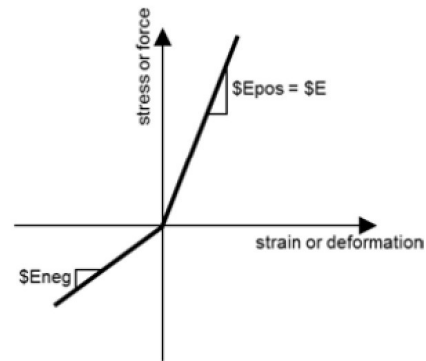
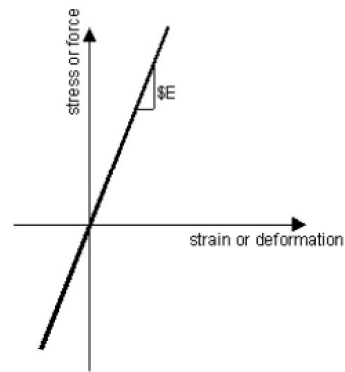
Δ_d = Lateral Displaement from Dynamic Analysis

Δ_y = Lateral Yield Displaement from Pushover Analysis

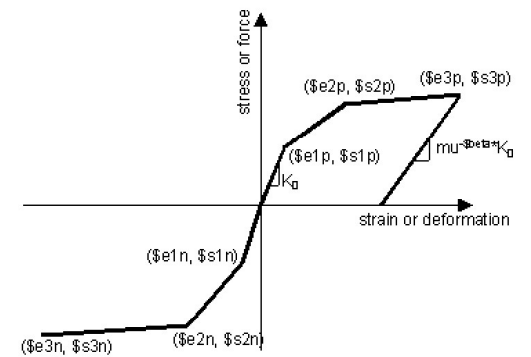
Δ_u = Lateral Ultimate Displaement from Pushover Analysis

$$P(t) = P_0 e^{-0.05\bar{\omega}t} \sin(\bar{\omega}t)$$

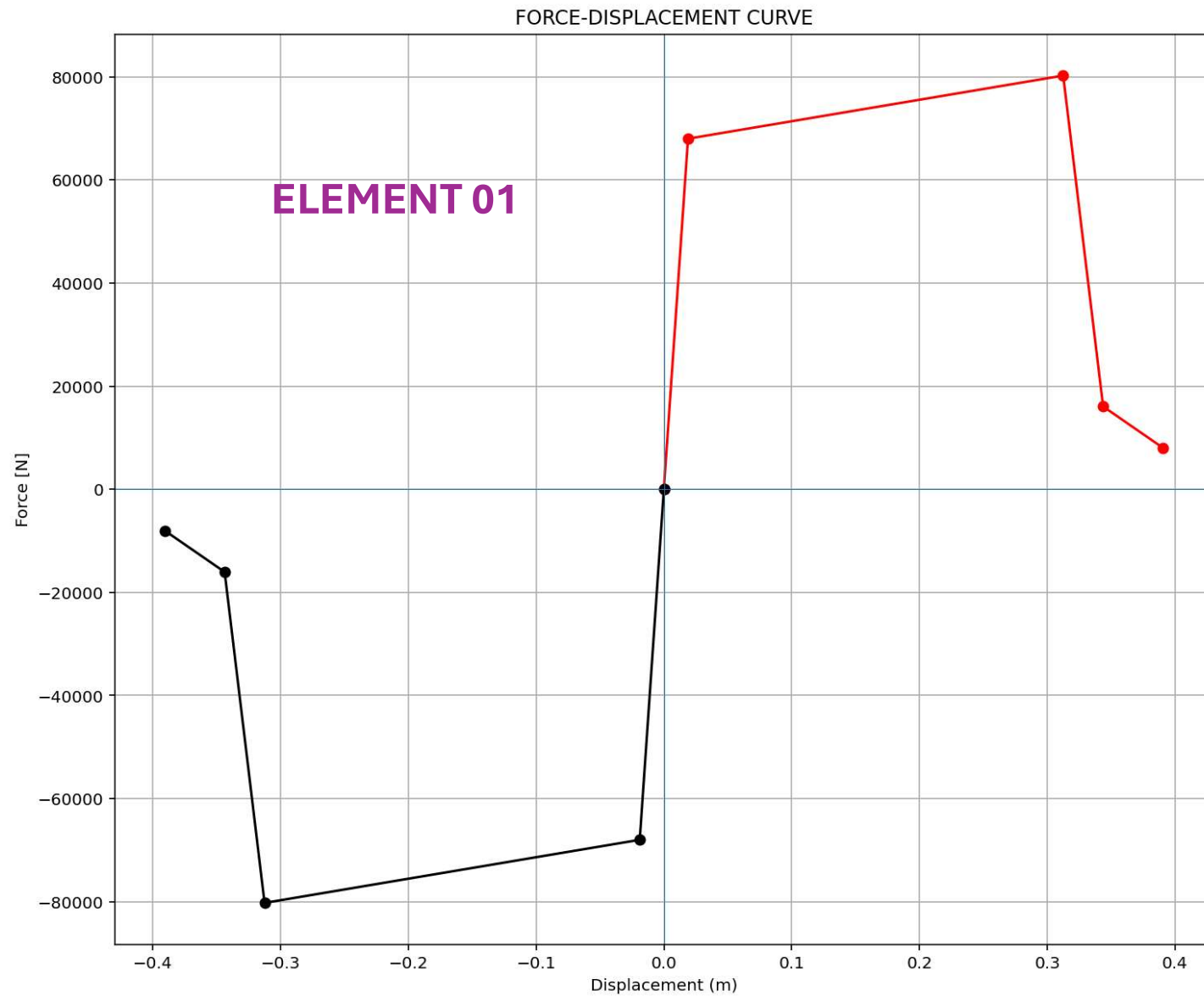
$$m\ddot{u} + c\dot{u} + ku = P(t)$$

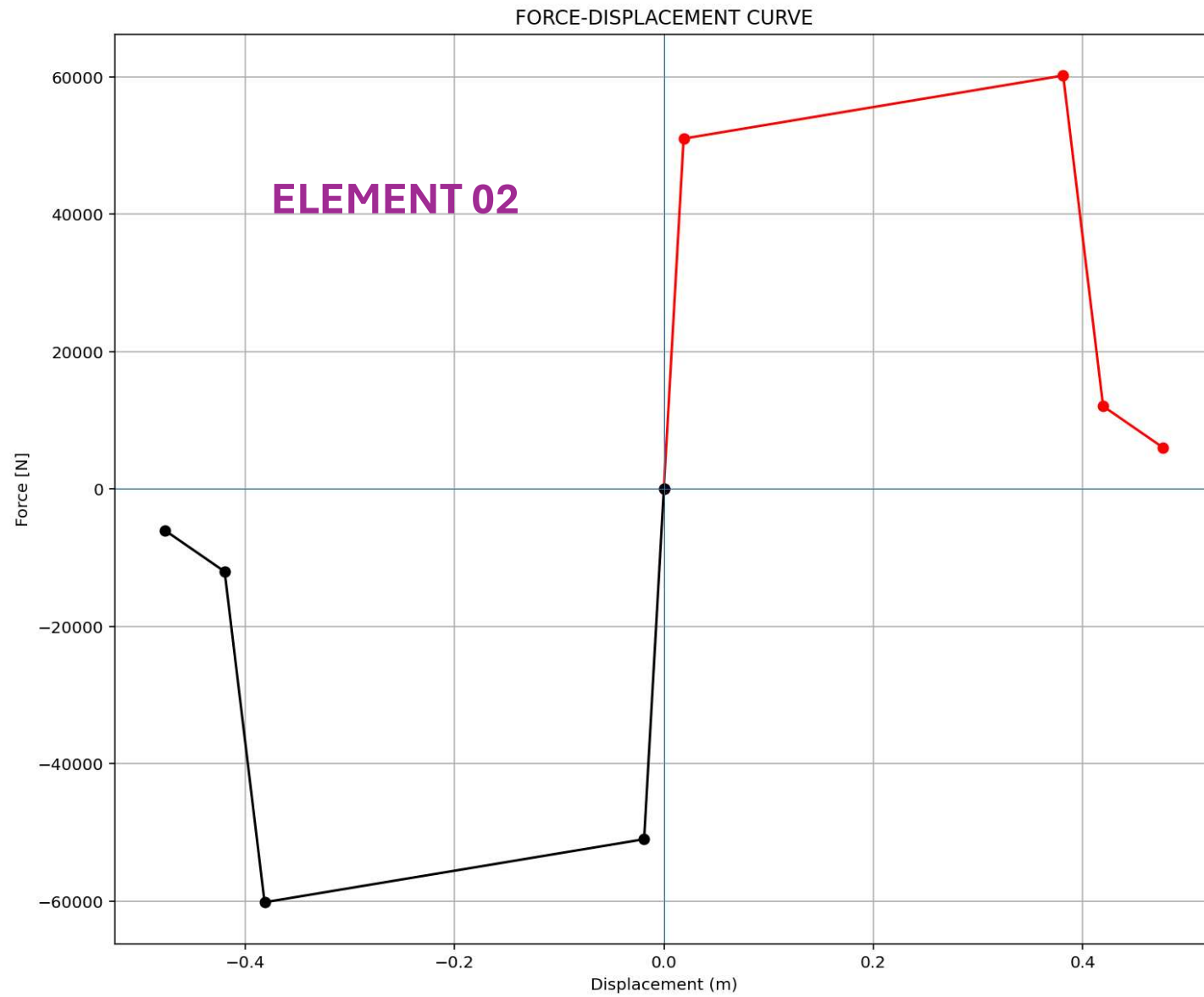


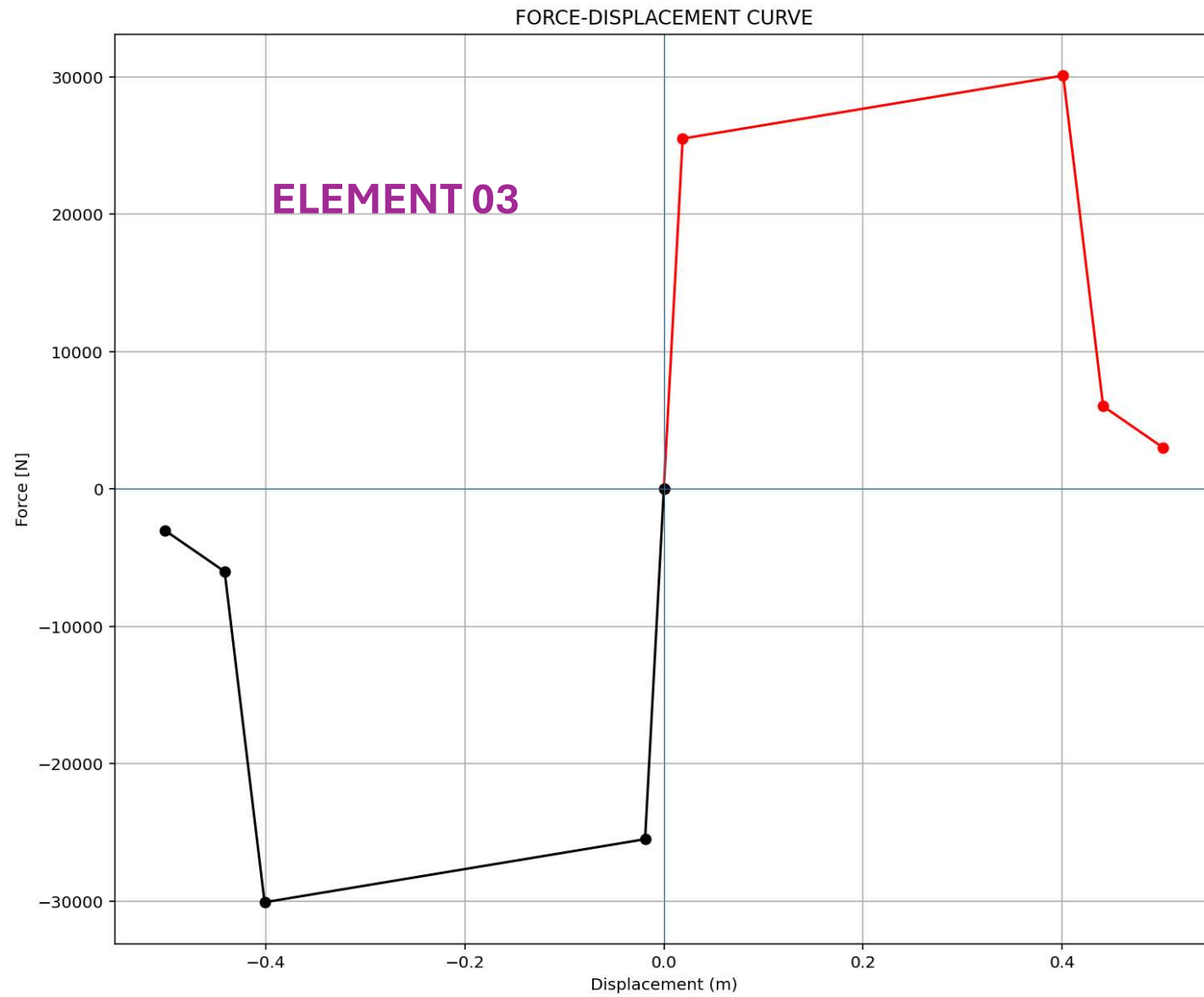
ELASTIC MATERIAL

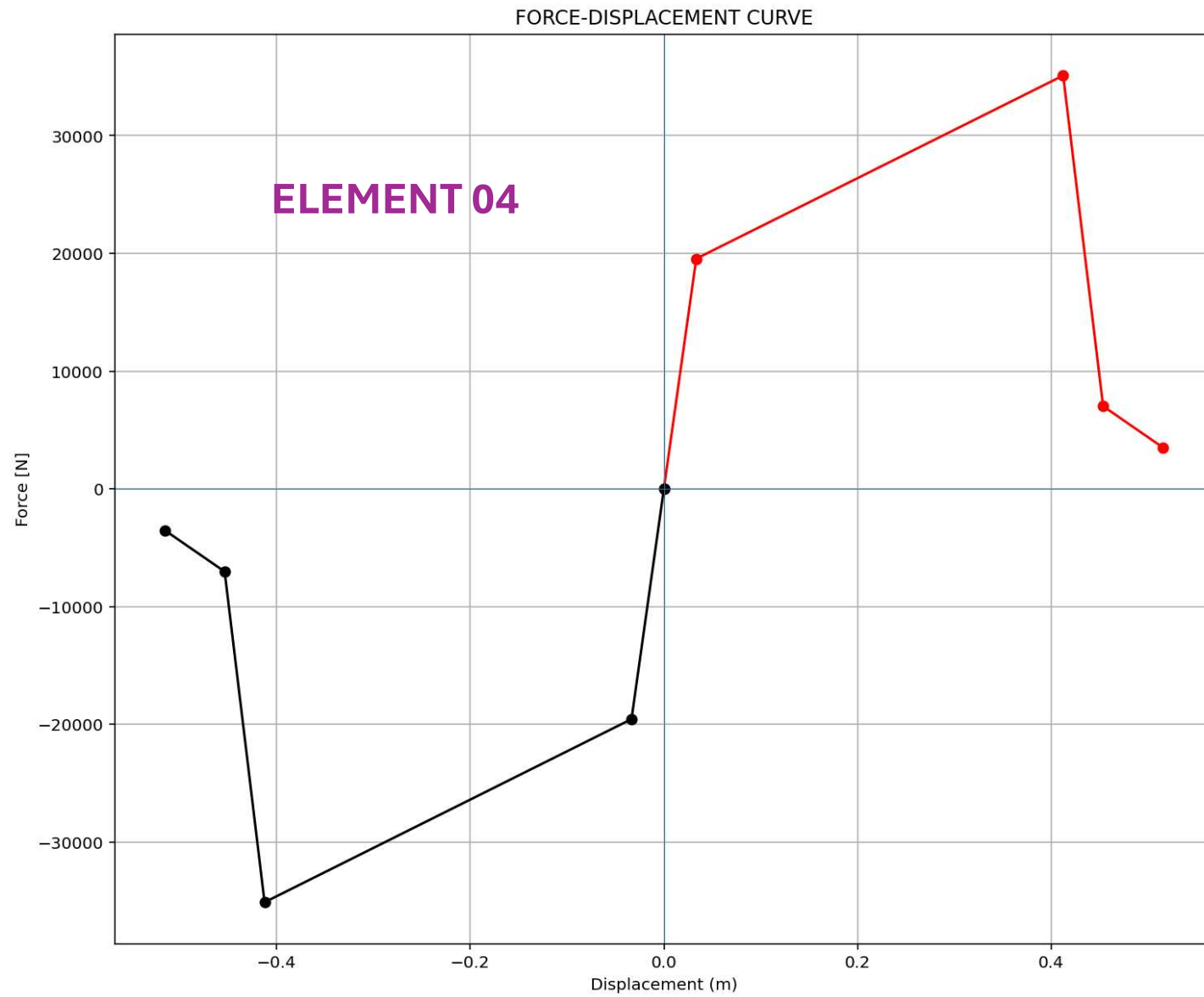


INELASTIC MATERIAL









STATIC ANALYSIS

Py (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\ DELL\Desktop\OPENSEES_FILES\+EIGHT_ANALY...3_SDOF\p(t)_POST-FIRE_SDOF_FOUR_ELEMENTS_8_ANA.py

P(t)_POST-FIRE_SDOF..._ELEMENTS_8_ANA.py X

```
1 #####
2 # >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST ME
3 # COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF A POST-FIRE SINGLE-DEG
4 # ELEMENTS STRUCTURE : AN OPENSEES FRAMEWORK FOR STATIC PUSHOVE
5 # STATIC TIME-HISTORY AND DYNAMIC TIME-HISTOR
6 # -----
7 # P(t) = P0 sin(wt)
8 # P(t) = P0 exp(-0.05wt) sin(wt)
9 # -----
10 # ASSESSMENT OF DUCTILITY DAMAGE INDICES FOR STRUCTURAL ELEMENTS AN
11 # OF ENERGY DISSIPATION CAPACITY IND
12 # -----
13 # THIS PYTHON SCRIPT WRITTEN BY SALAR DELAVAR GHASHGHAEE
14 # EMAIL: salar.d.ghashghaei@gmail.com
15 #####
16 """
17 Assume we have a structure where, after a fire, certain regions of
18 the structural members have lost their initial material properties
19 (yield strength and modulus of elasticity), and the entire column
20 member. Now, we assess the strength of this structure by performing eight
21
22 Nonlinear Seismic Performance Assessment of a SDOF:
23 An OpenSeesPy Framework for Material and Geometric Nonlinearity Under Stati
24
25 This OpenSeesPy script performs rigorous nonlinear static and dynamic analy
26 for performance-based earthquake engineering. The 2D model incorporates ba
27 (elastic-perfectly plastic or hysteretic steel with strain hardening)
28 and geometric nonlinearity (corotational truss formulation) to capture P-d
29 and large displacements-essential for collapse assessment.
30
31 Eight analysis protocols are implemented:
32 (1) [PERIOD] : Structural Period
33 (2) [STATIC] : Gravity load analysis establishing dead load state
34 (3) [PUSHOVER] : Displacement-controlled pushover generating full capacity
```

FORCE-DISPLACEMENT CURVE

Displacement (m)	Force (N)	Path Type
-0.4	-10000	Loading
-0.3	-80000	Loading
0.0	0	Loading
0.3	80000	Loading
0.4	10000	Unloading
0.35	15000	Unloading
0.3	18000	Unloading

IPython Console

Files

Help

Variable Explorer

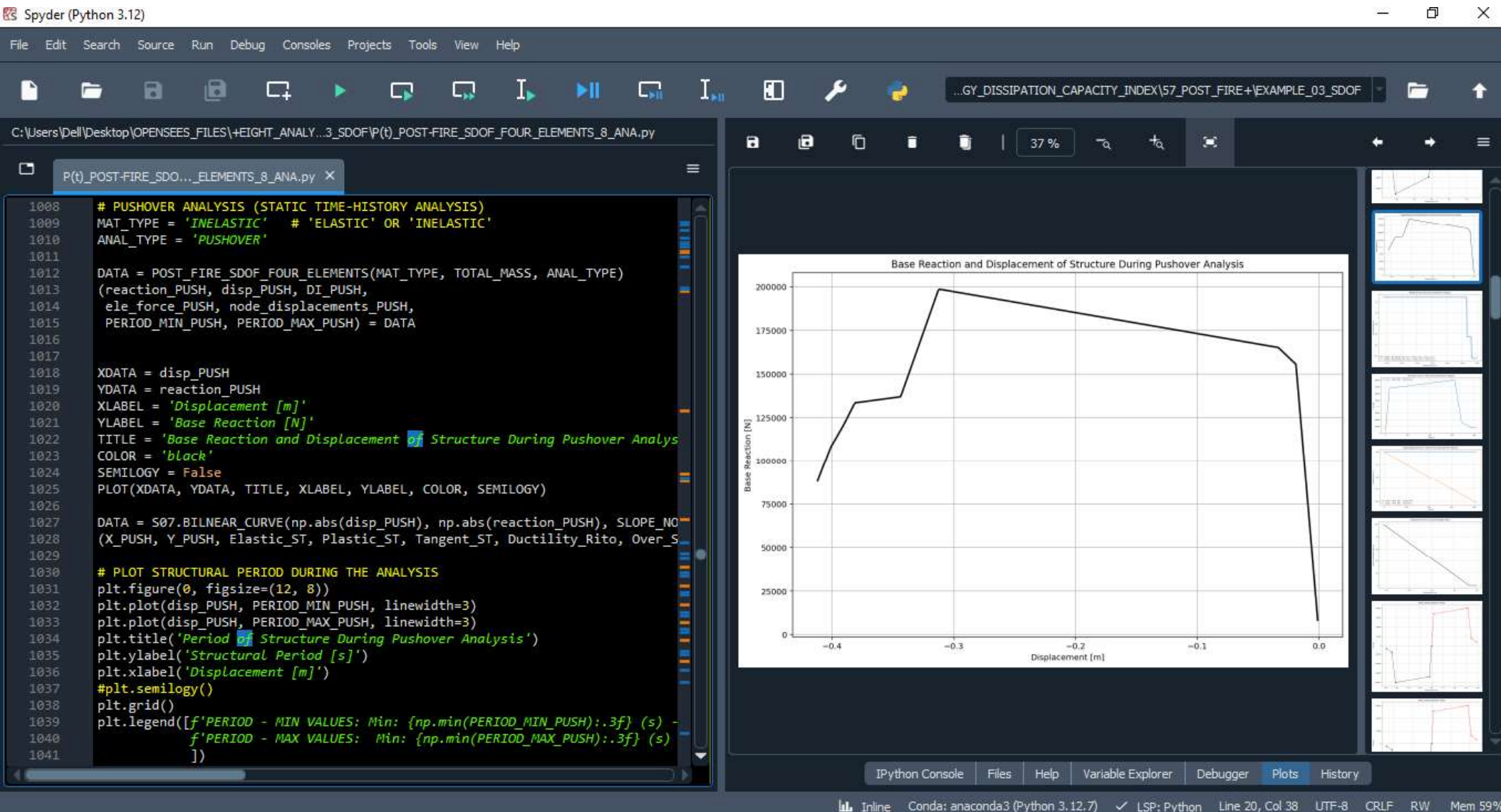
Debugger

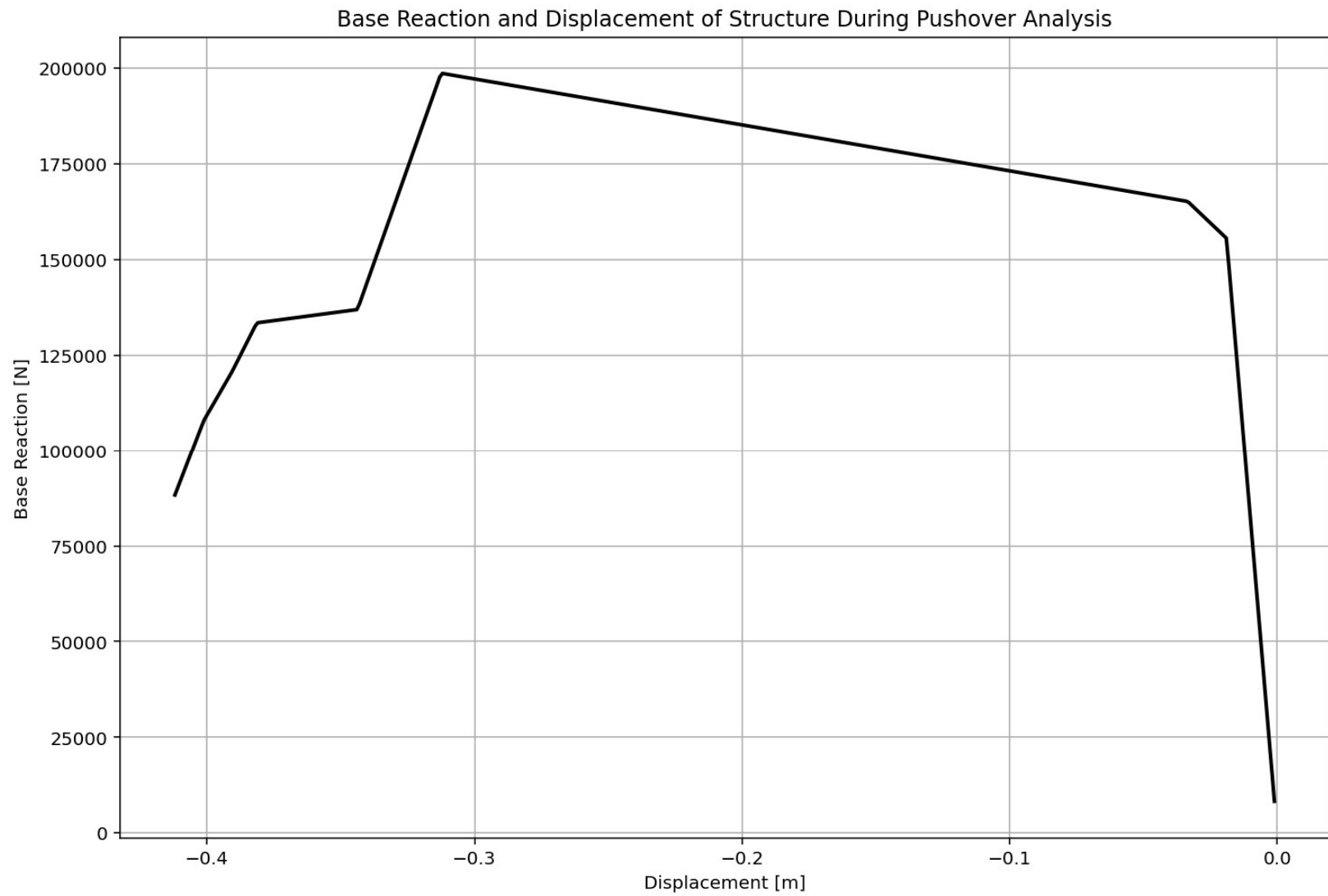
Plots

History

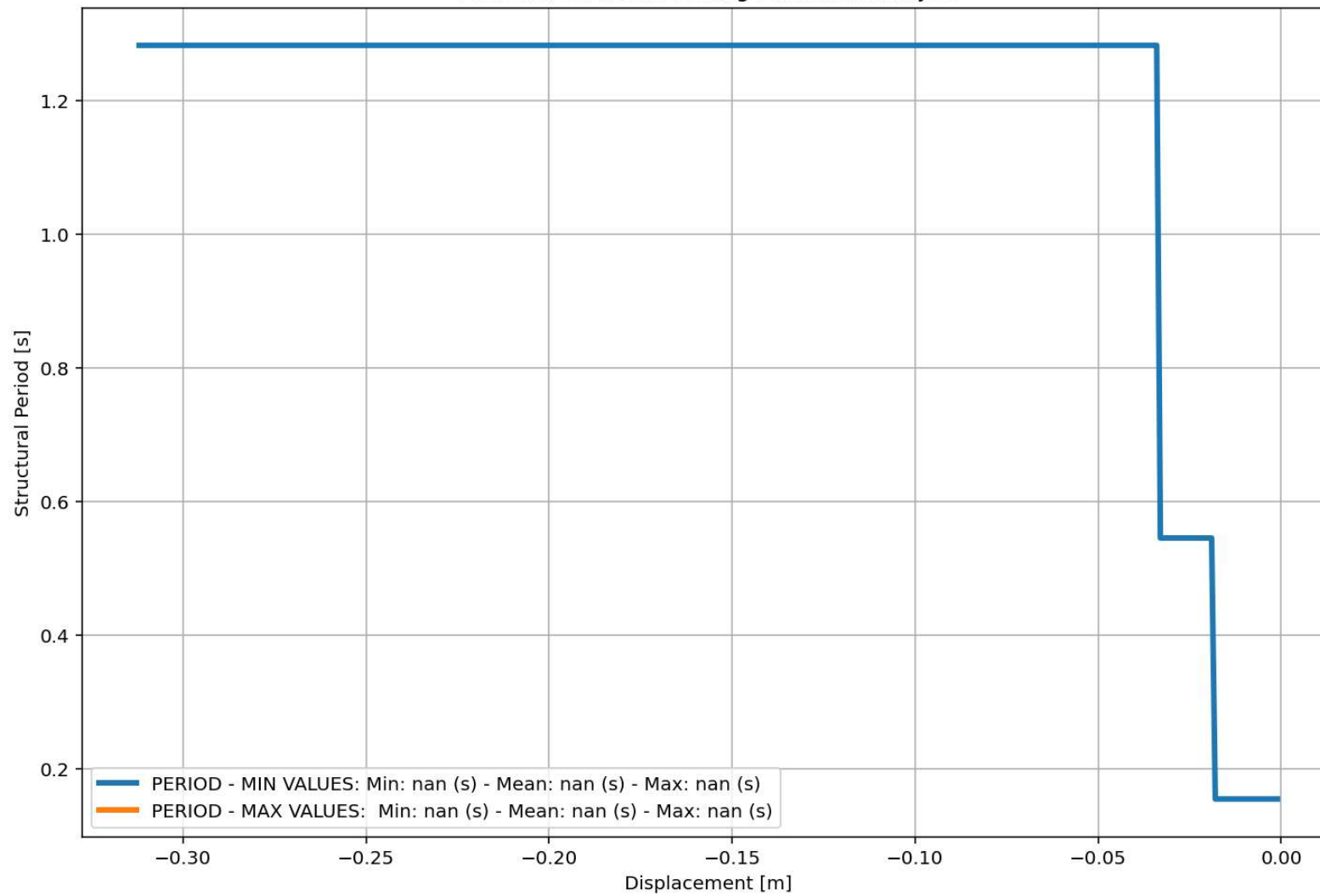
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 20, Col 38 UTF-8 CRLF RW Mem 59%

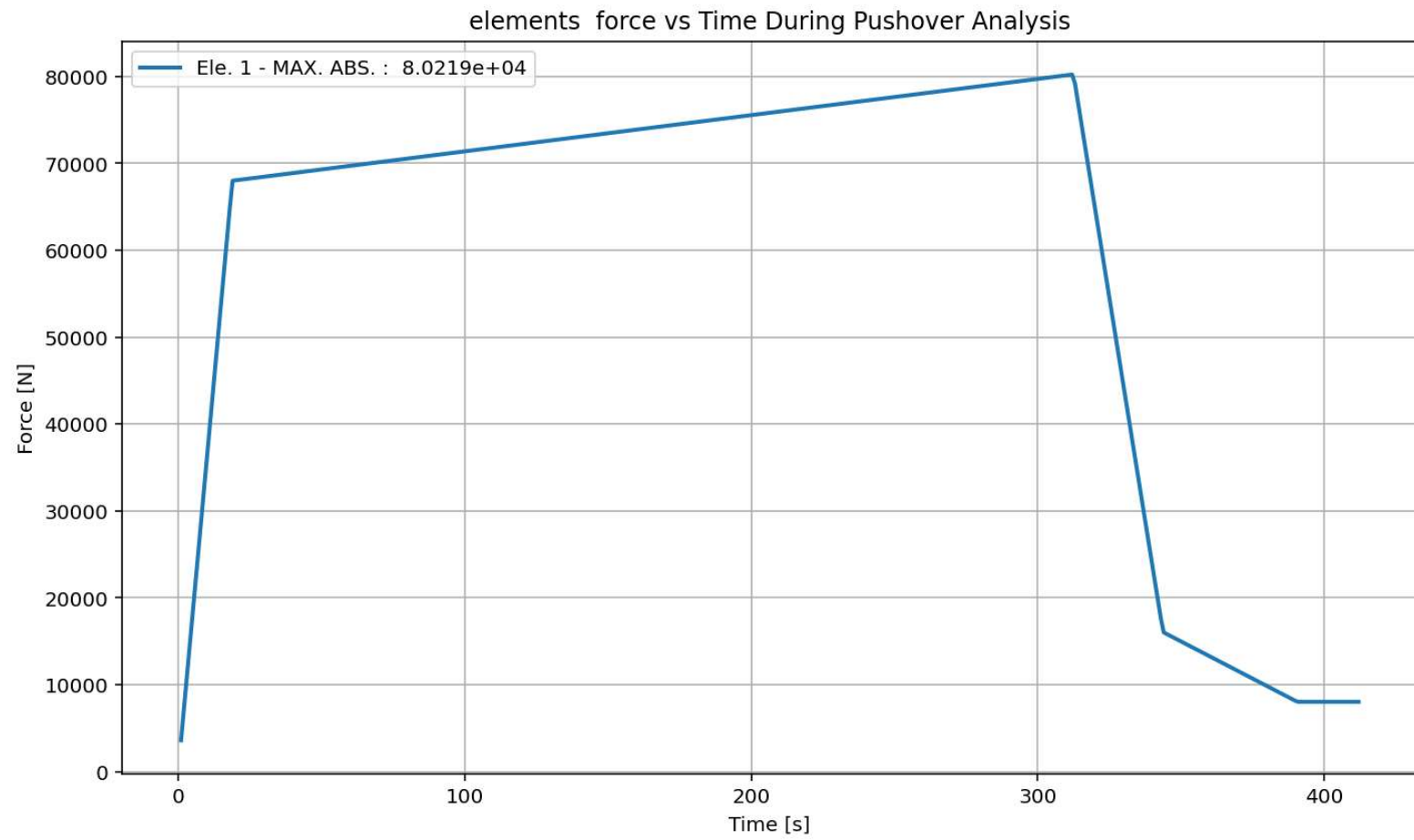
PUSHOVER ANALYSIS

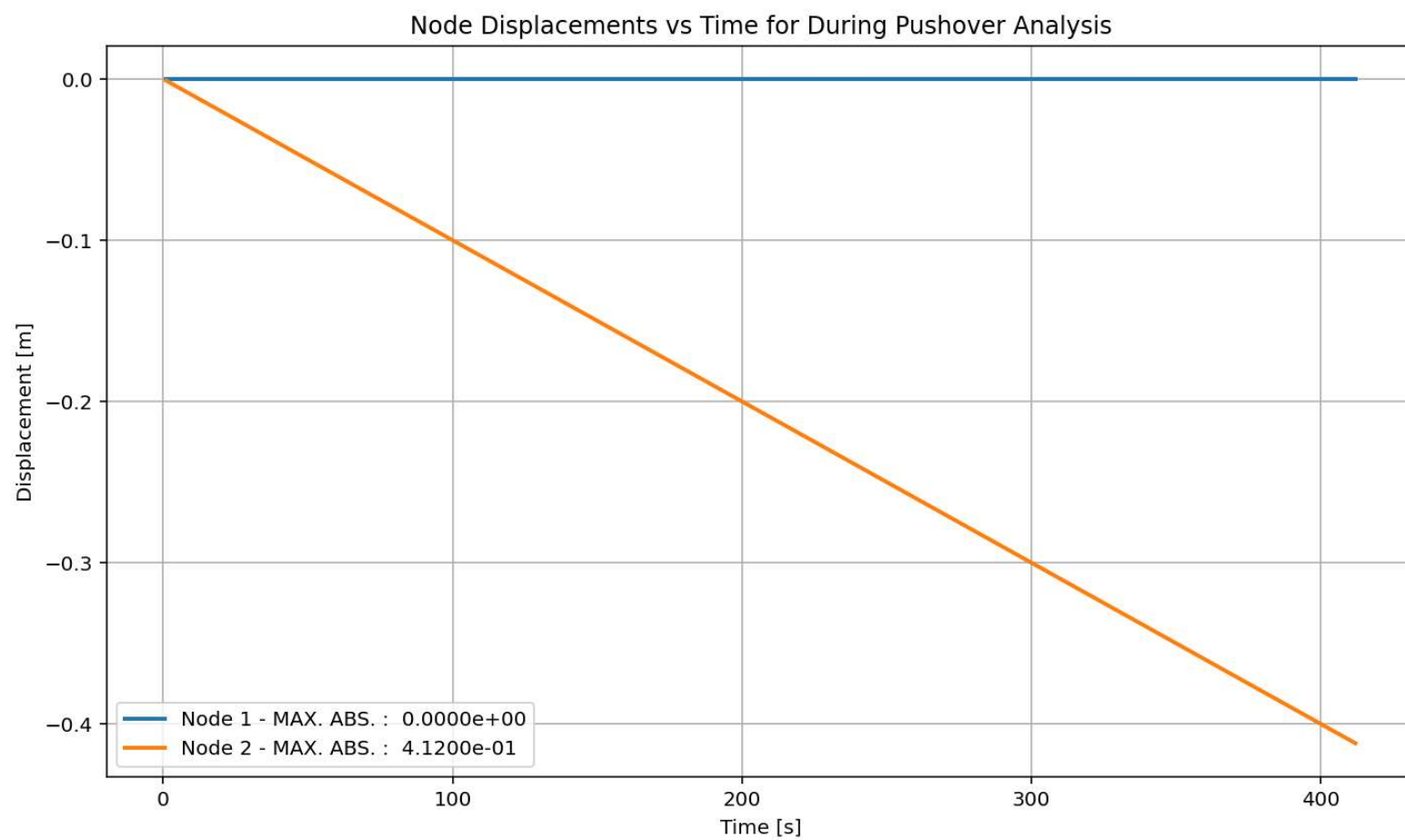


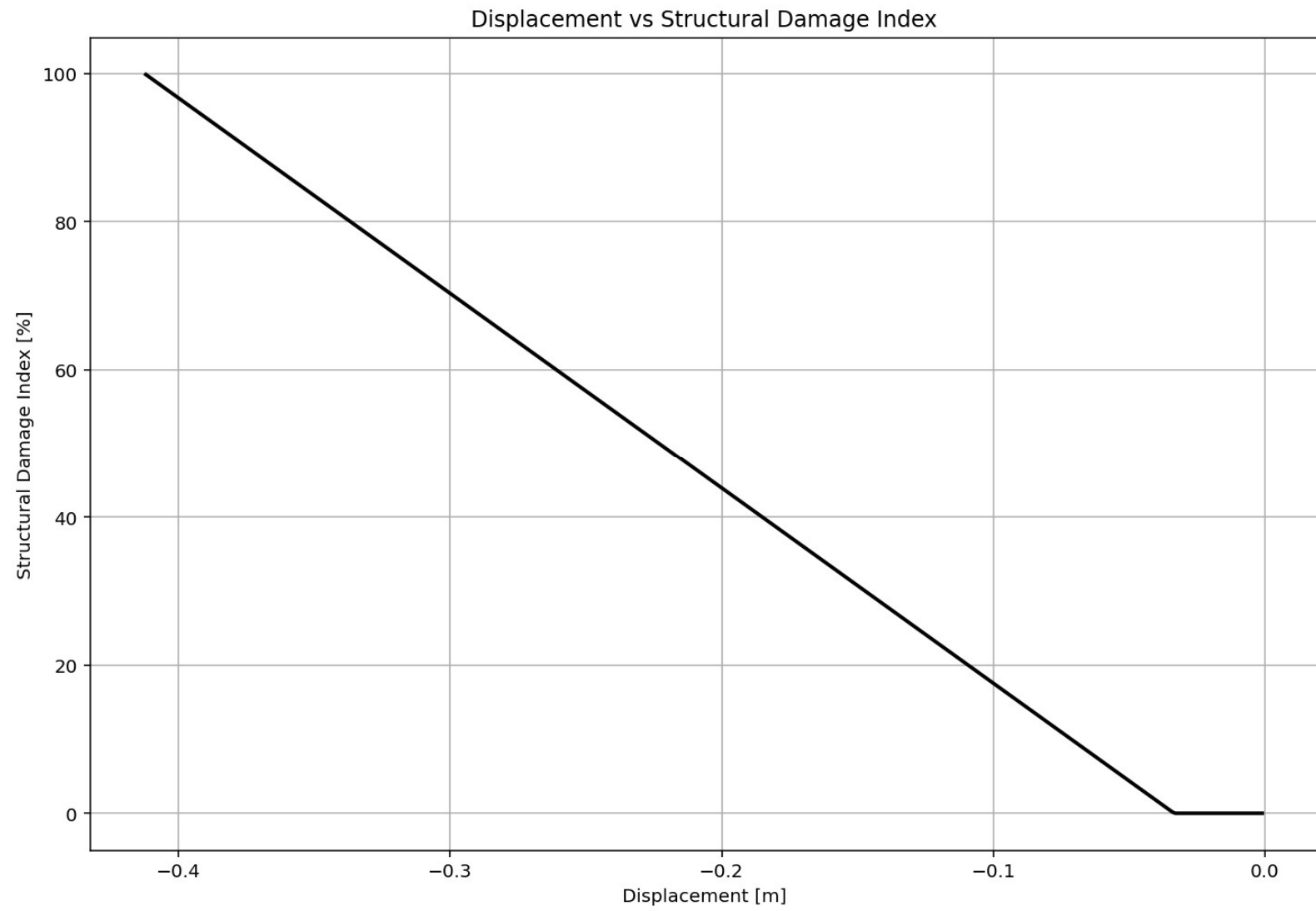


Period of Structure During Pushover Analysis







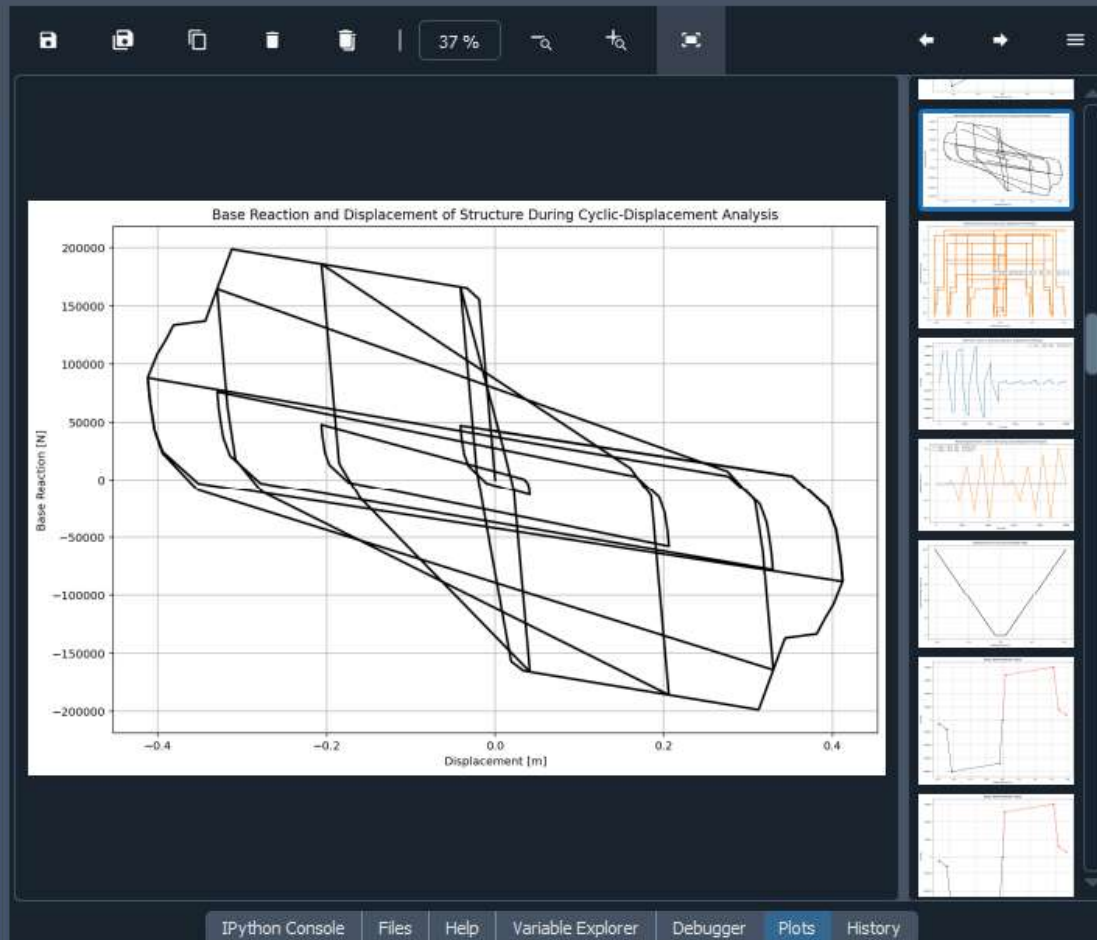


CYCLIC DISPLACEMENT ANALYSIS

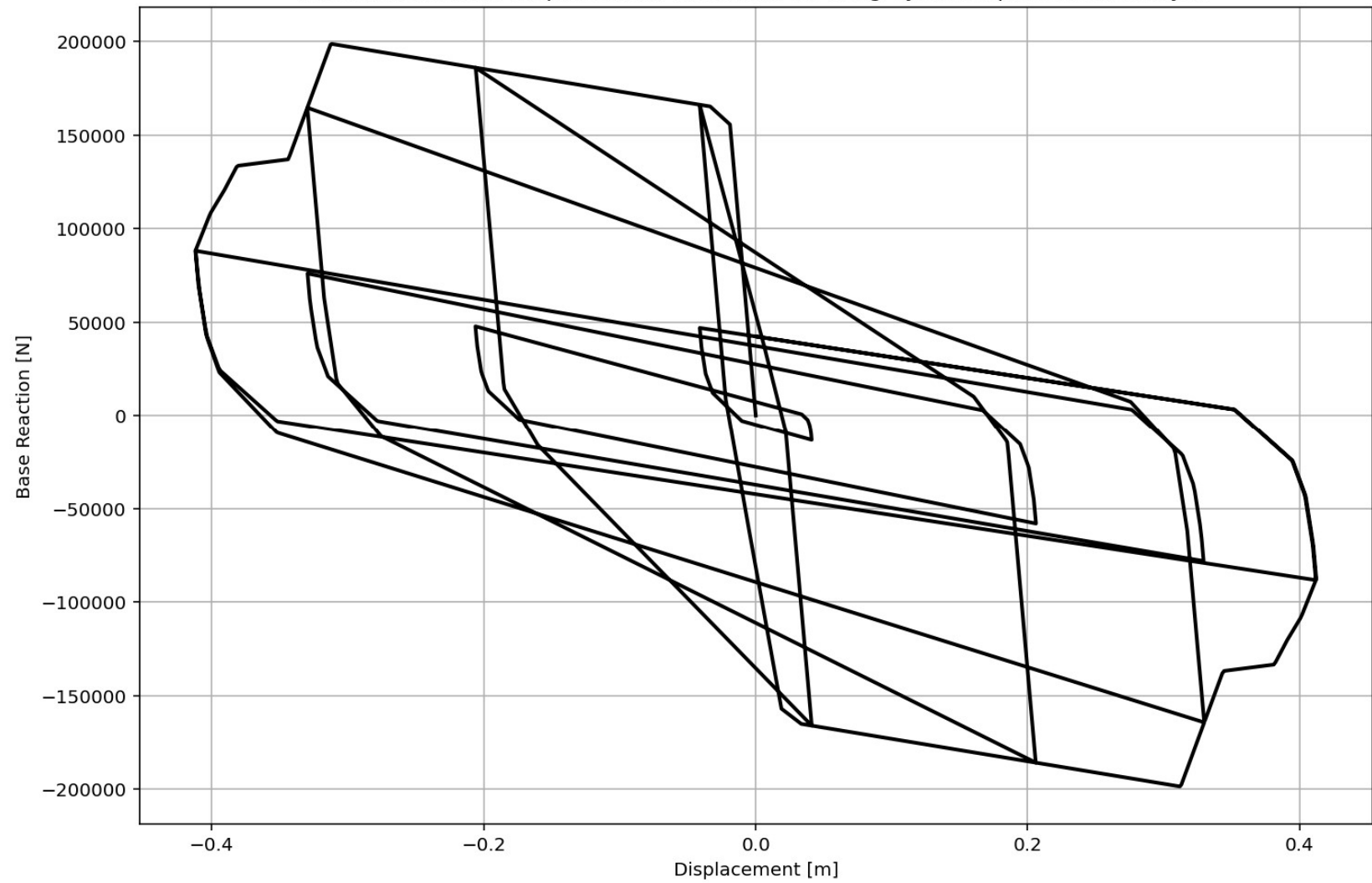
```

1066 # CYCLIC DISPLACEMENT ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
1067 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1068 ANAL_TYPE = 'CYCLIC_DISPLACEMENT'
1069
1070 DATA = POST_FIRE_SDOF_FOUR_ELEMENTS(MAT_TYPE, TOTAL_MASS, ANAL_TYPE)
1071 (reaction_CP, disp_CP, DI_CP,
1072 ele_force_CP, node_displacements_CP,
1073 PERIOD_MIN_CP, PERIOD_MAX_CP) = DATA
1074
1075
1076 XDATA = disp_CP
1077 YDATA = reaction_CP
1078 XLABEL = 'Displacement [m]'
1079 YLABEL = 'Base Reaction [N]'
1080 TITLE = 'Base Reaction and Displacement of Structure During Cyclic-Displace
1081 COLOR = 'black'
1082 SEMIOLOGY = False
1083 PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMIOLOGY)
1084
1085 # PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
1086 plt.figure(0, figsize=(12, 8))
1087 plt.plot(disp_CP, PERIOD_MIN_CP, linewidth=3)
1088 plt.plot(disp_CP, PERIOD_MAX_CP, linewidth=3)
1089 plt.title('Period of Structure During Cyclic Displacement Analysis')
1090 plt.ylabel('Structural Period [s]')
1091 plt.xlabel('Displacement [m]')
1092 #plt.semilogy()
1093 plt.grid()
1094 plt.legend([f'PERIOD - MIN VALUES: Min: {np.min(PERIOD_MIN_CP):.3f} (s) - M
1095 f'PERIOD - MAX VALUES: Min: {np.min(PERIOD_MAX_CP):.3f} (s) - M
1096 ])
1097 plt.show()
1098
1099

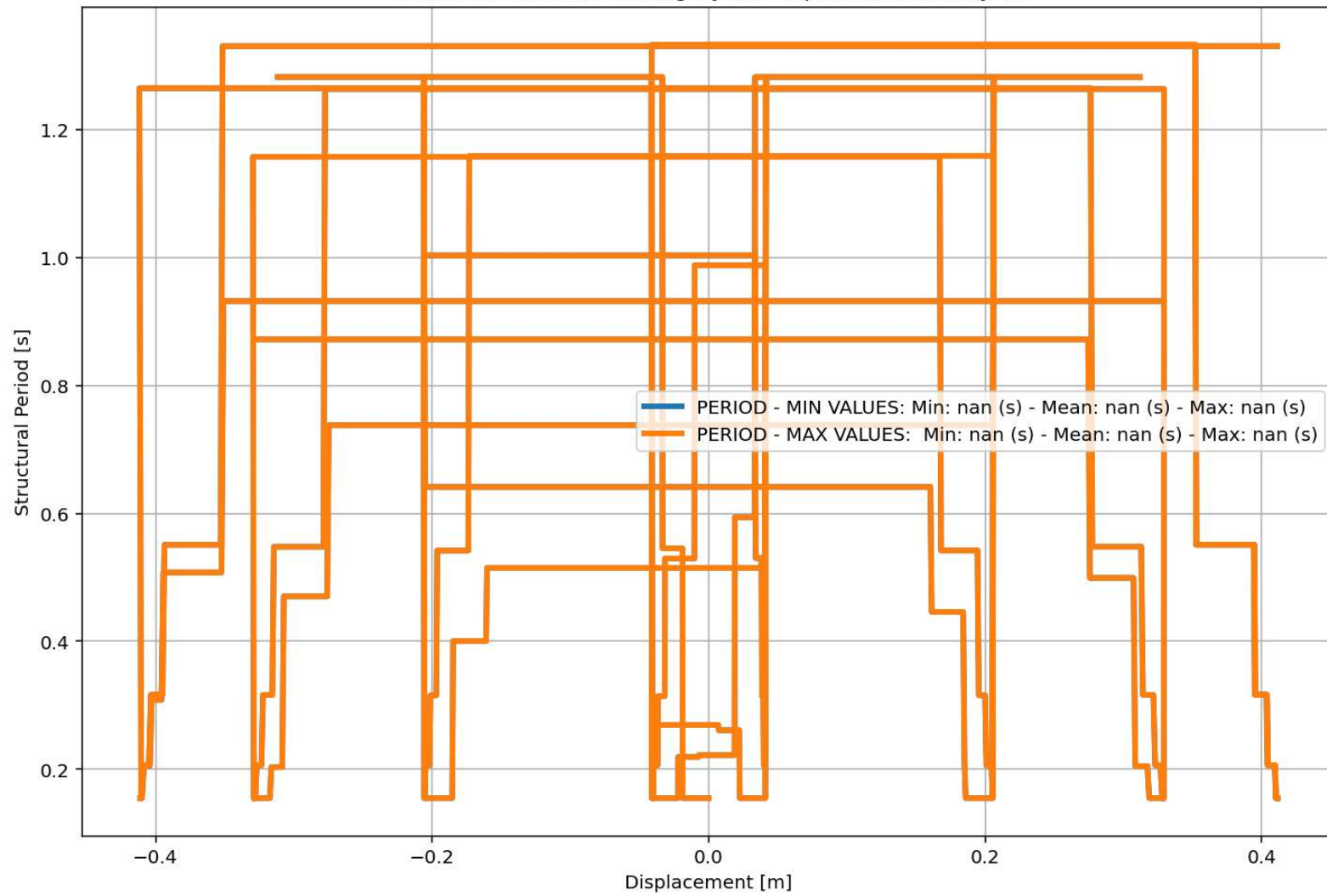
```

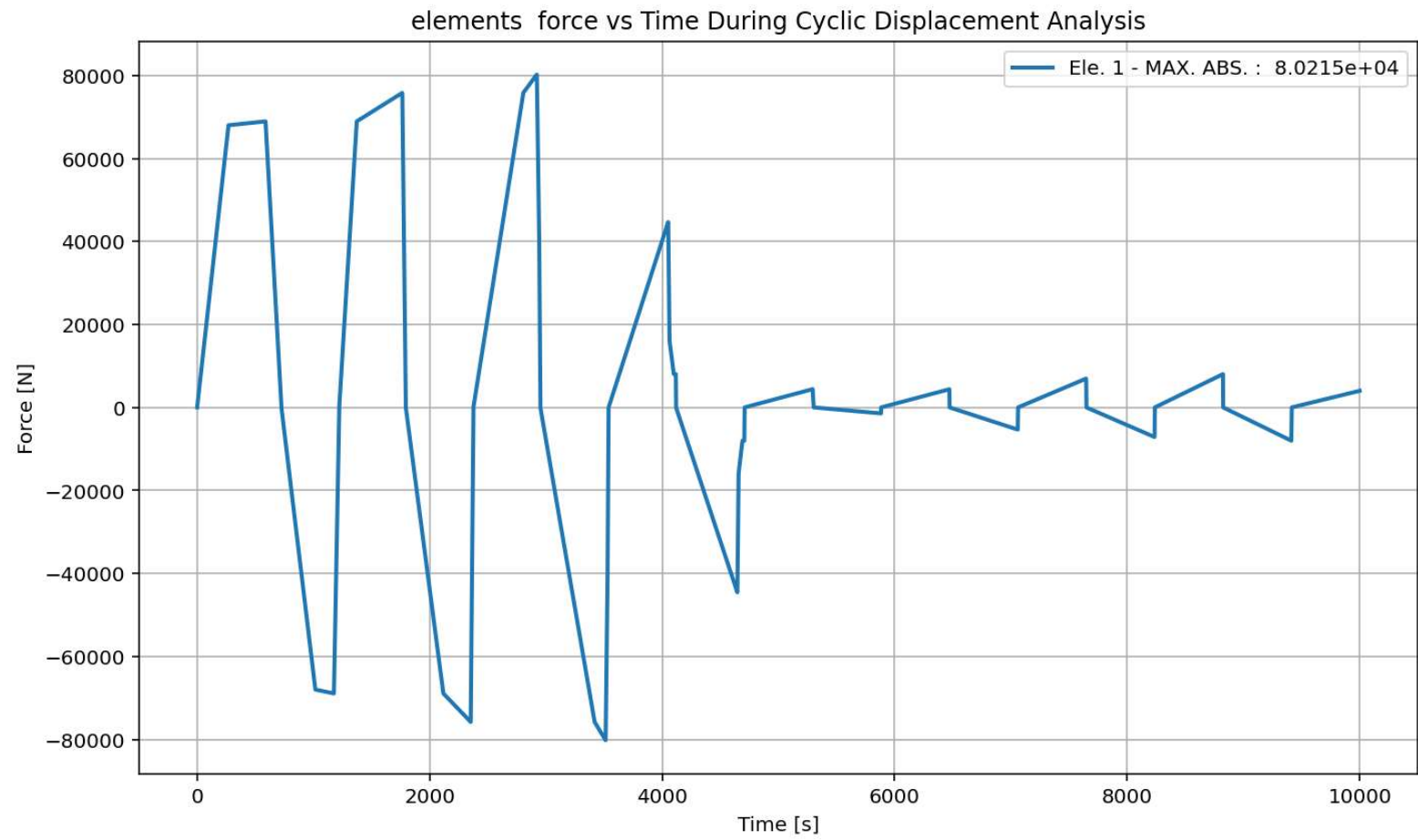


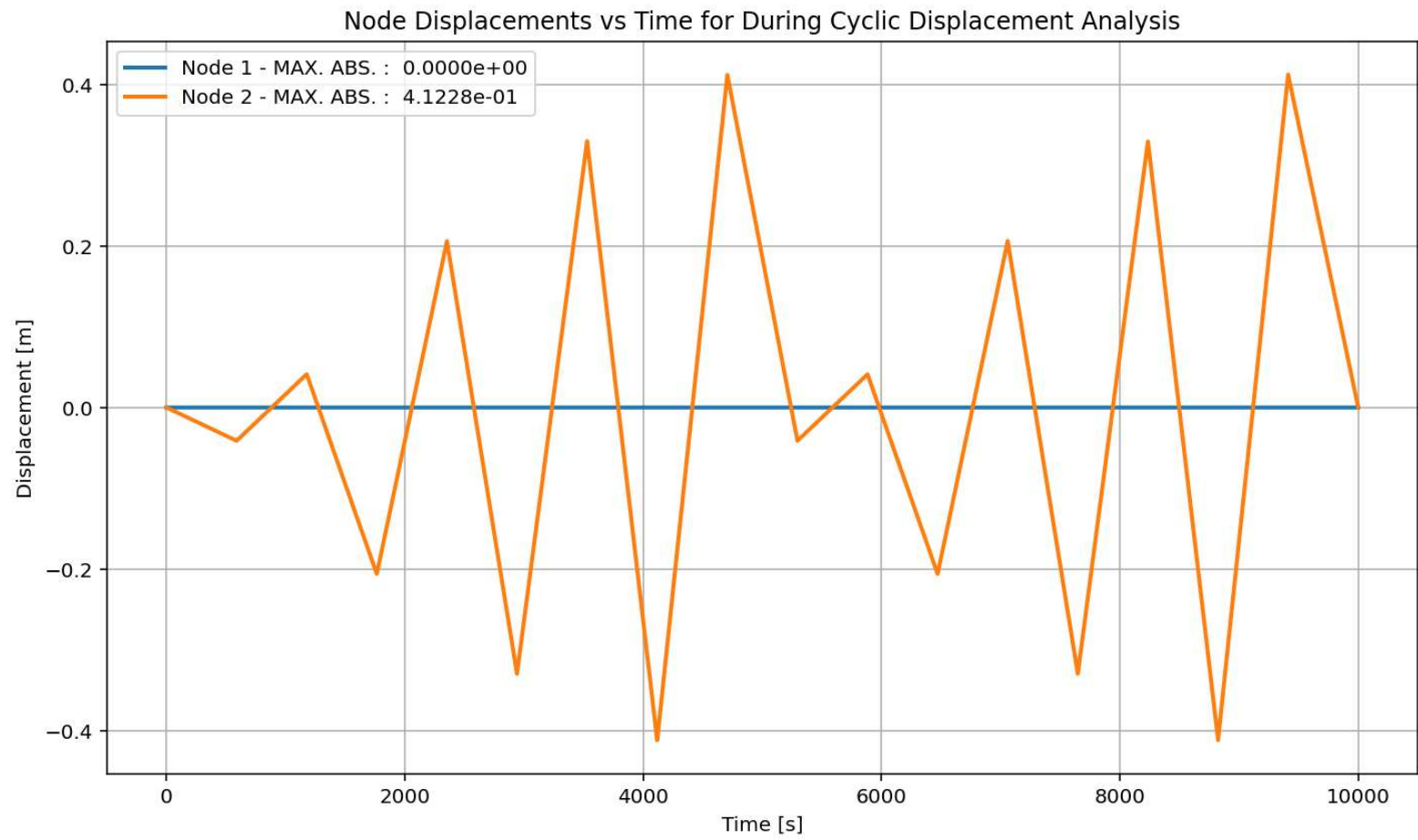
Base Reaction and Displacement of Structure During Cyclic-Displacement Analysis

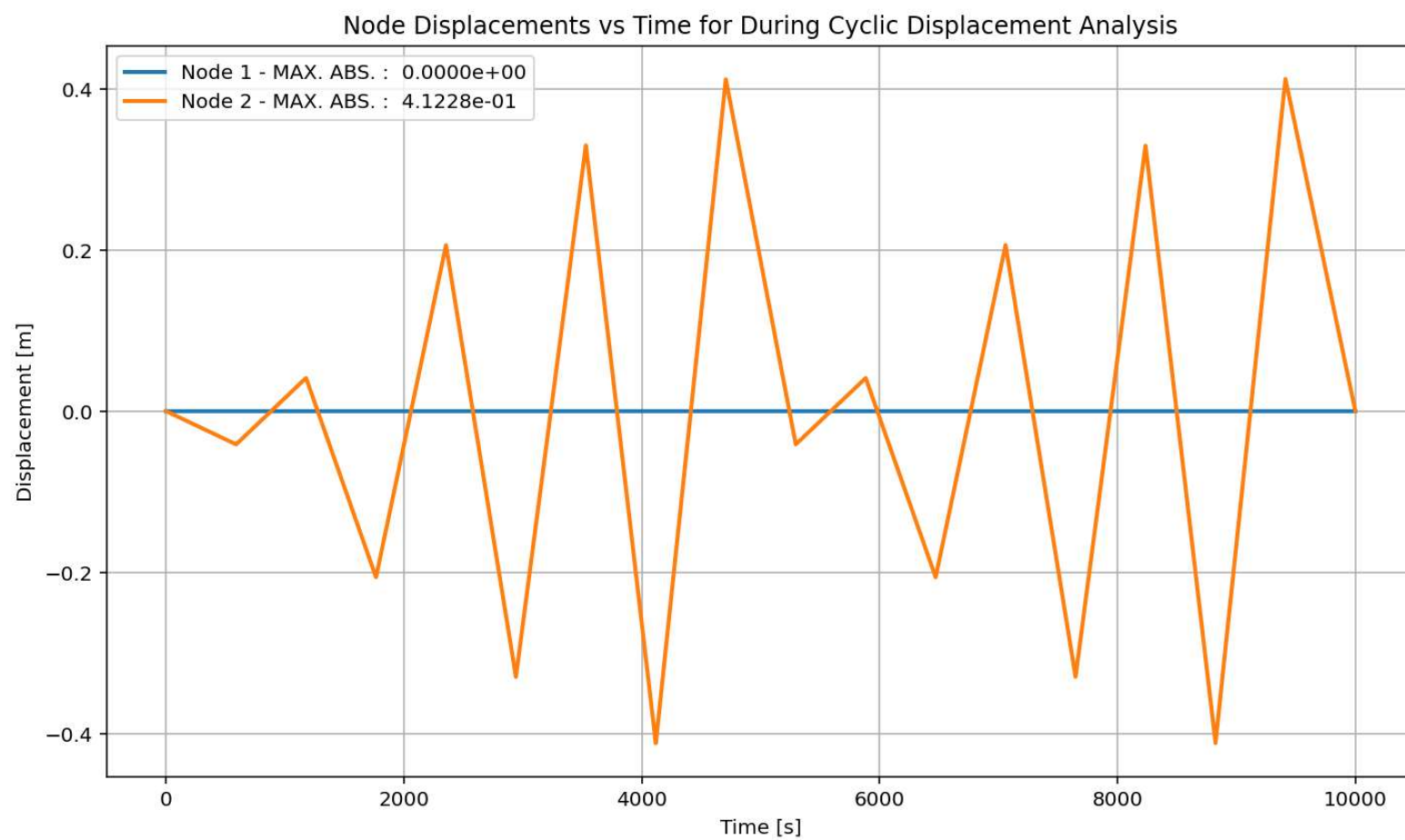


Period of Structure During Cyclic Displacement Analysis

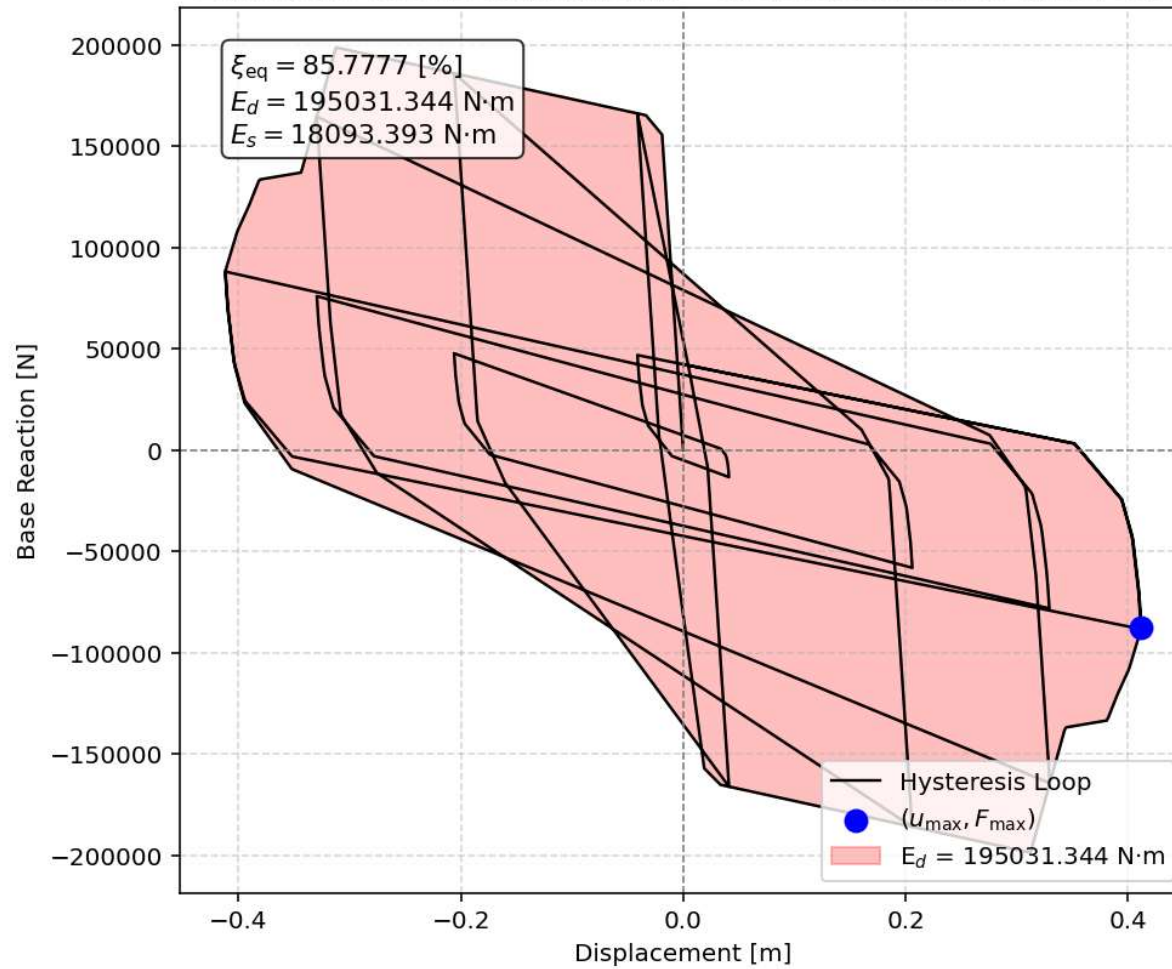




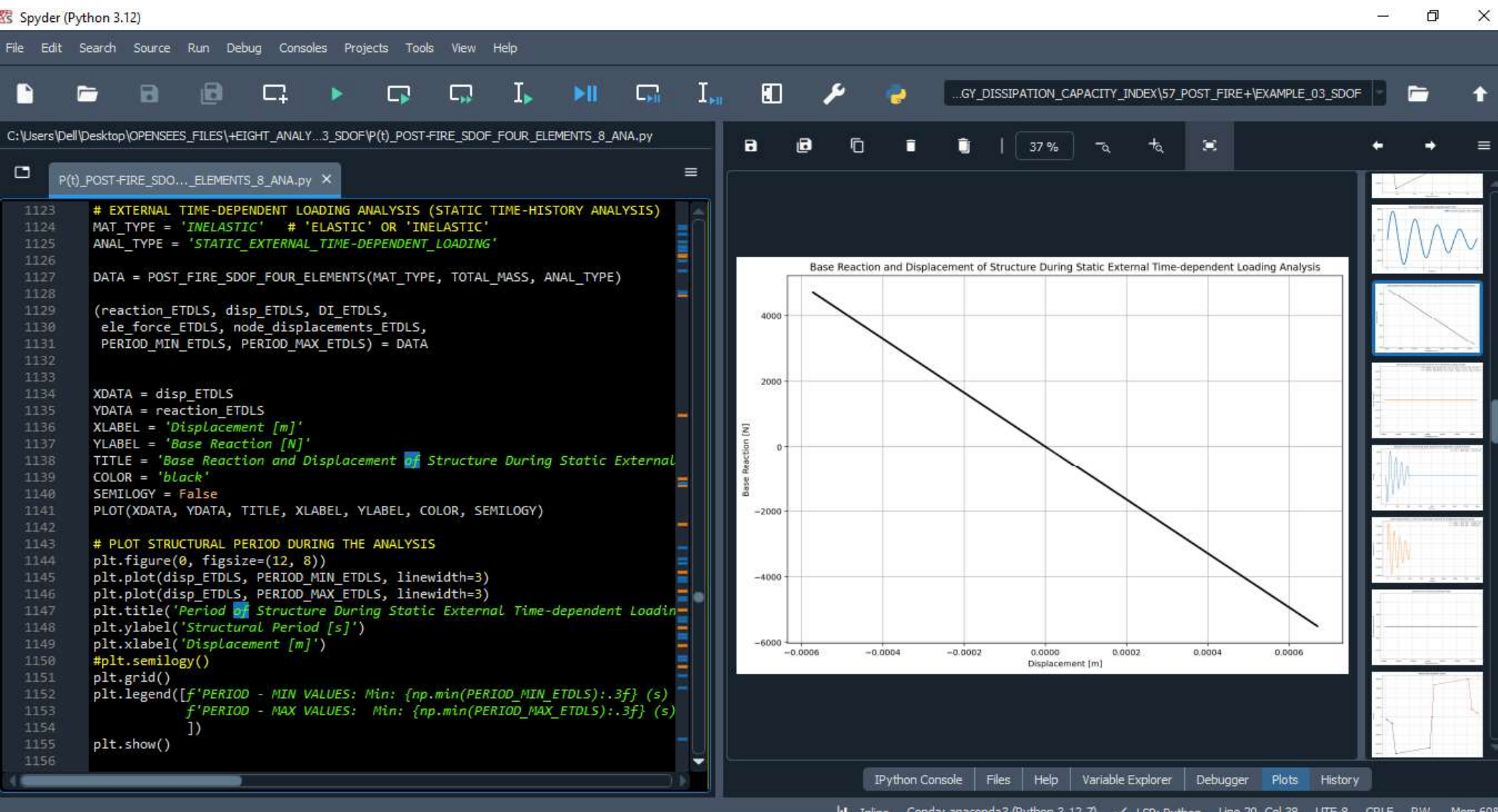


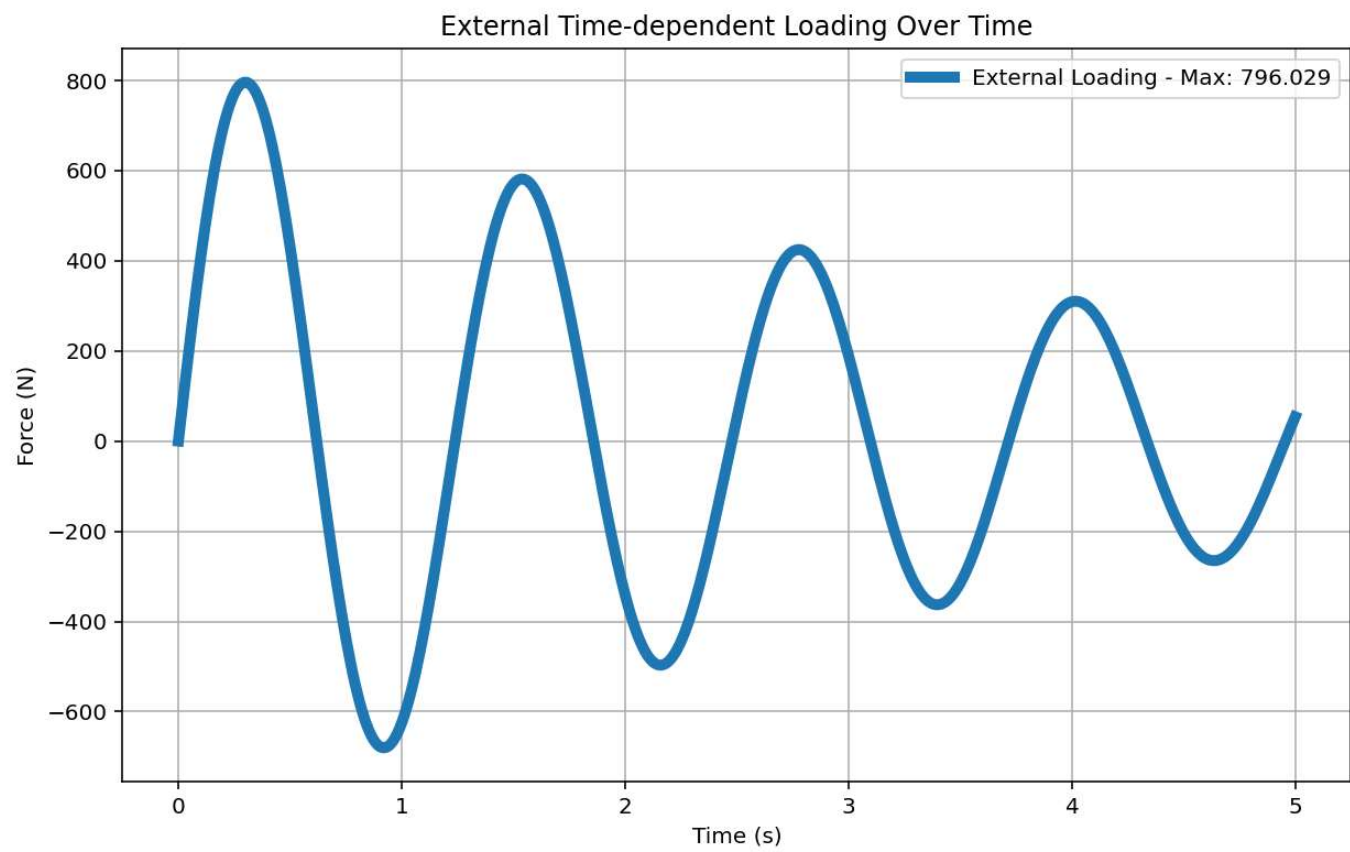


Equivalent viscous damping ratio - Cyclic Displacement Hysteresis

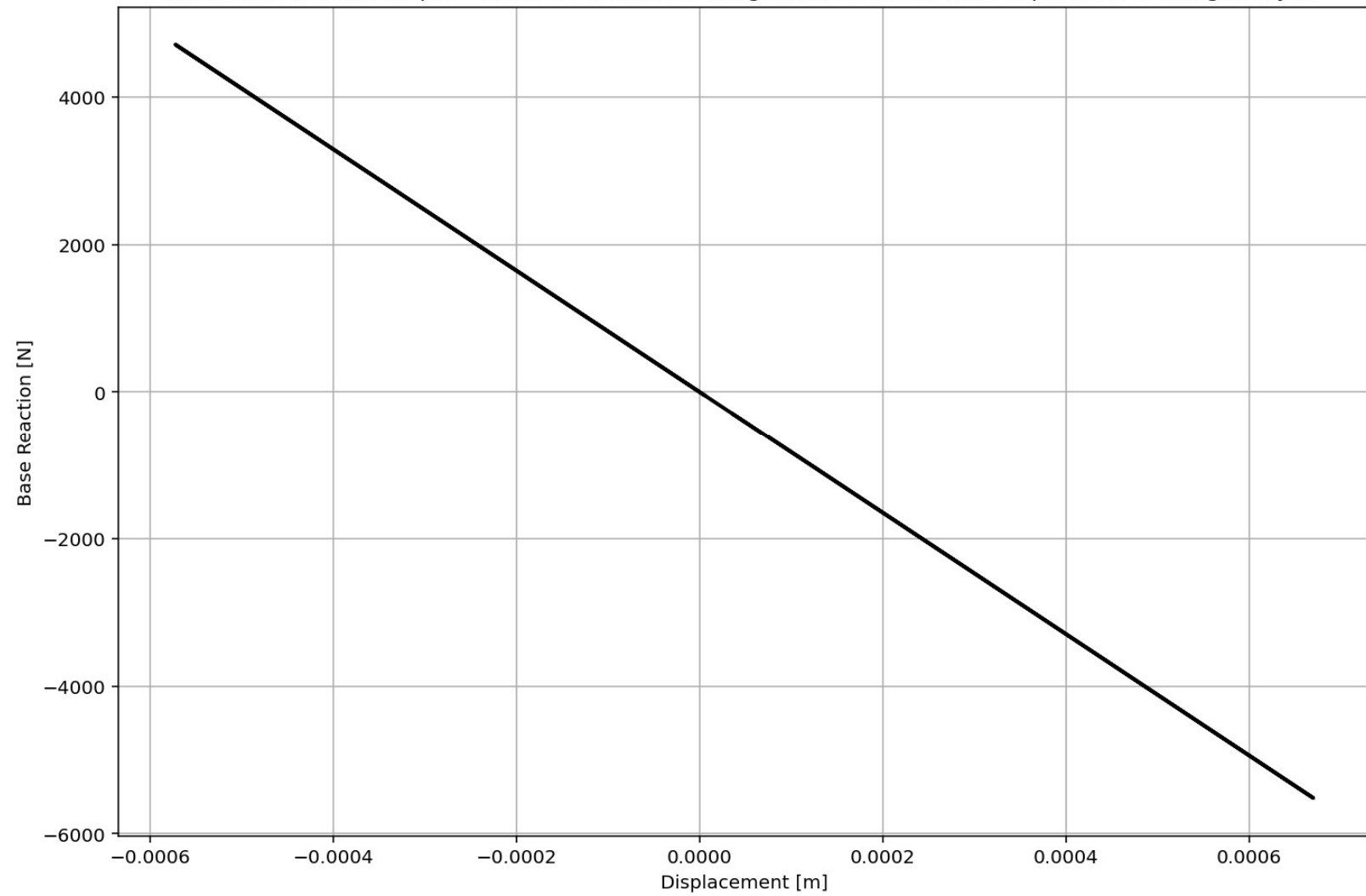


STATIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS

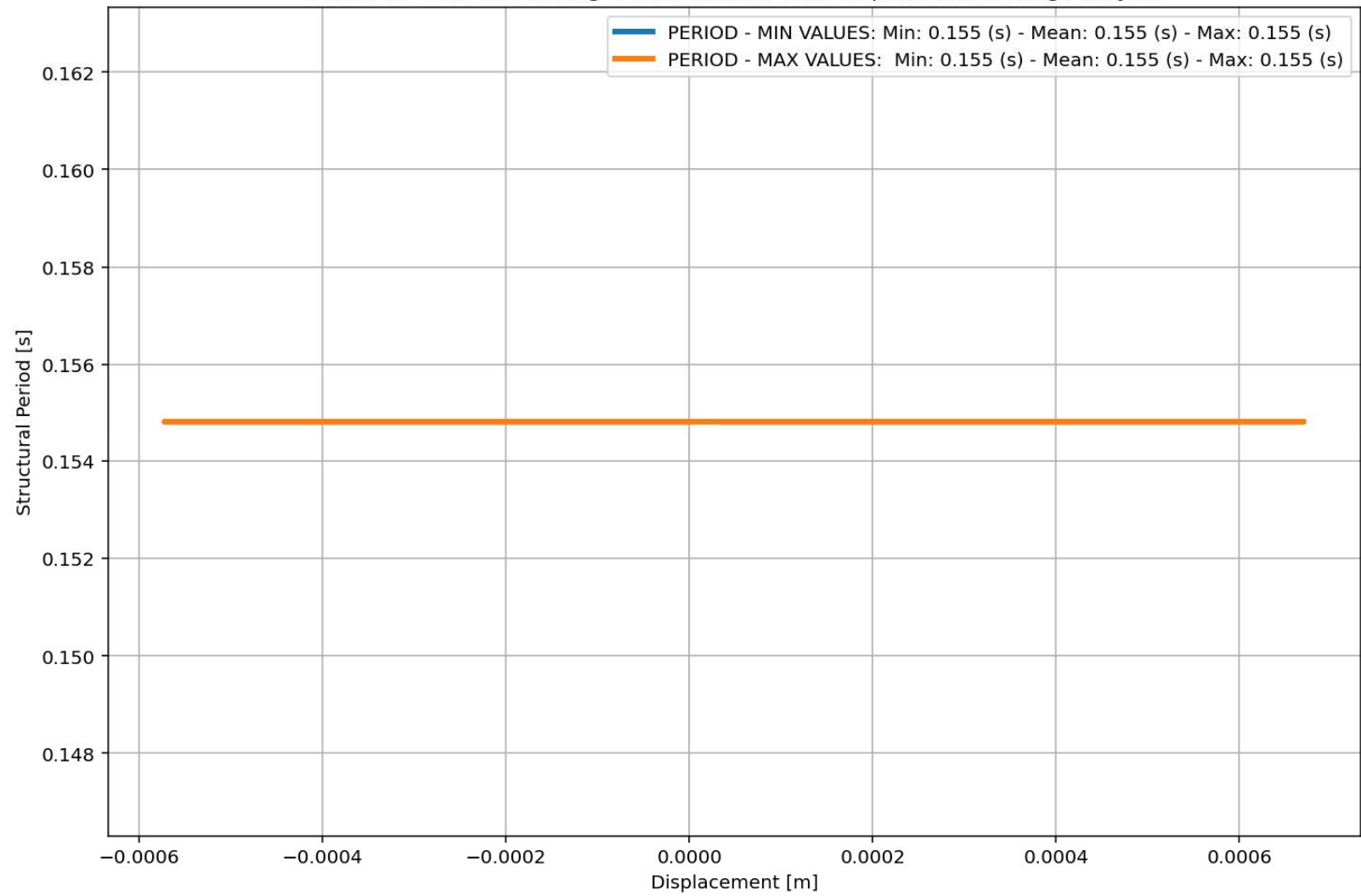


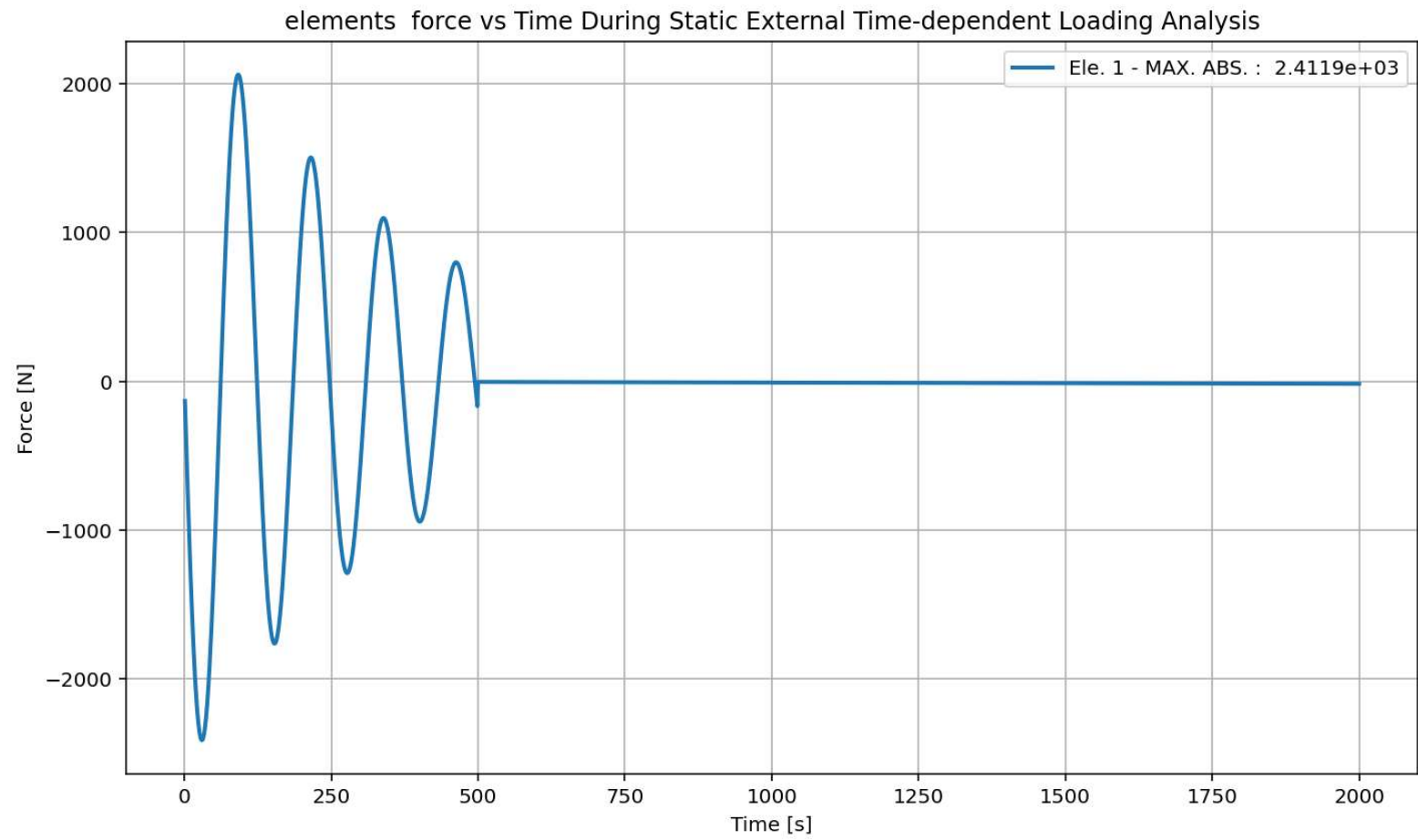


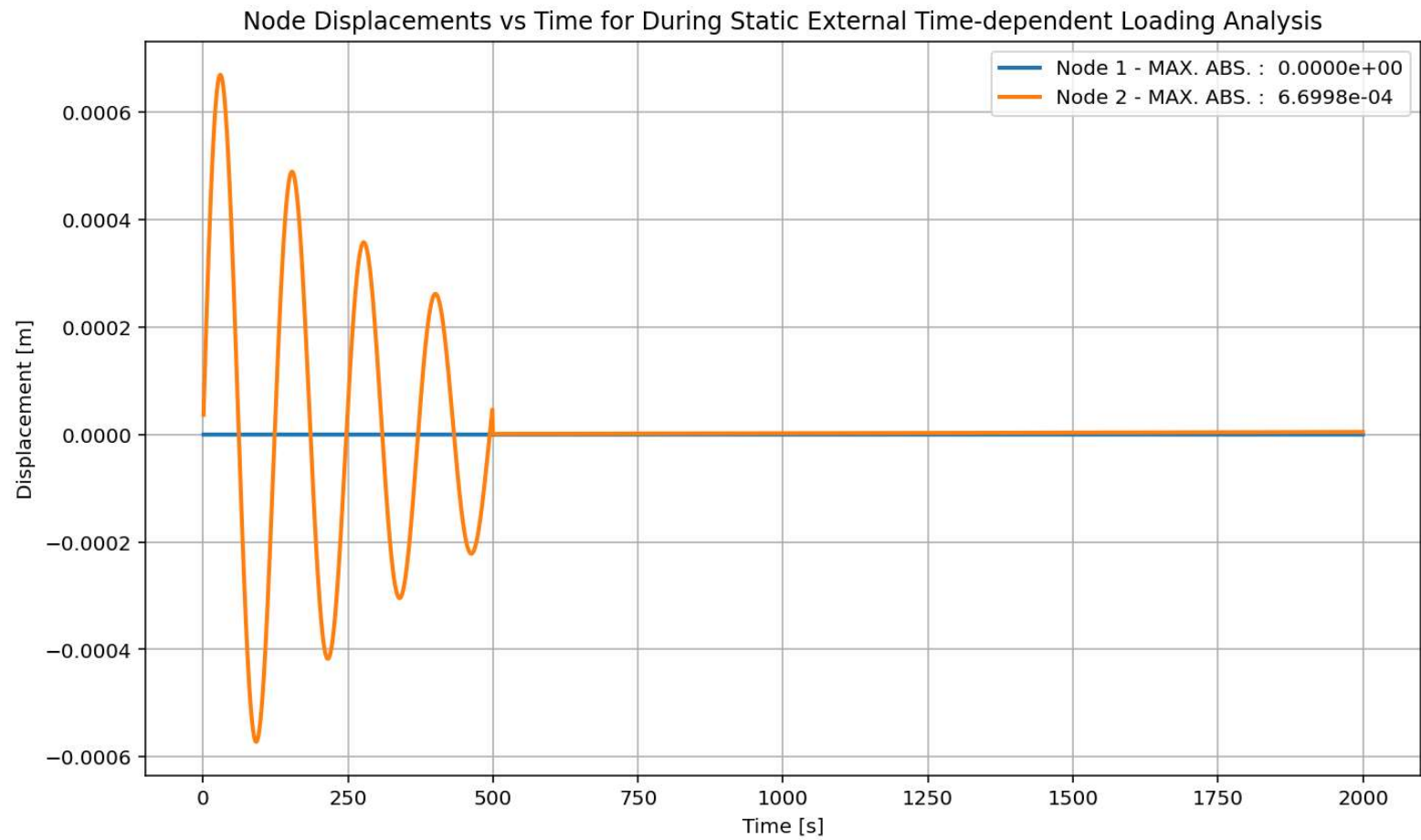
Base Reaction and Displacement of Structure During Static External Time-dependent Loading Analysis

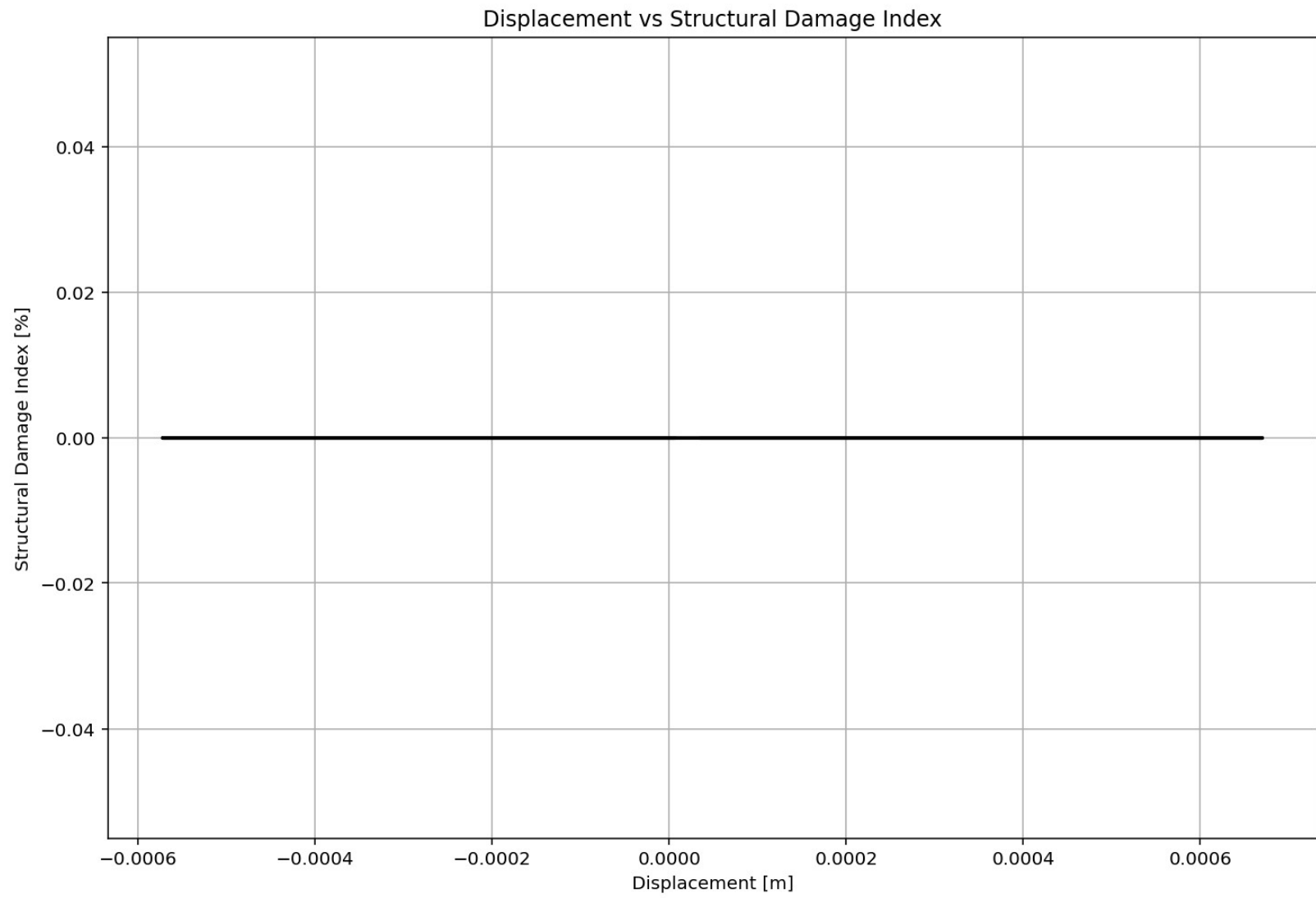


Period of Structure During Static External Time-dependent Loading Analysis







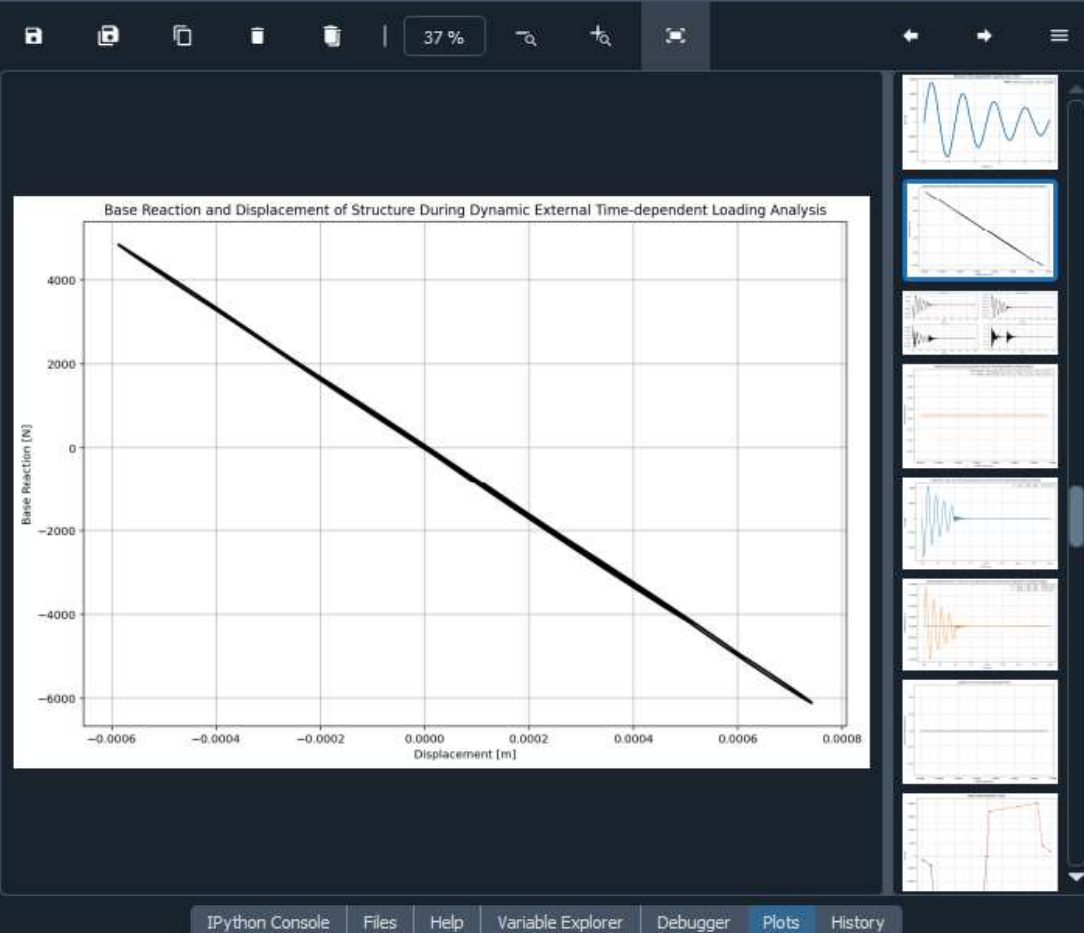


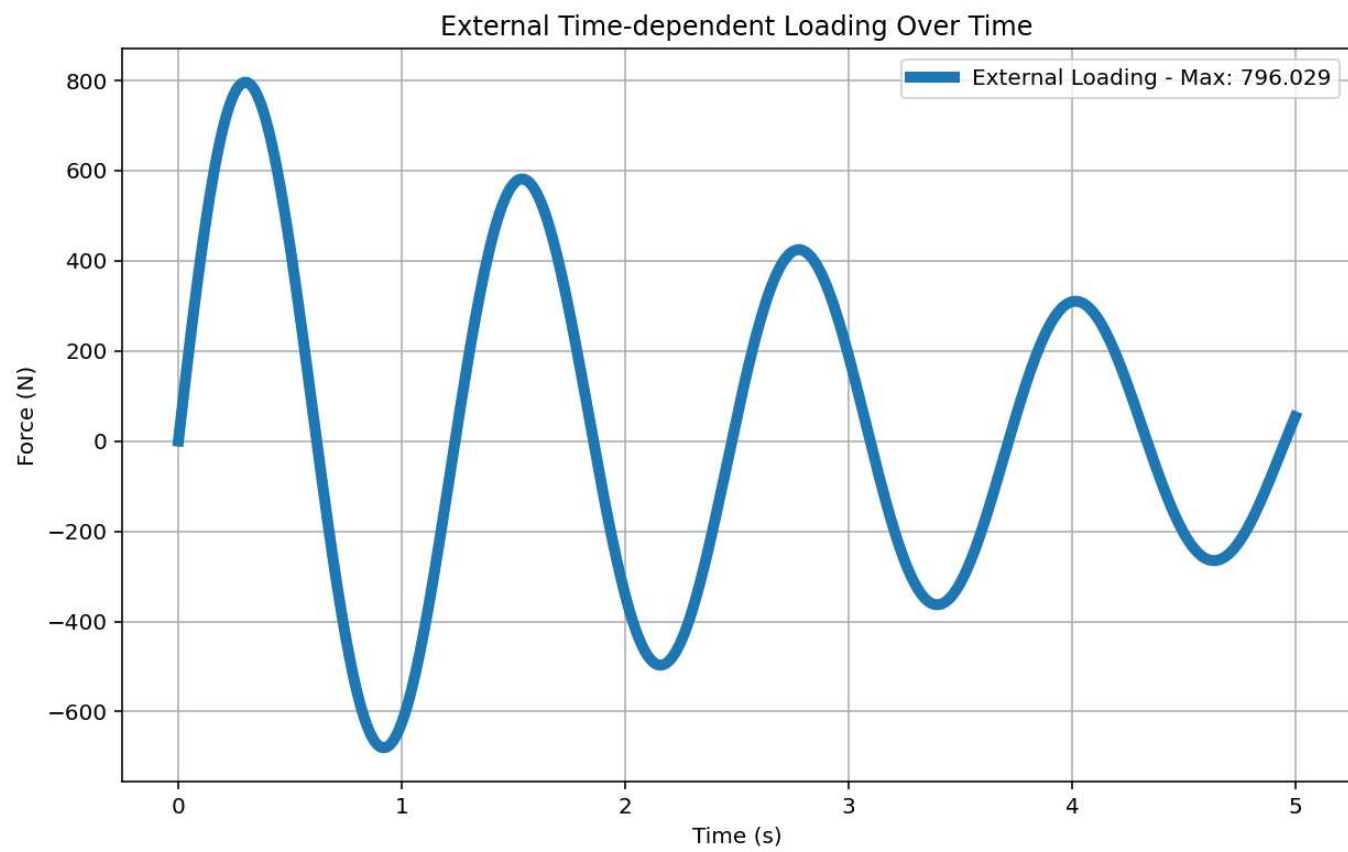
DYNAMIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS

```

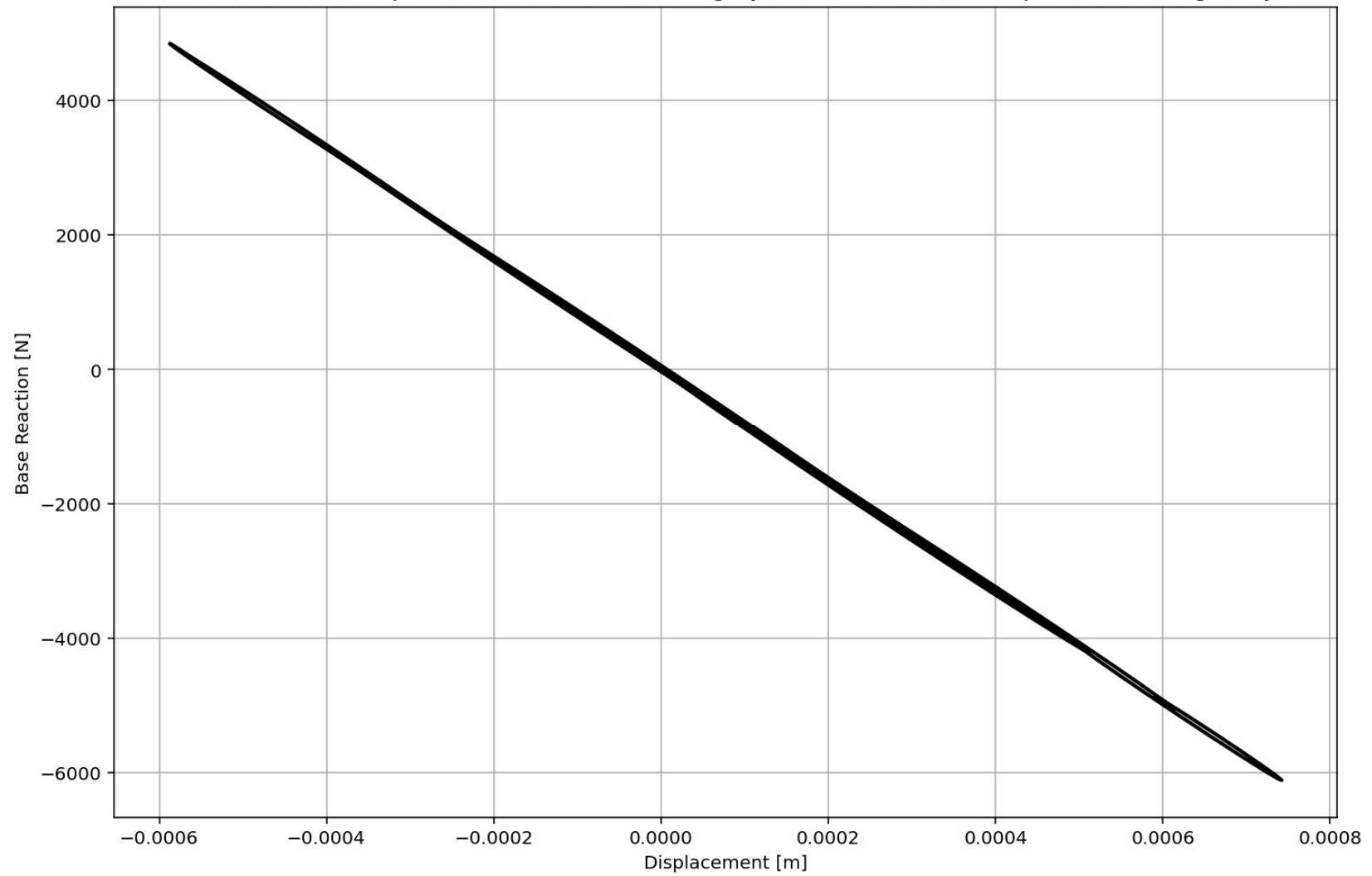
1181 # EXTERNAL TIME-DEPENDENT LOADING ANALYSIS (DYNAMIC TIME-HISTORY ANALYSIS)
1182 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1183 ANAL_TYPE = 'DYNAMIC_EXTERNAL_TIME-DEPENDENT_LOADING'
1184
1185 DATA = POST_FIRE_SDOF_FOUR_ELEMENTS(MAT_TYPE, TOTAL_MASS, ANAL_TYPE)
1186
1187 (time_ETDLD, reaction_ETDLD, disp_ETDLD, velo_ETDLD, acc_ETDLD, DI_ETDLD,
1188 ele_force_ETDLD, node_displacements_ETDLD,
1189 stiffness, PERIOD, damping_ratio,
1190 PERIOD_MIN_ETDLD, PERIOD_MAX_ETDLD) = DATA
1191
1192
1193
1194 XDATA = disp_ETDLD
1195 YDATA = reaction_ETDLD
1196 XLABEL = 'Displacement [m]'
1197 YLABEL = 'Base Reaction [N]'
1198 TITLE = 'Base Reaction and Displacement of Structure During Dynamic External Time-dependent Loading Analysis'
1199 COLOR = 'black'
1200 SEMILOGY = False
1201 PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
1202
1203 PLOT_TIME_HISTORY(time_ETDLD, reaction_ETDLD, disp_ETDLD, velo_ETDLD, acc_ETDLD)
1204
1205 # PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
1206 plt.figure(0, figsize=(12, 8))
1207 plt.plot(disp_ETDLD, PERIOD_MIN_ETDLD, linewidth=3)
1208 plt.plot(disp_ETDLD, PERIOD_MAX_ETDLD, linewidth=3)
1209 plt.title('Period of Structure During Dynamic External Time-dependent Loading Analysis')
1210 plt.ylabel('Structural Period [s]')
1211 plt.xlabel('Displacement [m]')
1212 #plt.semilogy()
1213 plt.grid()
1214 plt.legend(['PERIOD - MIN VALUES: Min: {np.min(PERIOD_MIN_ETDLD):.3f} (s)'])

```

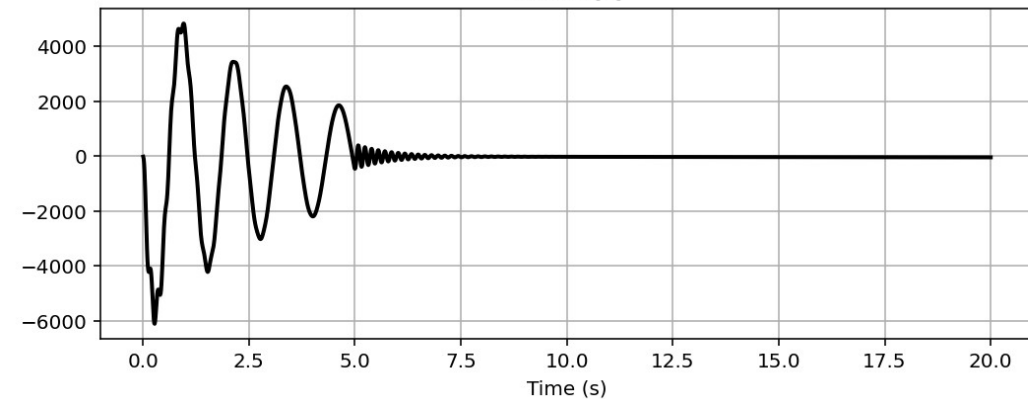




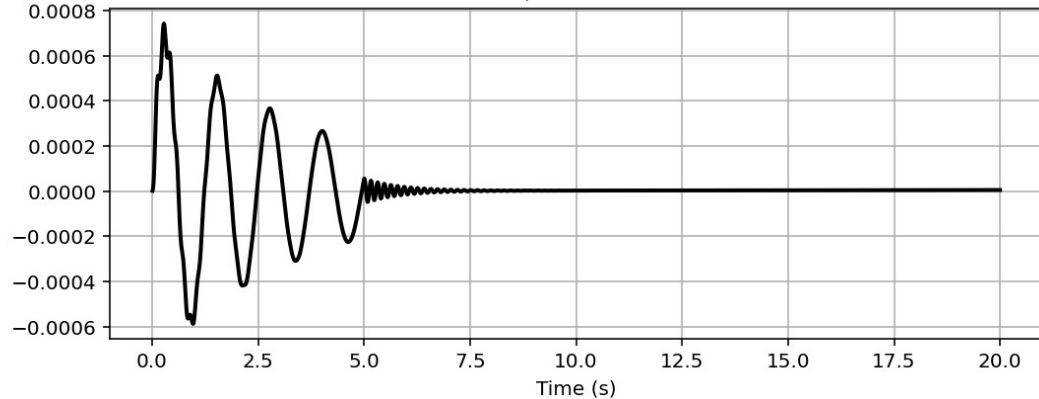
Base Reaction and Displacement of Structure During Dynamic External Time-dependent Loading Analysis



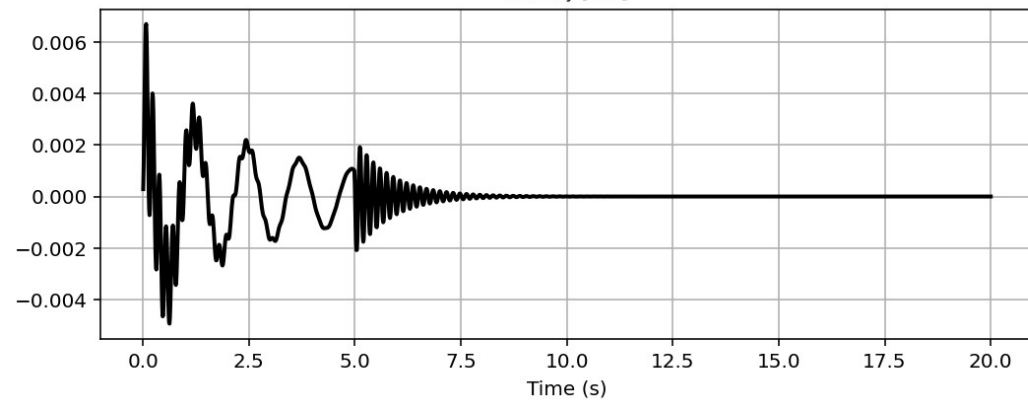
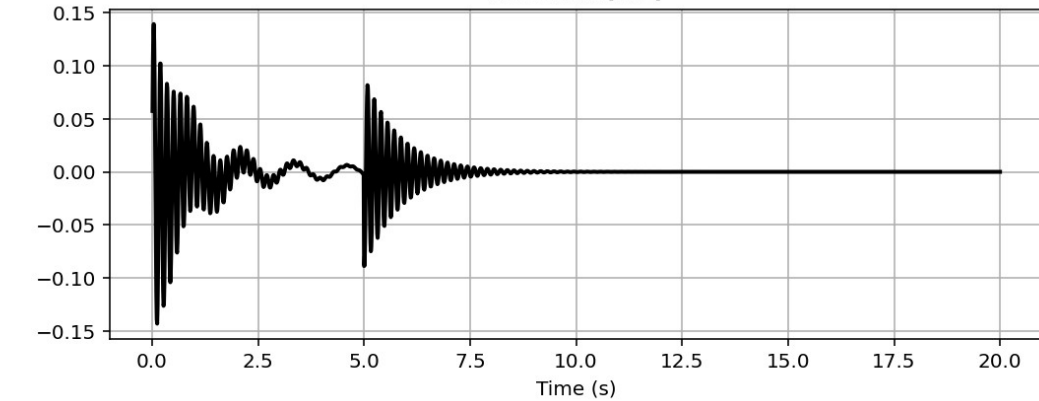
Reaction [N]



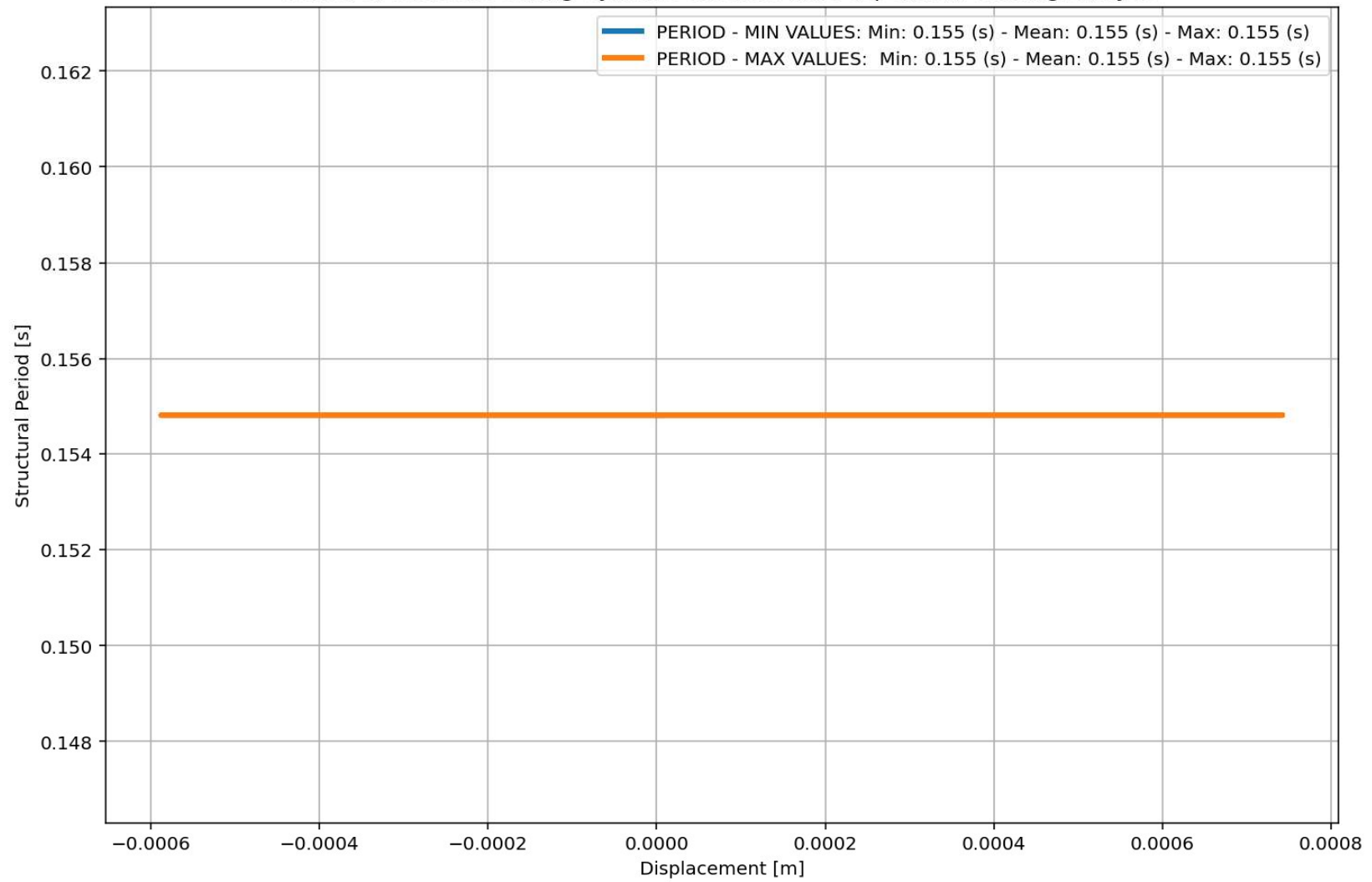
Displacement [m]

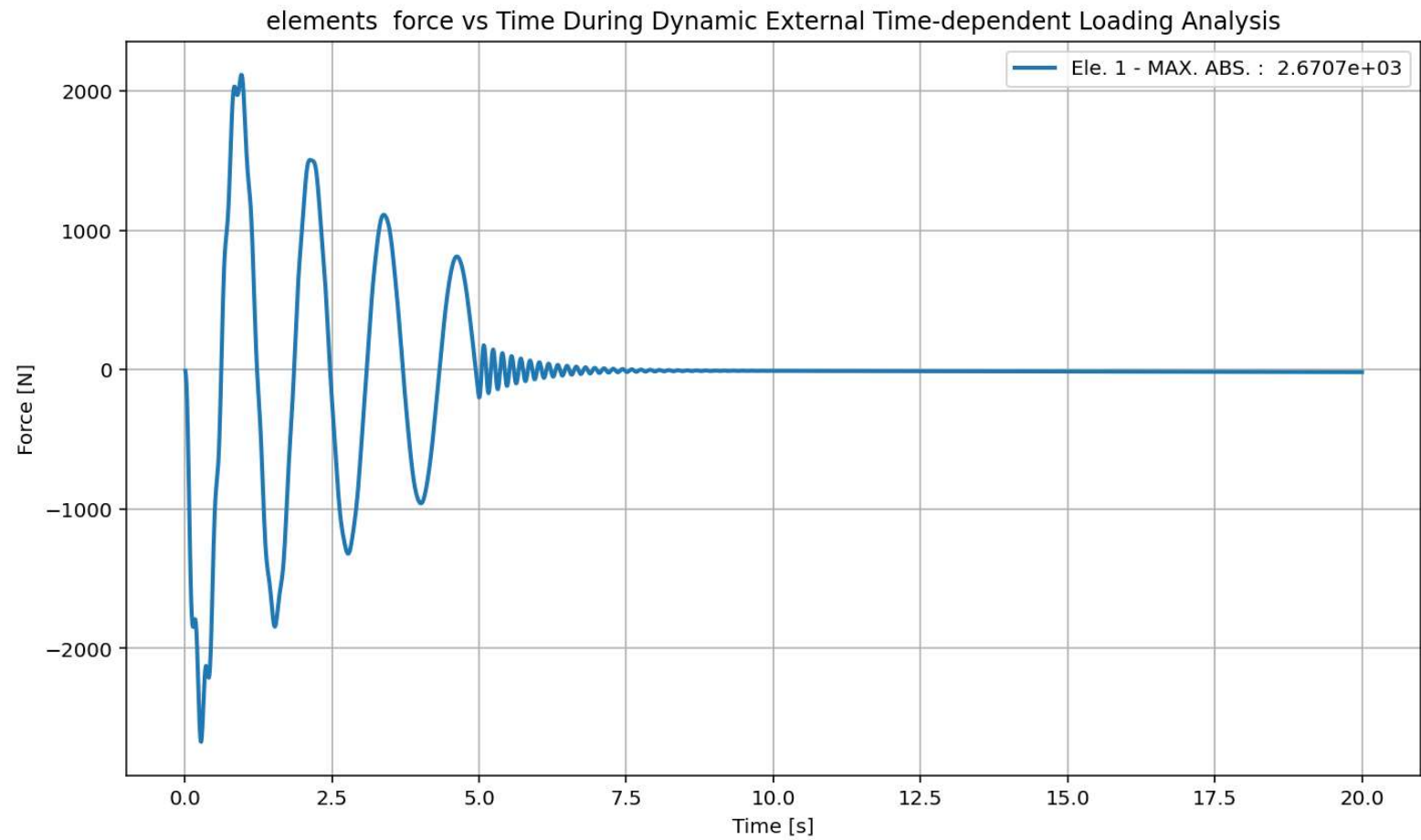


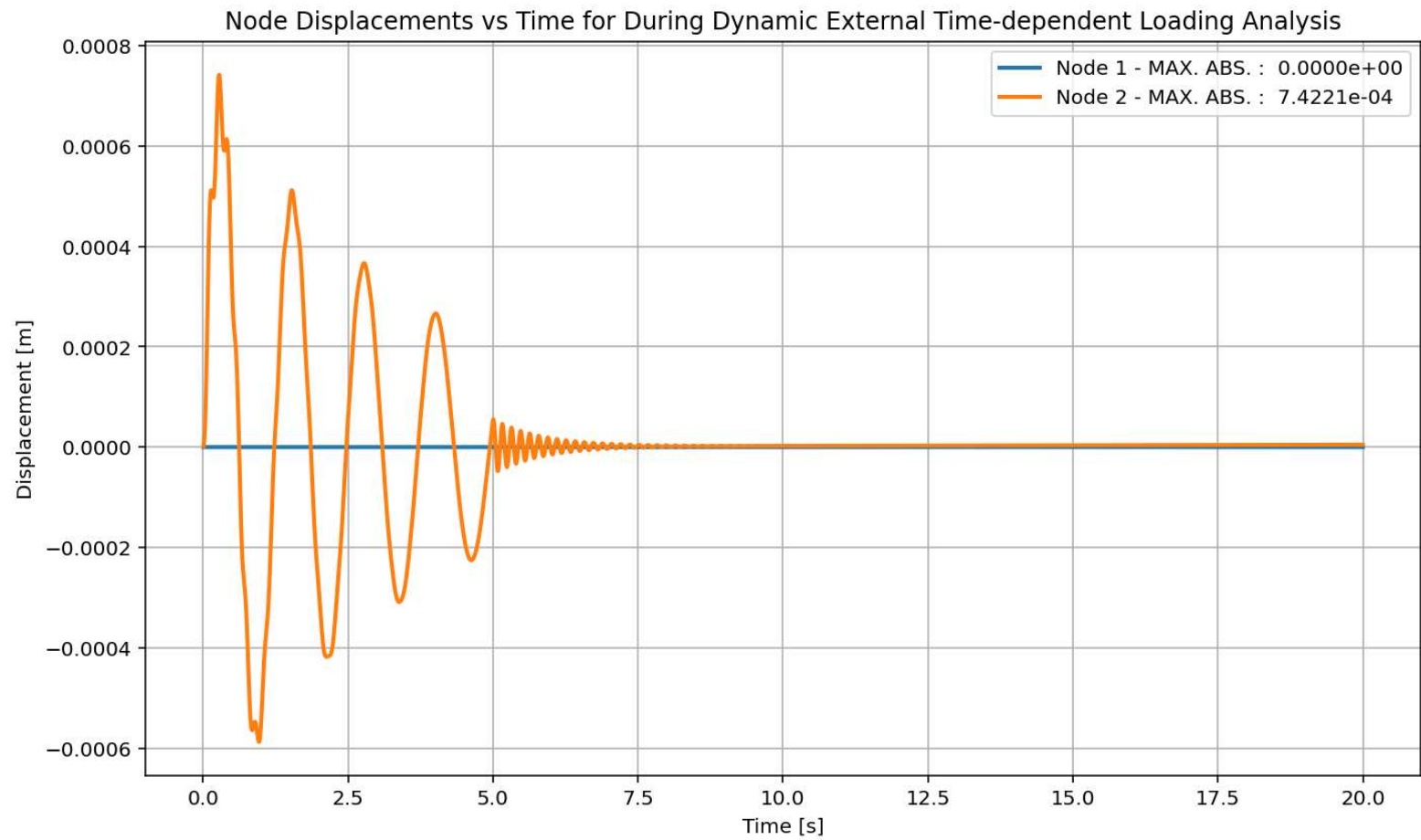
Velocity [m/s]

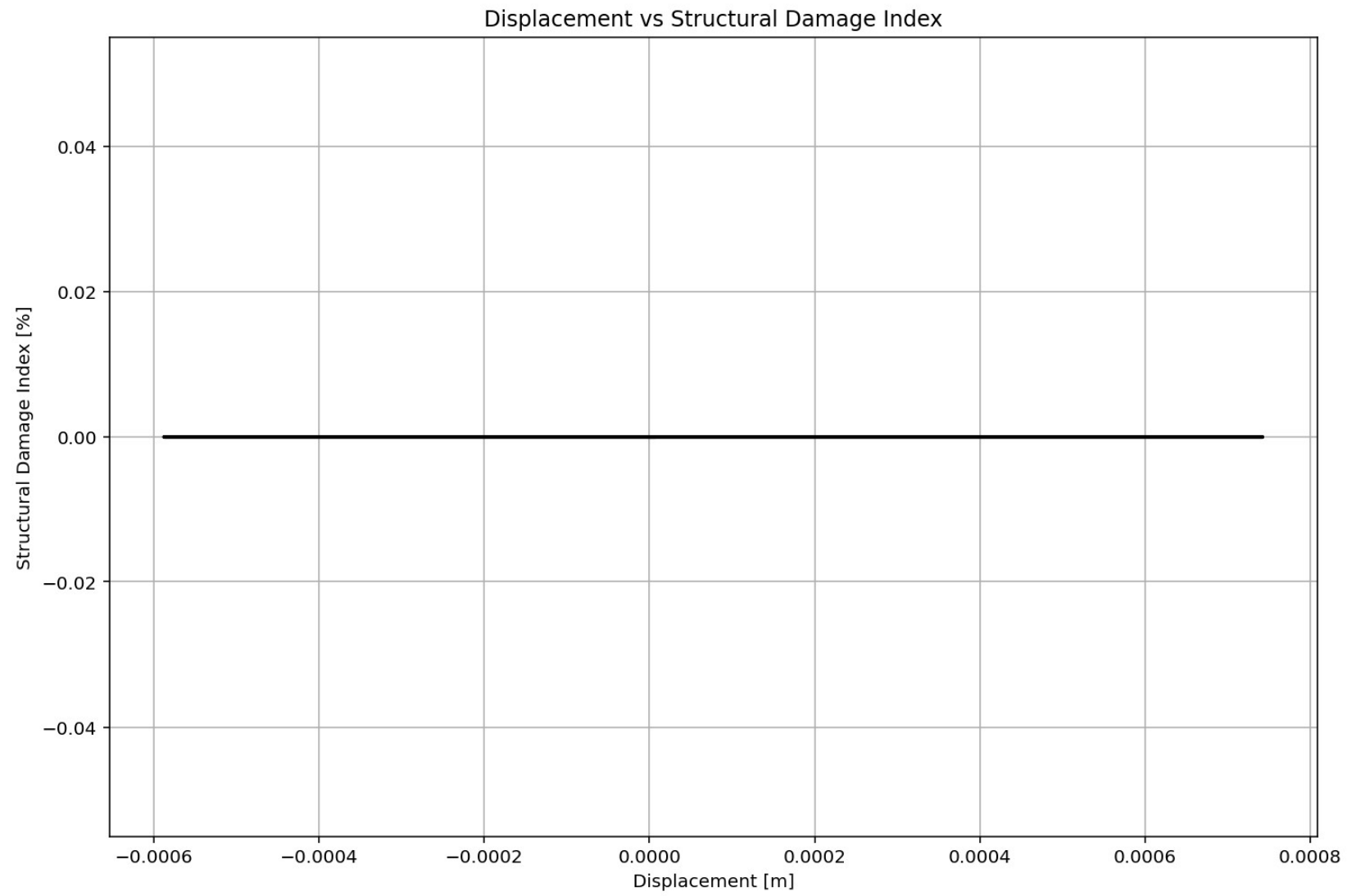
Acceleration [m/s²]

Period of Structure During Dynamic External Time-dependent Loading Analysis









FREE-VIBRATION ANALYSIS

Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\Dell\Desktop\OPENSEES_FILES\HEIGHT_ANALY...3_SDOF\p(t)_POST-FIRE_SDOF_FOUR_ELEMENTS_8_ANA.py

P(t)_POST-FIRE_SDO..._ELEMENTS_8_ANA.py

```
1235 # FREE-VIBRATION ANALYSIS (DYNAMIC TIME-HISTORY ANALYSIS)
1236 MAT_TYPE = 'INELASTIC' # 'ELASTIC' OR 'INELASTIC'
1237 ANAL_TYPE = 'FREE-VIBRATION'
1238 DATA = POST_FIRE_SDOF_FOUR_ELEMENTS(MAT_TYPE, TOTAL_MASS, ANAL_TYPE)
1239
1240 (time_FV, reaction_FV, disp_FV, velo_FV, acc_FV, DI_FV,
1241  ele_force_FV, node_displacements_FV,
1242  stiffness_FV, PERIOD_FV, damping_ratio_FV,
1243  PERIOD_MIN_FV, PERIOD_MAX_FV) = DATA
1244
1245 XDATA = disp_FV
1246 YDATA = reaction_FV
1247 XLABEL = 'Displacement [m]'
1248 YLABEL = 'Base Reaction [N]'
1249 TITLE = 'Base Reaction and Displacement of Structure During Free-vibration
1250 COLOR = 'black'
1251 SEMILOGY = False
1252 PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
1253
1254 PLOT_TIME_HISTORY(time_FV, reaction_FV, disp_FV, velo_FV, acc_FV)
1255
1256 # PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
1257 plt.figure(0, figsize=(12, 8))
1258 plt.plot(time_FV, PERIOD_MIN_FV, linewidth=3)
1259 plt.plot(time_FV, PERIOD_MAX_FV, linewidth=3)
1260 plt.title('Period of Structure During Free-vibration Analysis')
1261 plt.ylabel('Structural Period [s]')
1262 plt.xlabel('Time [s]')
1263 plt.semilogy()
1264 plt.grid()
1265 plt.legend([f'PERIOD - MIN VALUES: Min: {np.min(PERIOD_MIN_FV):.3f} (s) - M
1266            f'PERIOD - MAX VALUES: Min: {np.min(PERIOD_MAX_FV):.3f} (s) - M
1267            ])
1268 plt.show()
```

Base Reaction and Displacement of Structure During Free-vibration Analysis

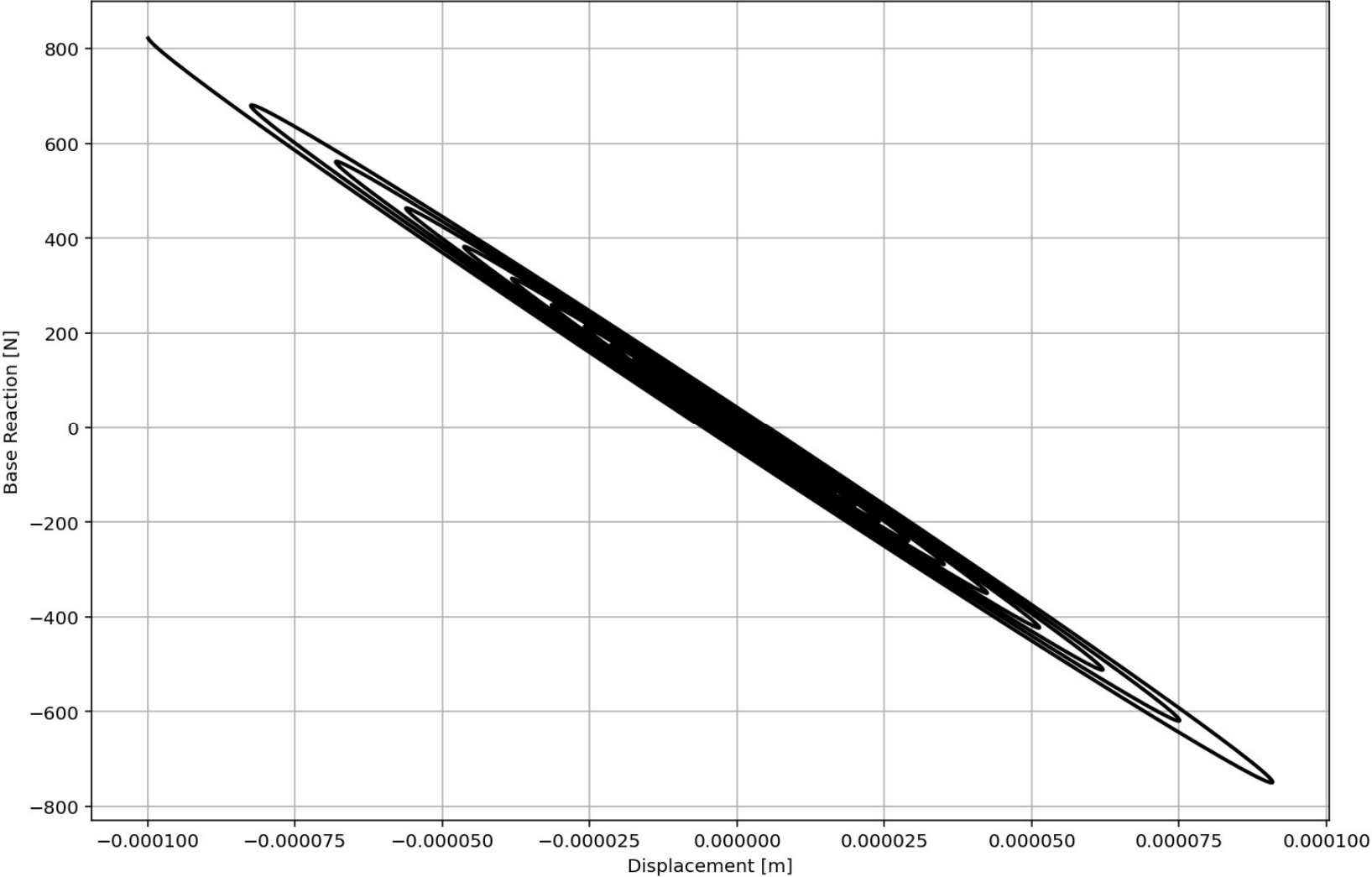
Base Reaction [N]

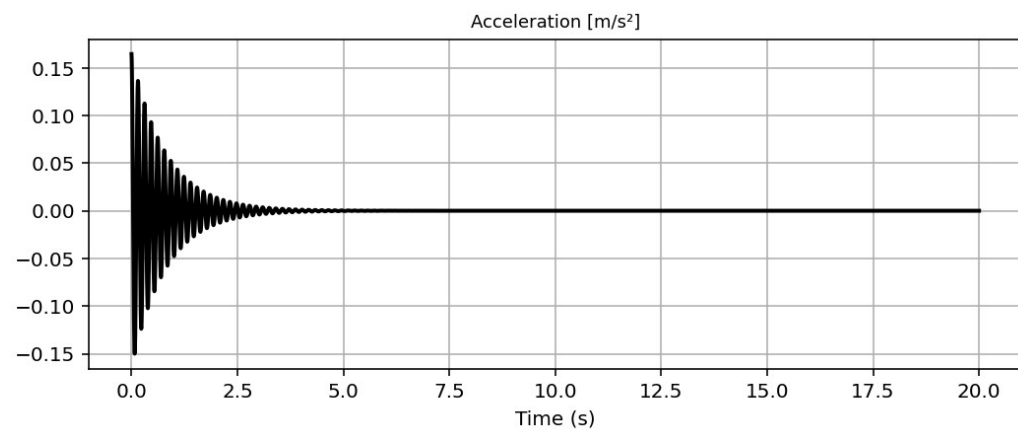
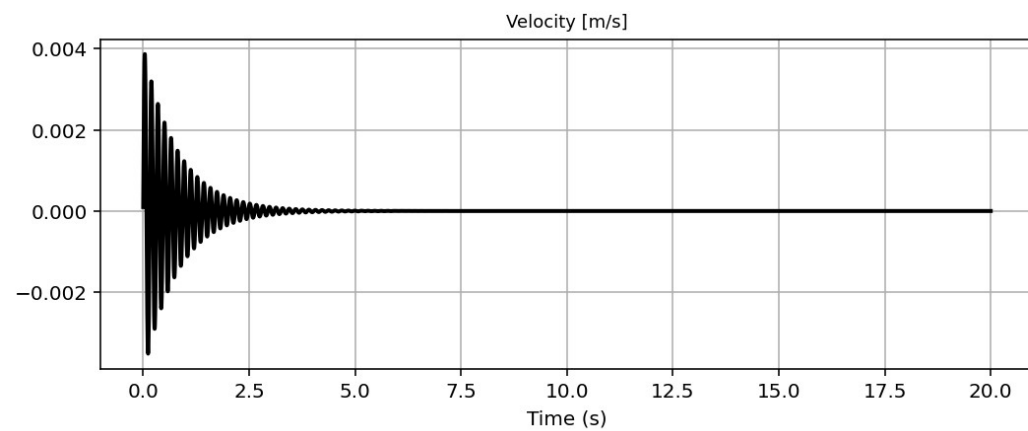
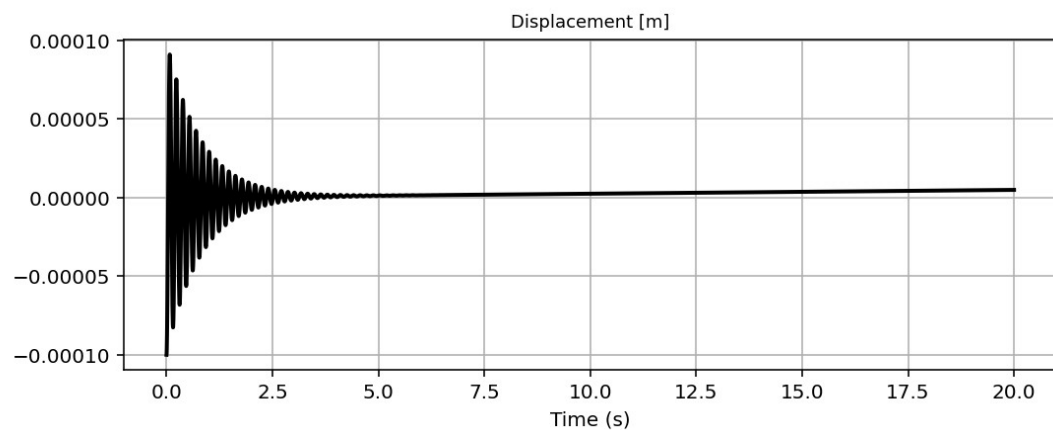
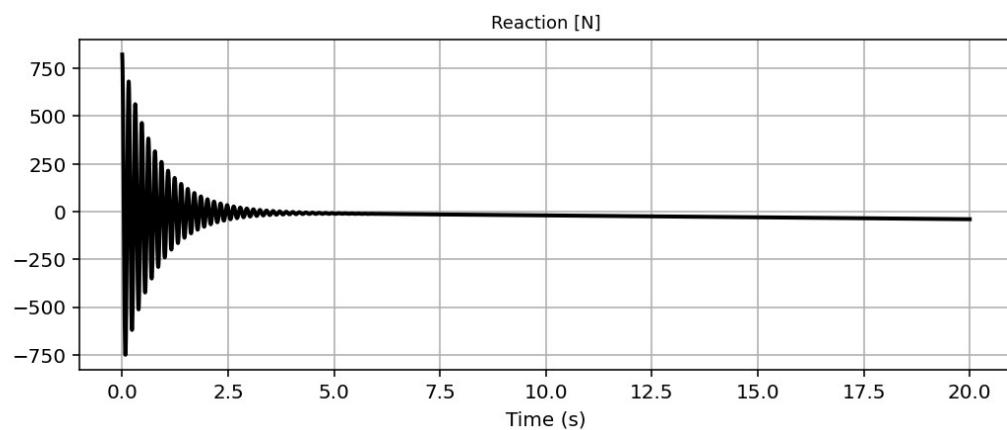
Displacement [m]

IPython Console Files Help Variable Explorer Debugger Plots History

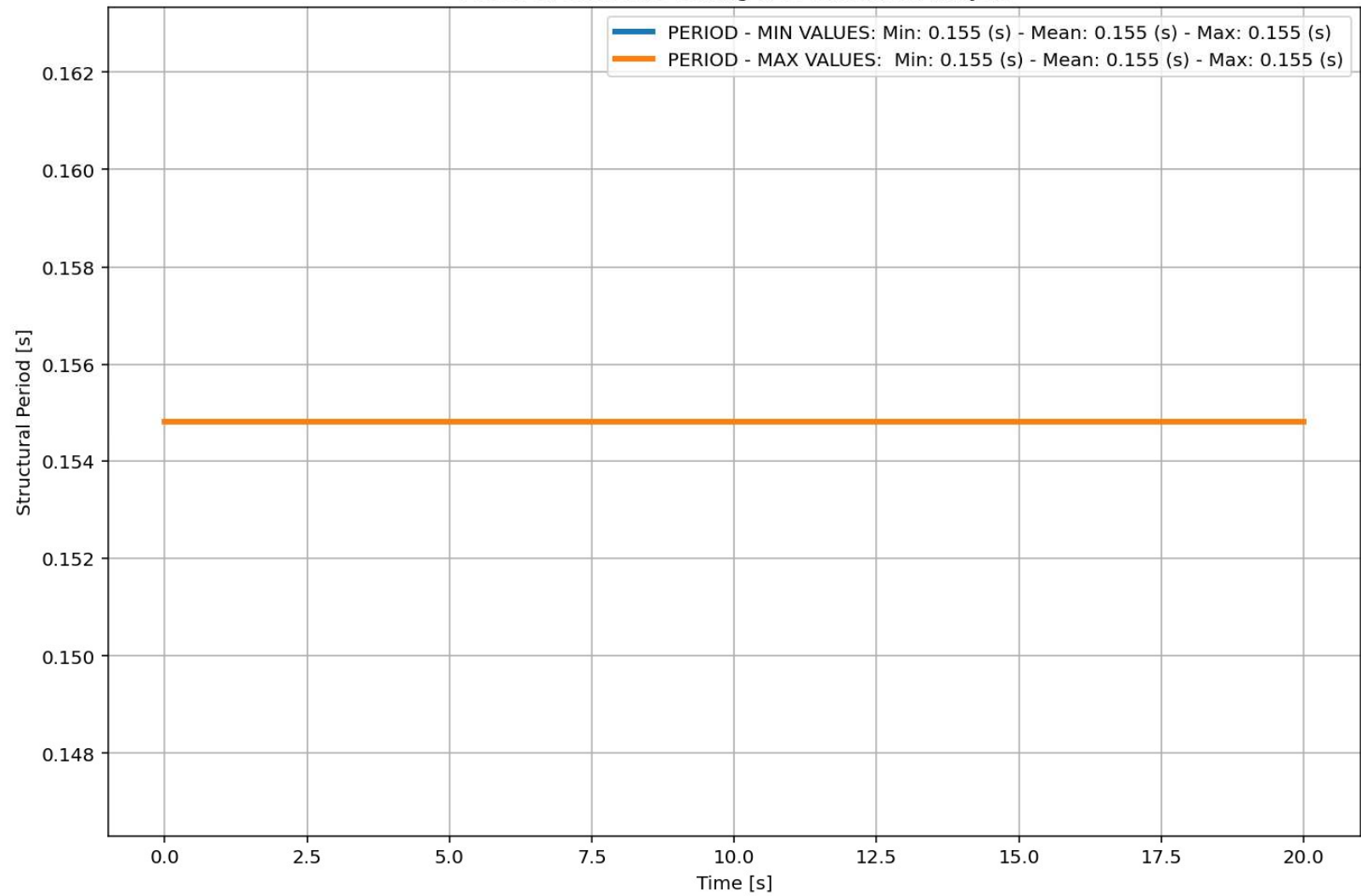
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 20, Col 38 UTF-8 CRLF RW Mem 60%

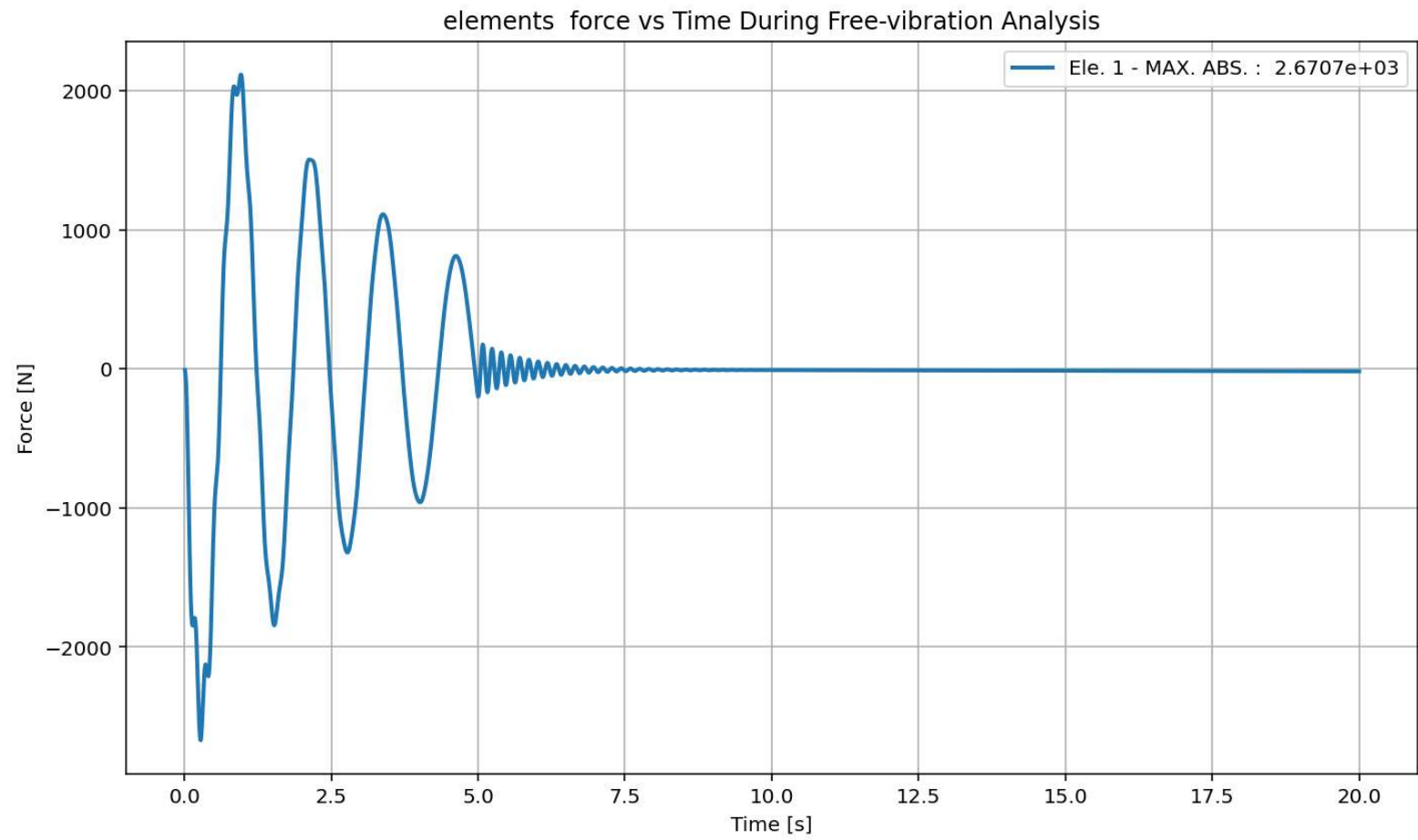
Base Reaction and Displacement of Structure During Free-vibration Analysis

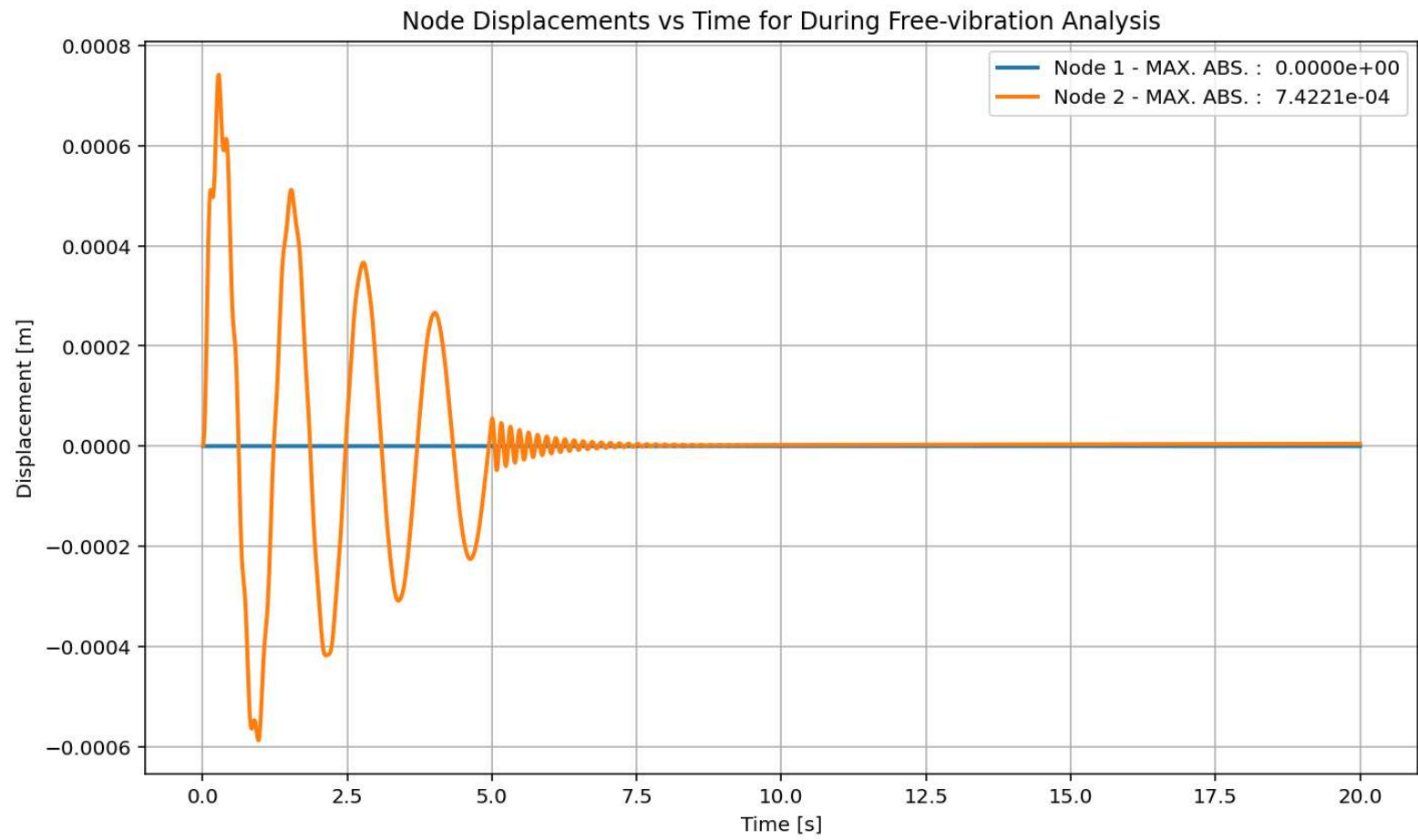


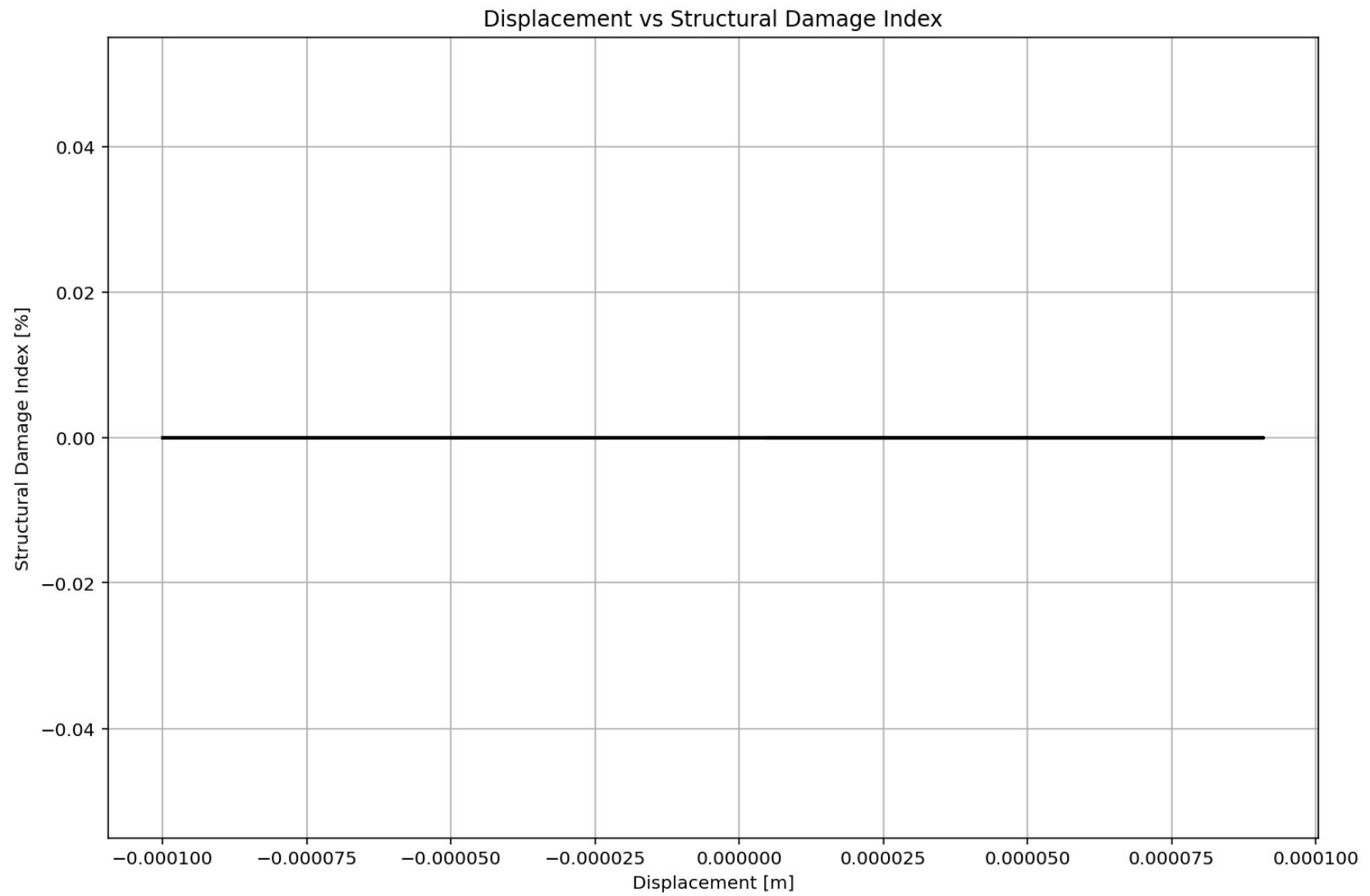


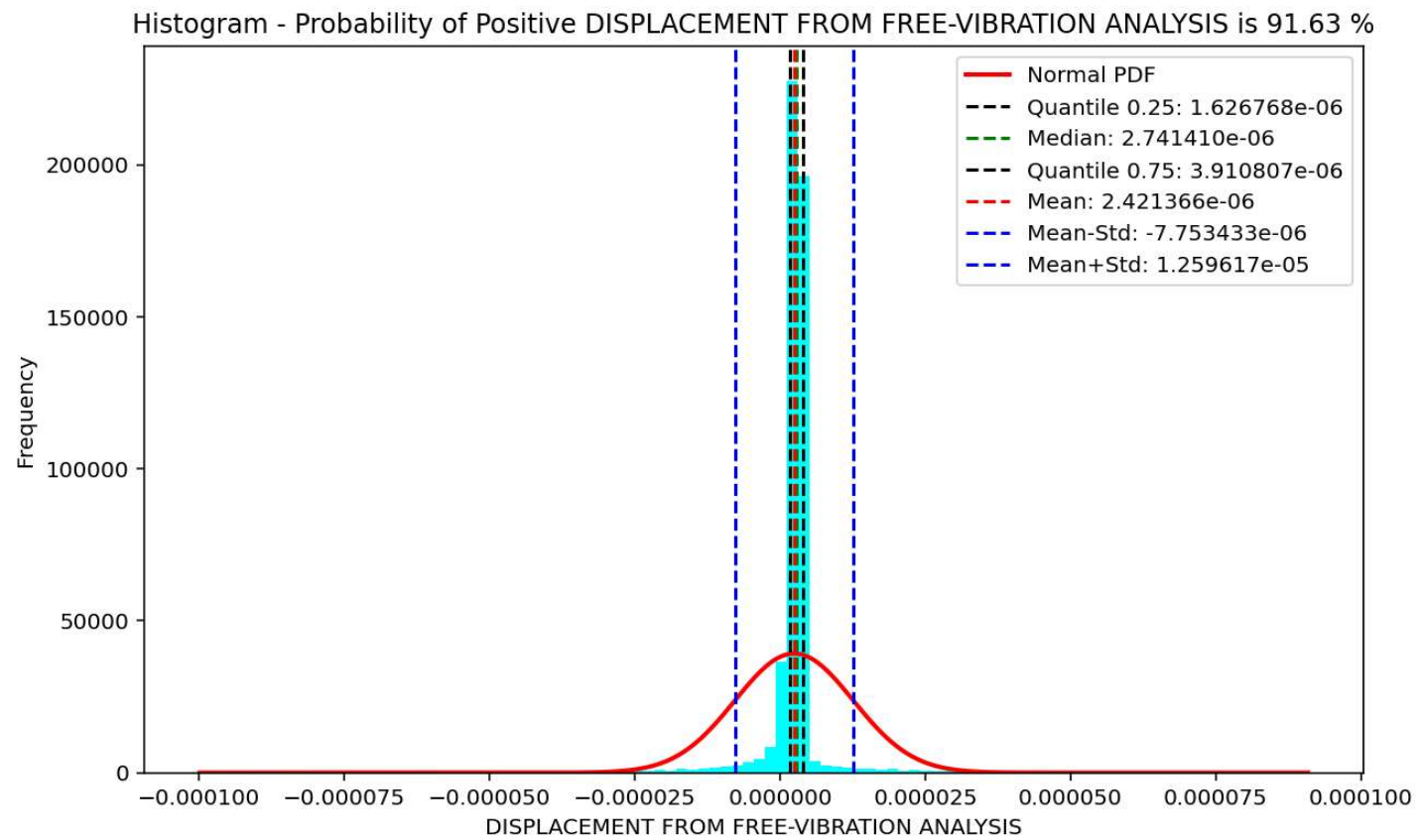
Period of Structure During Free-vibration Analysis

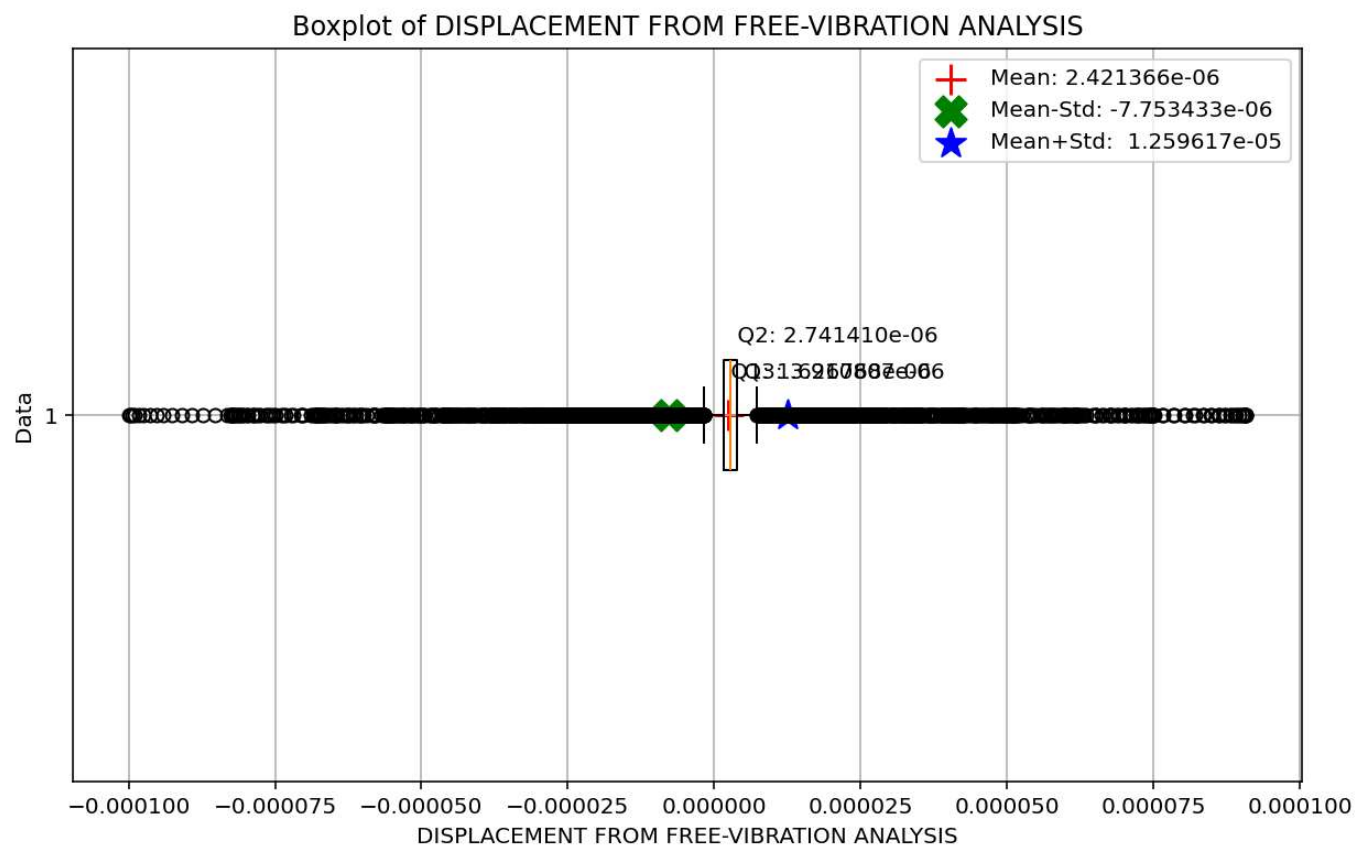




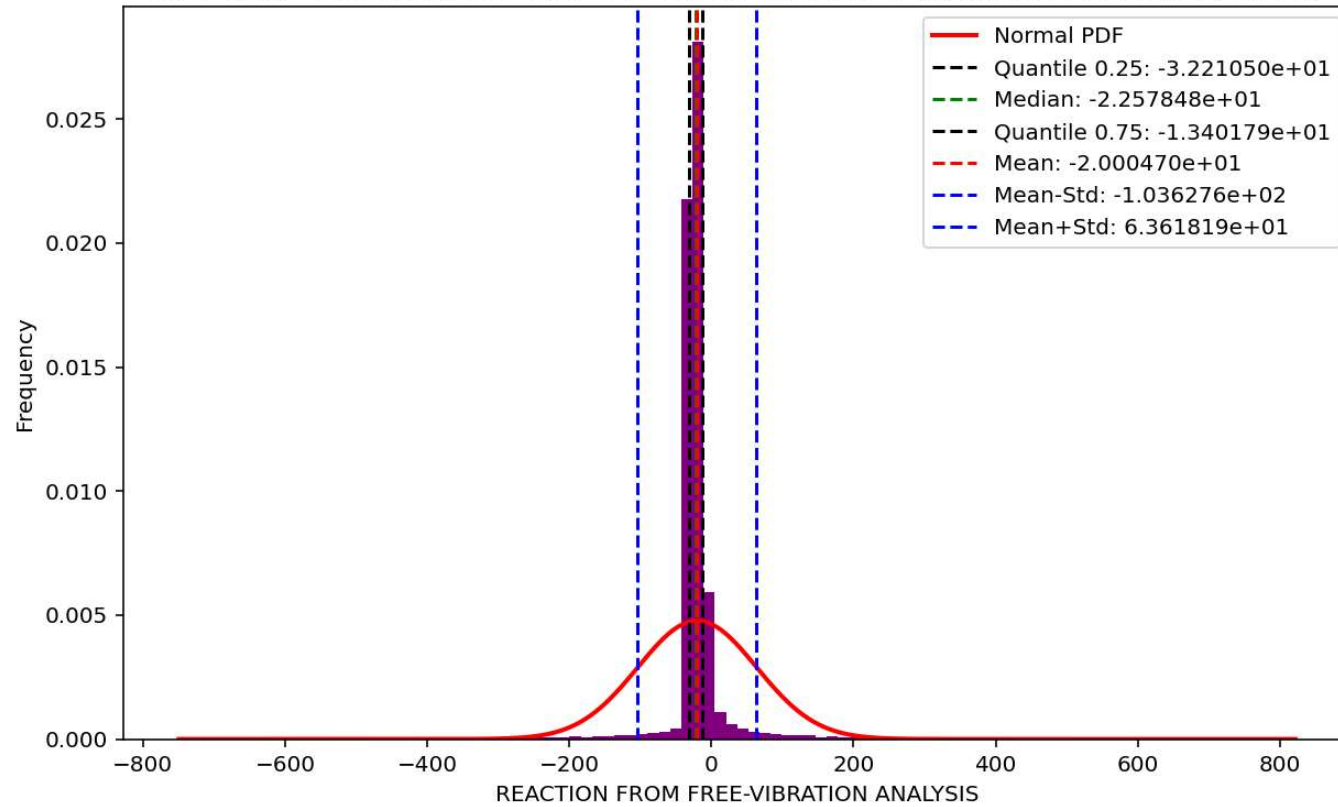


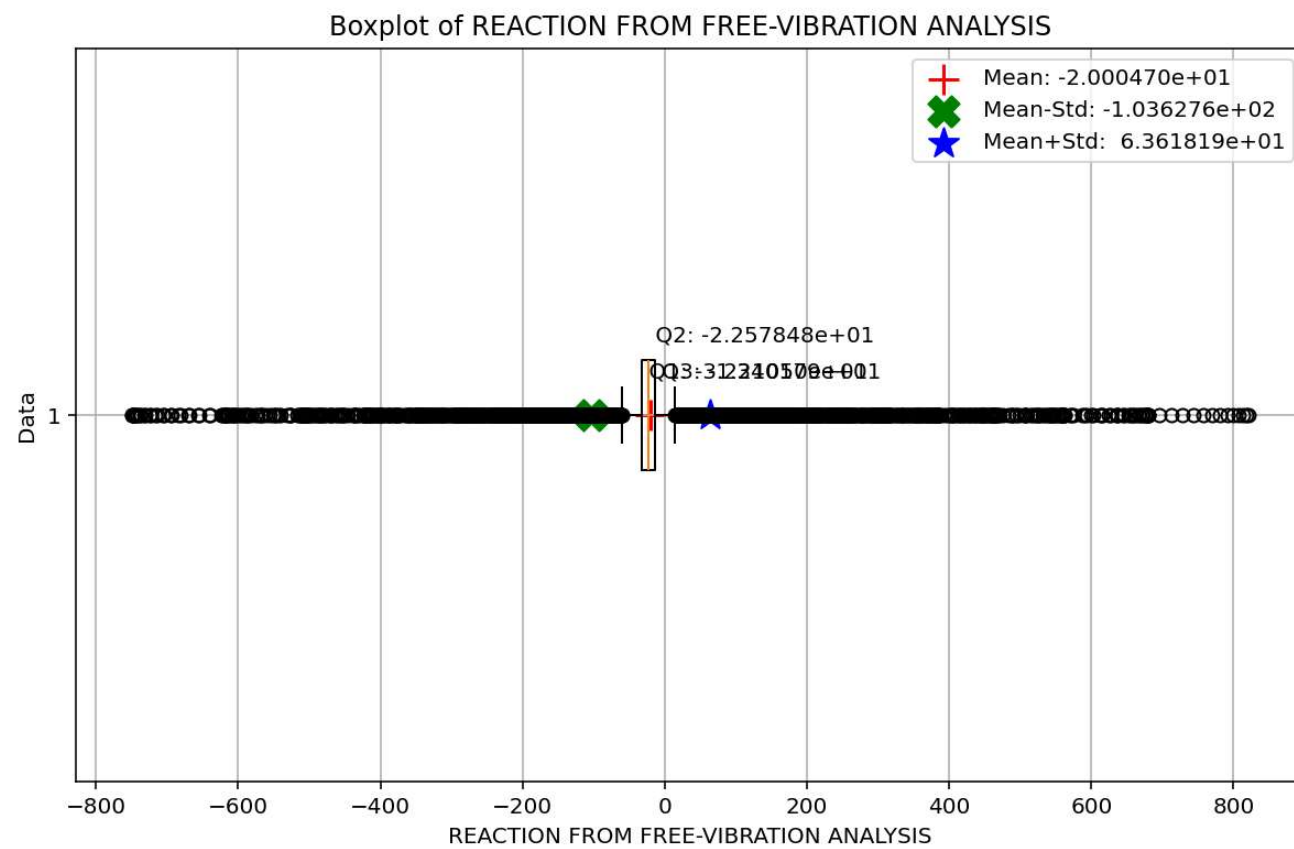




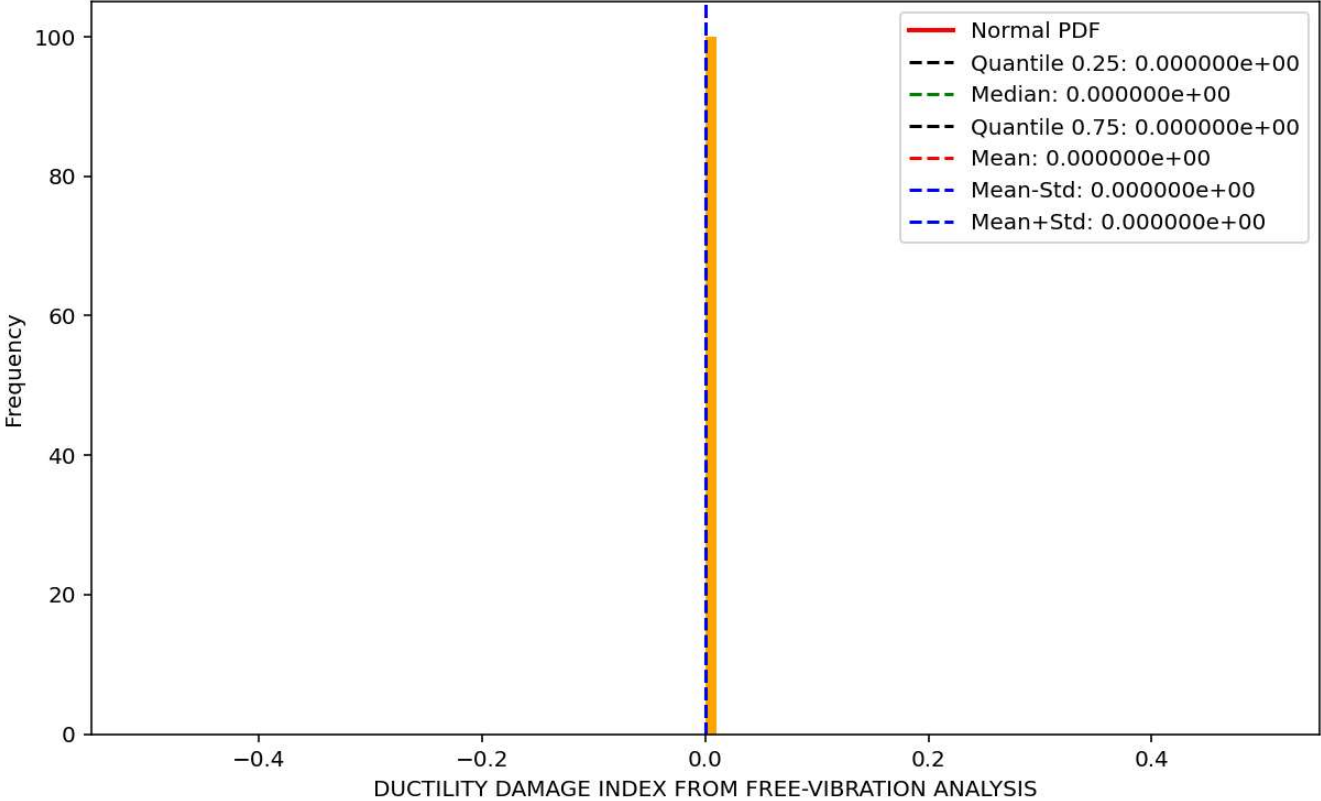


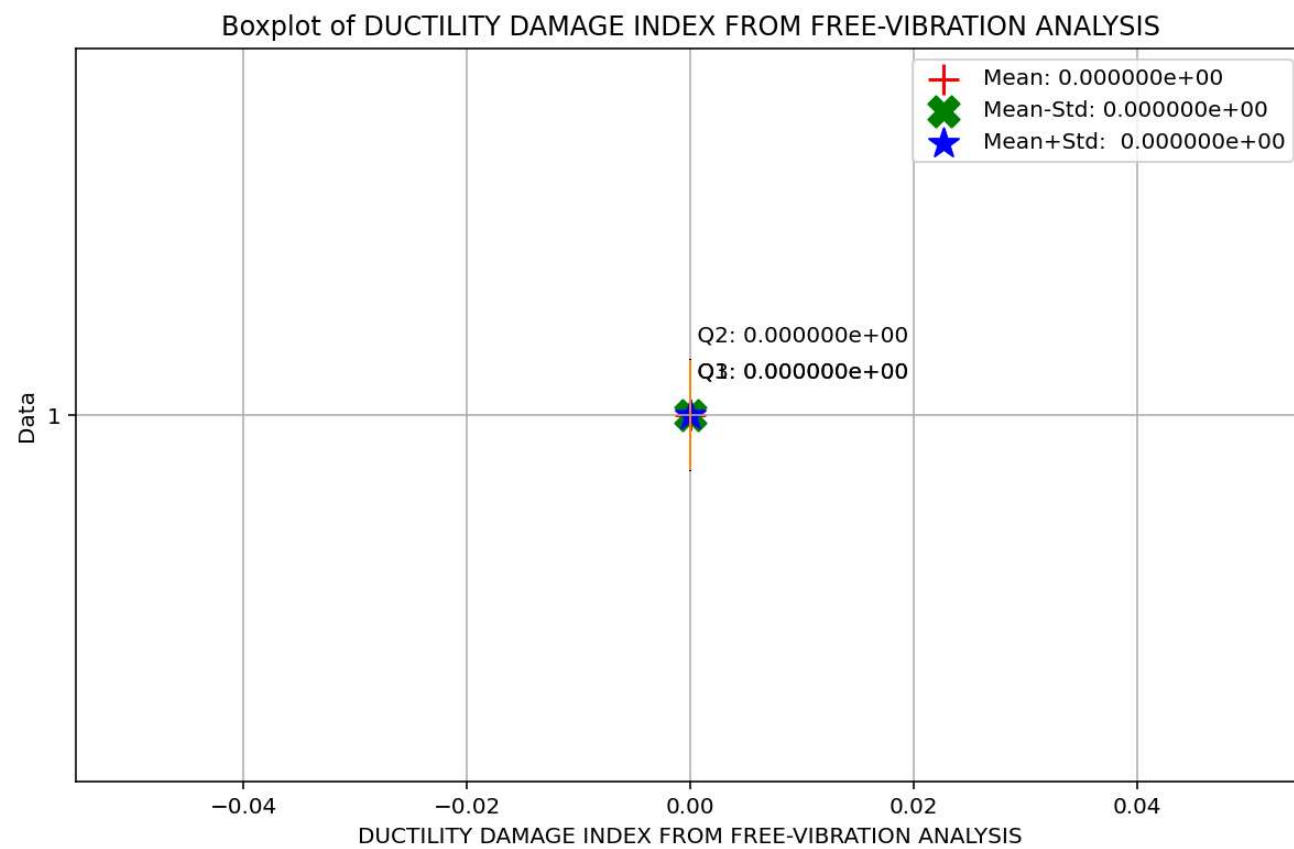
Histogram - Probability of Positive REACTION FROM FREE-VIBRATION ANALYSIS is 8.40 %

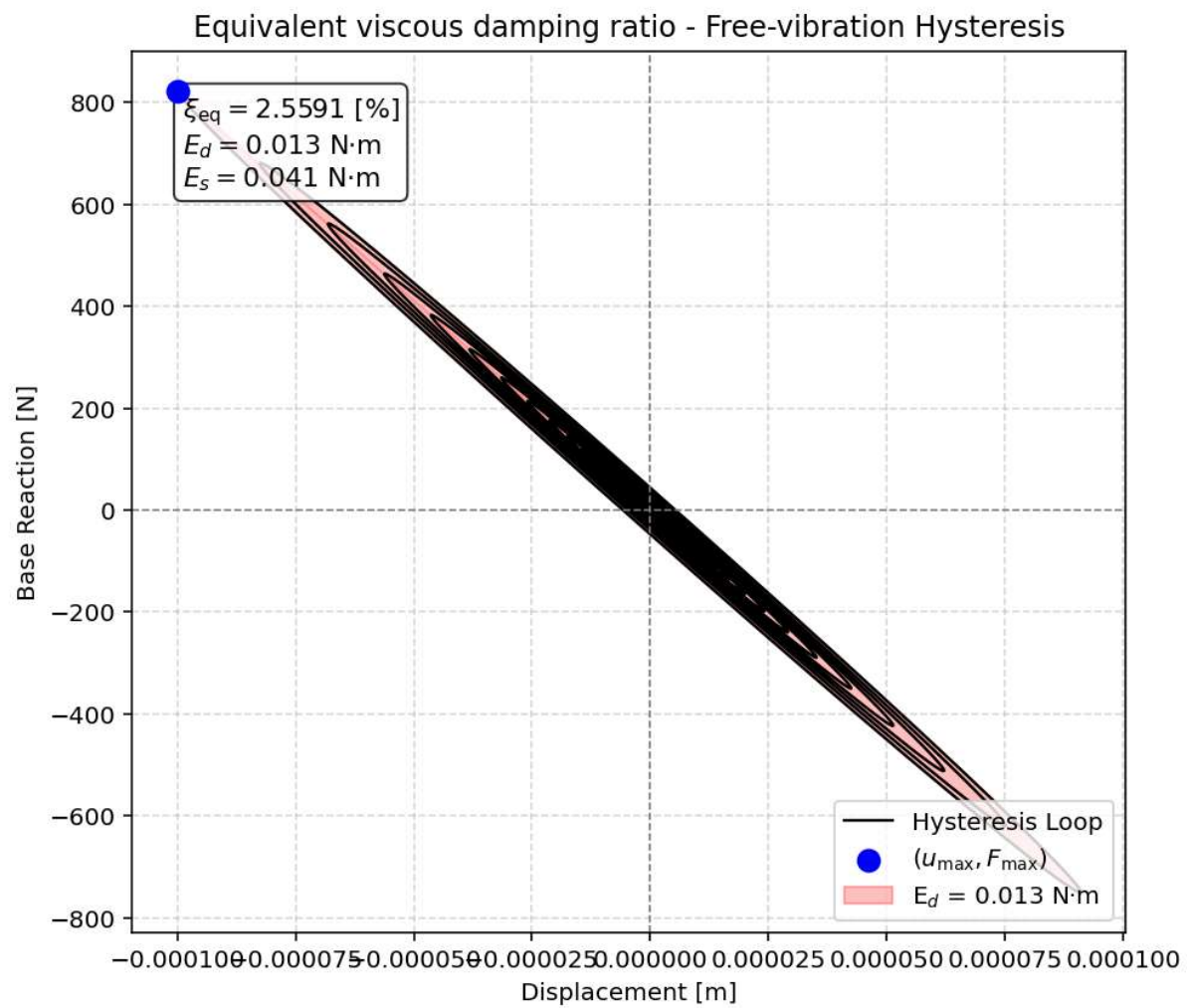




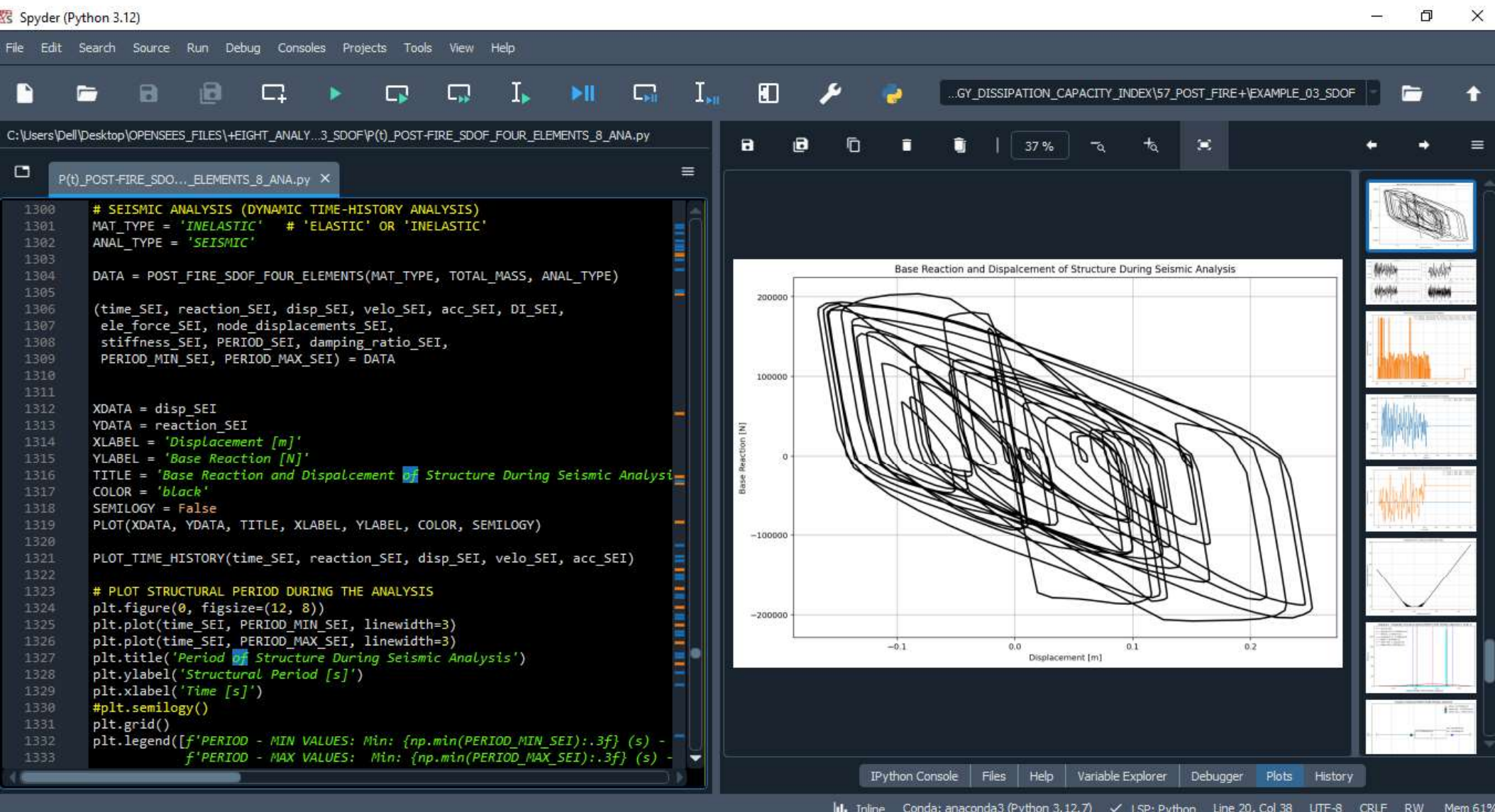
Histogram - Probability of Positive DUCTILITY DAMAGE INDEX FROM FREE-VIBRATION ANALYSIS is 0.00 %

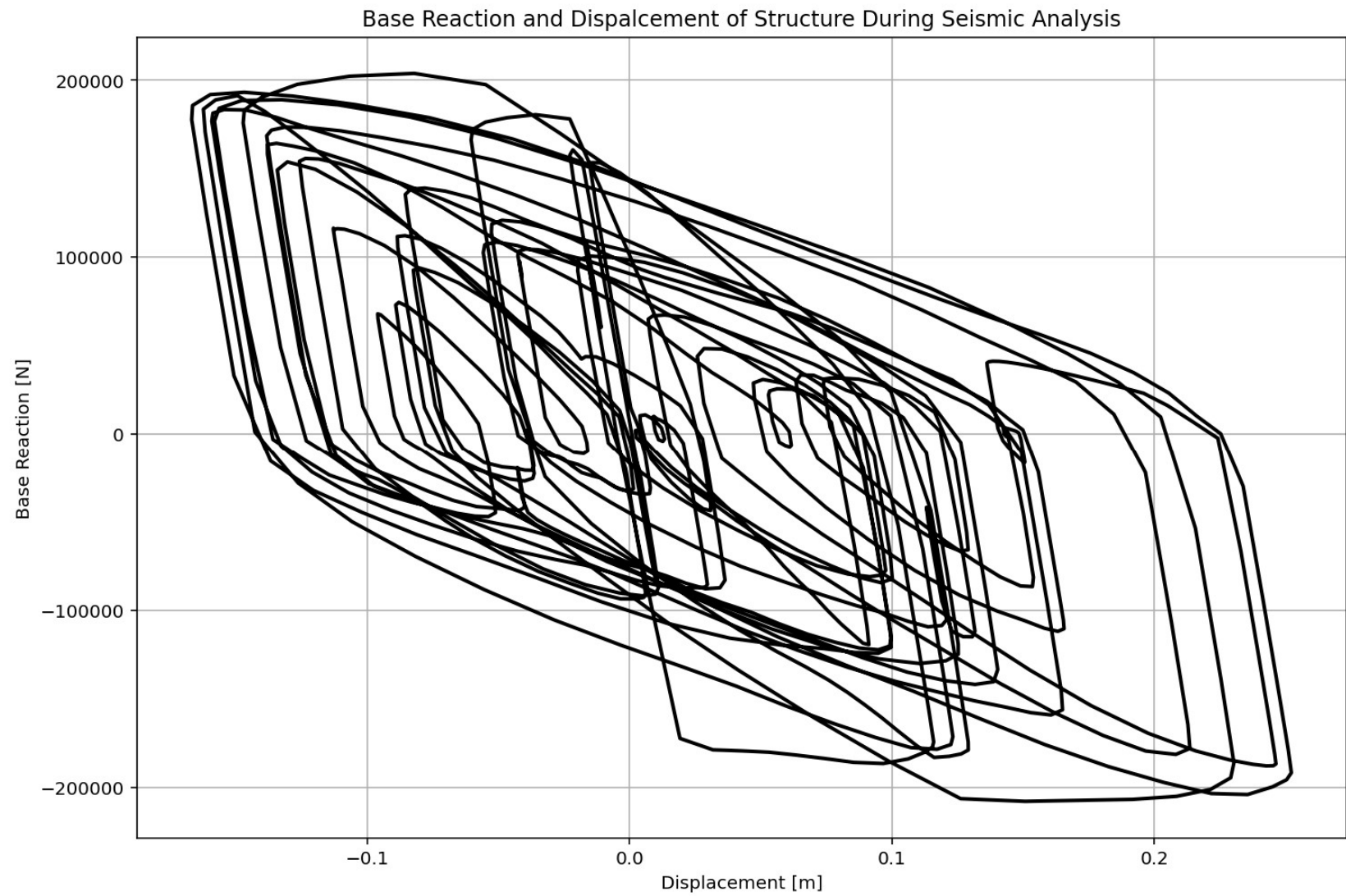


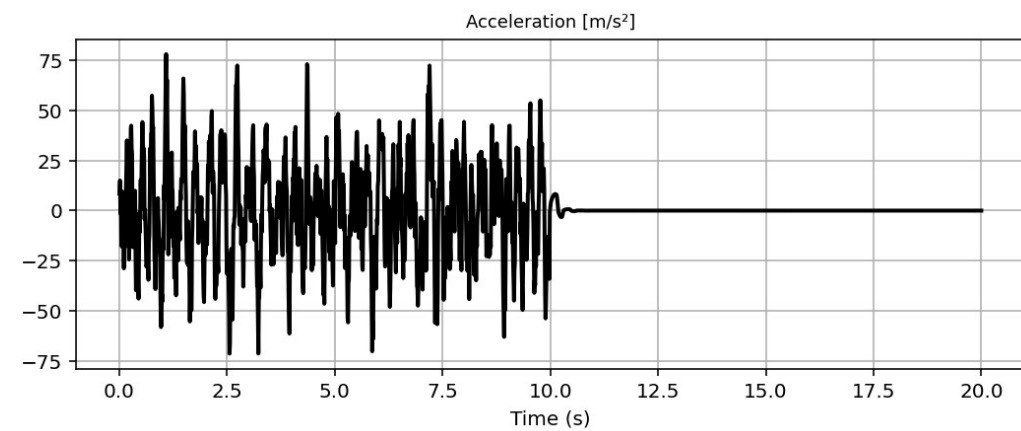
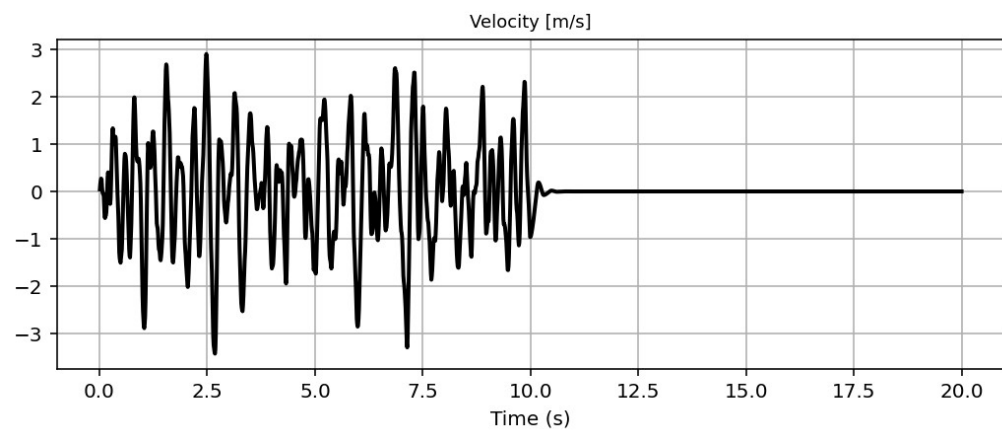
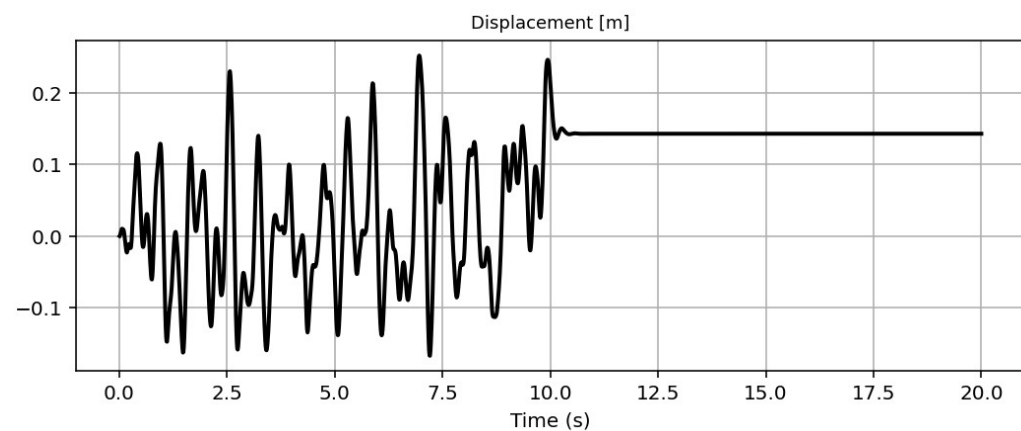
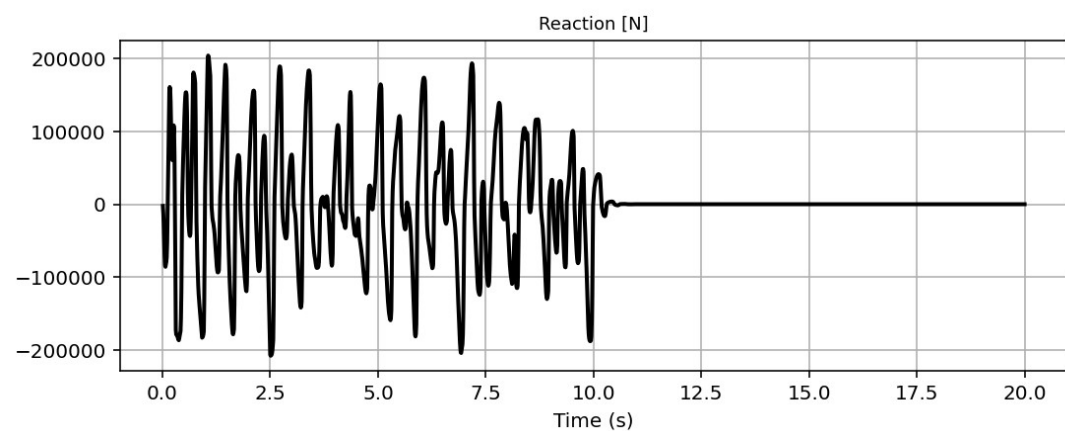




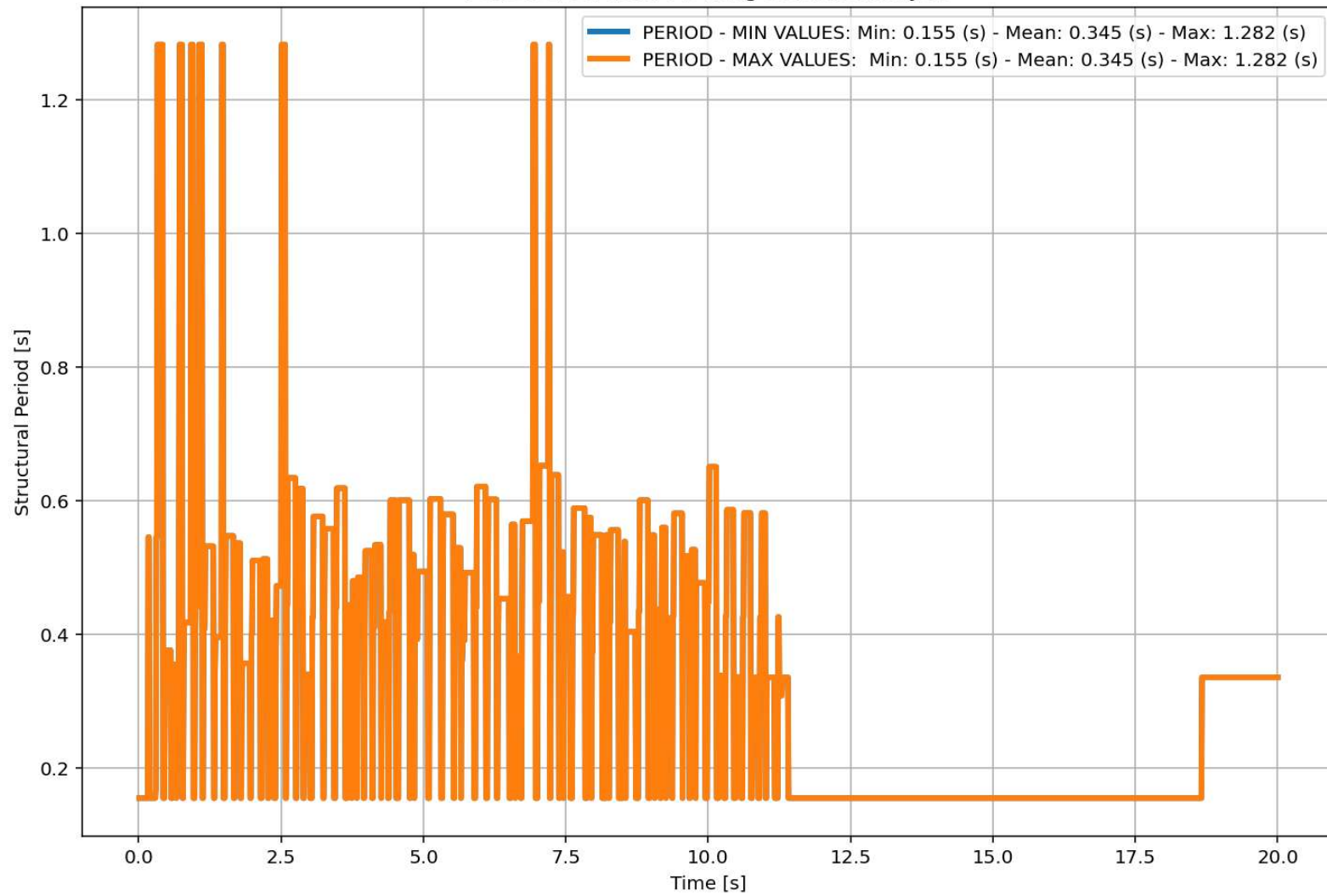
SEISMIC ANALYSIS

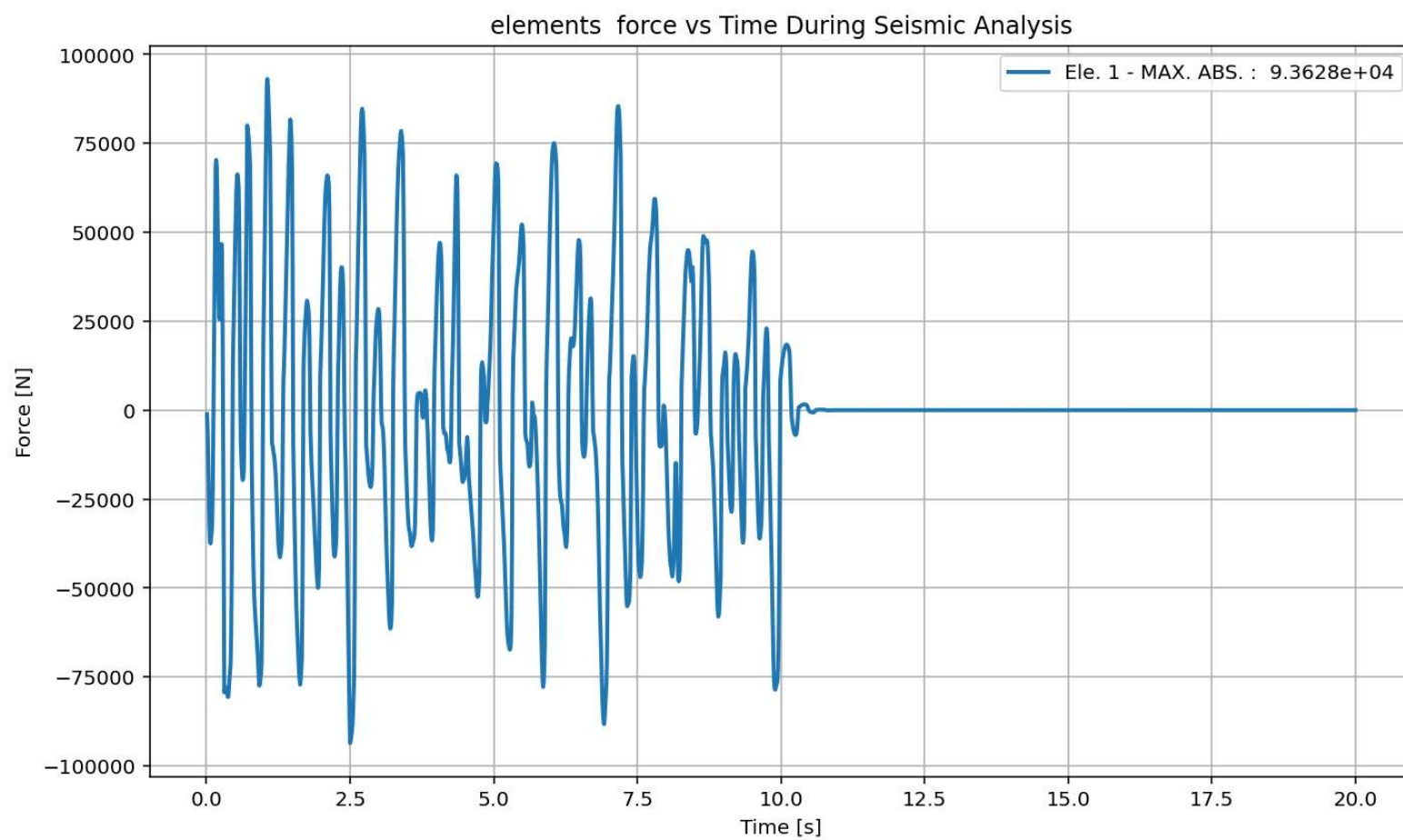




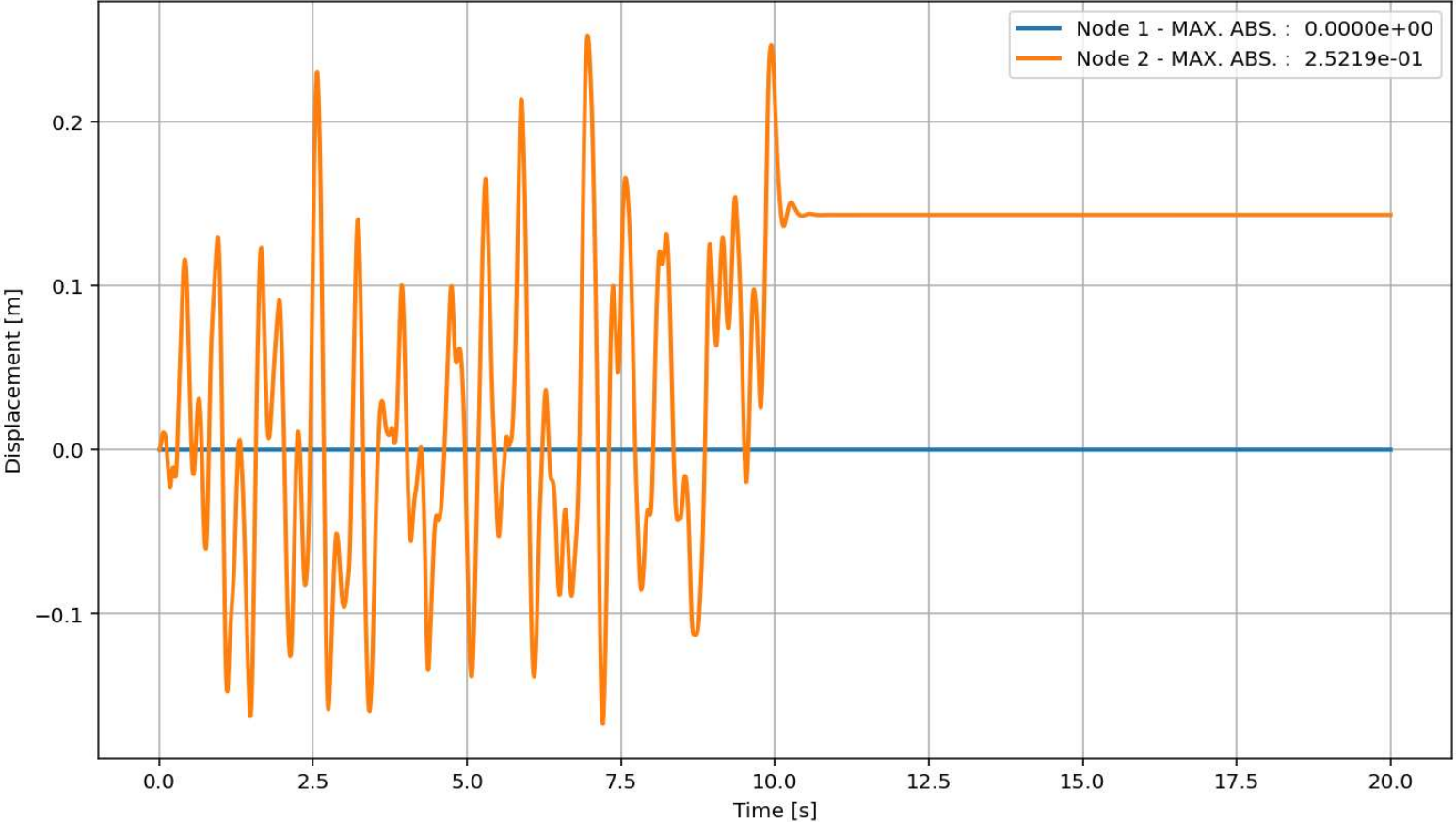


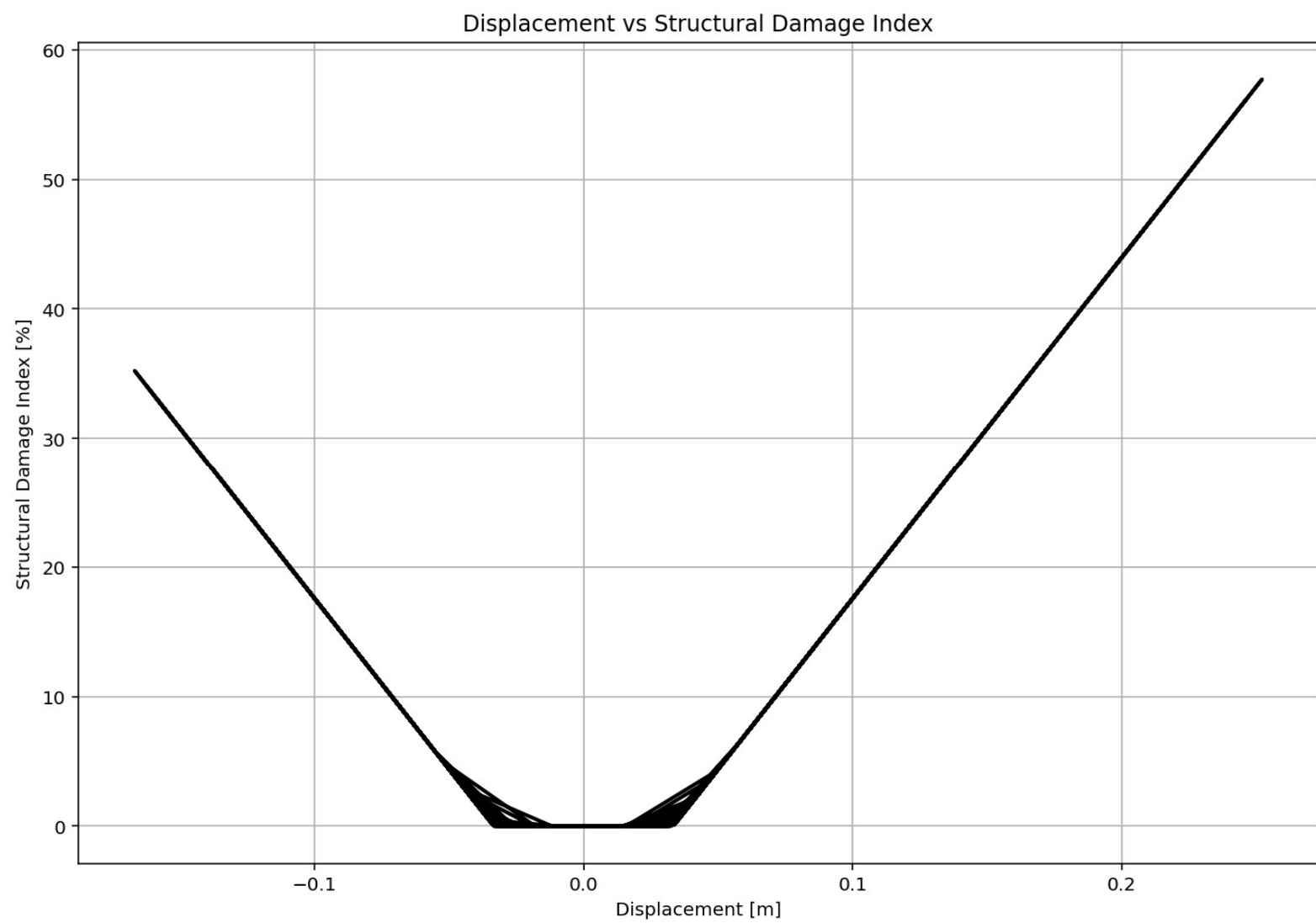
Period of Structure During Seismic Analysis

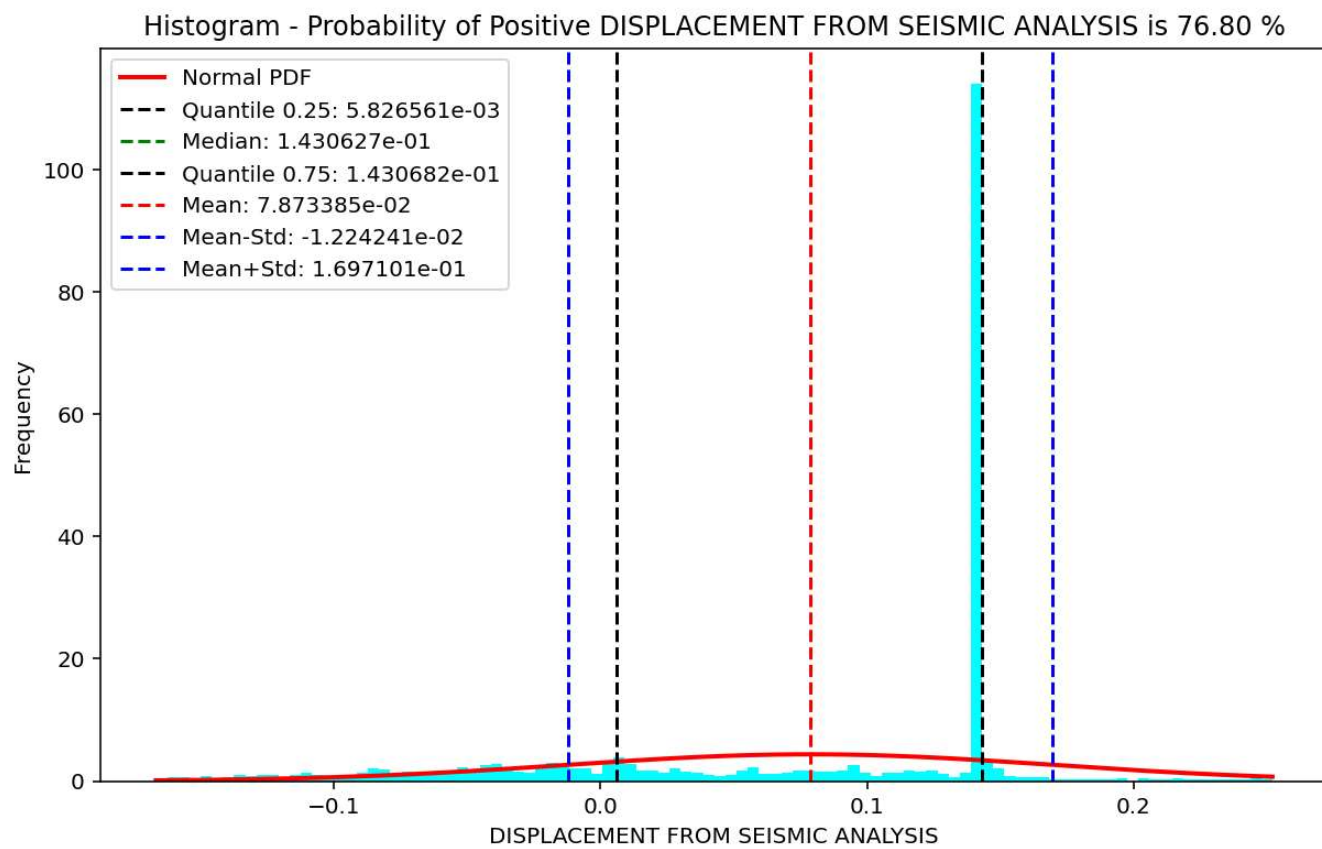


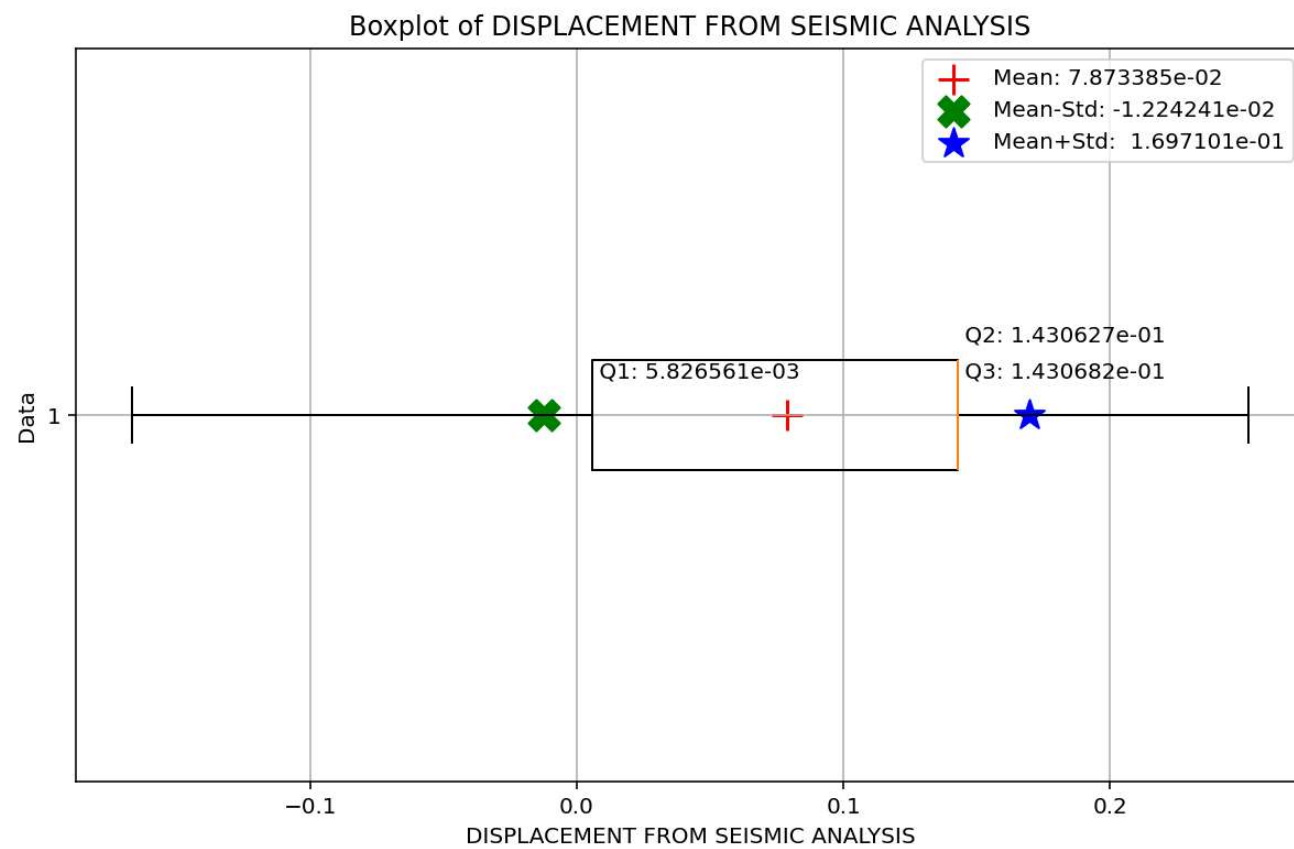


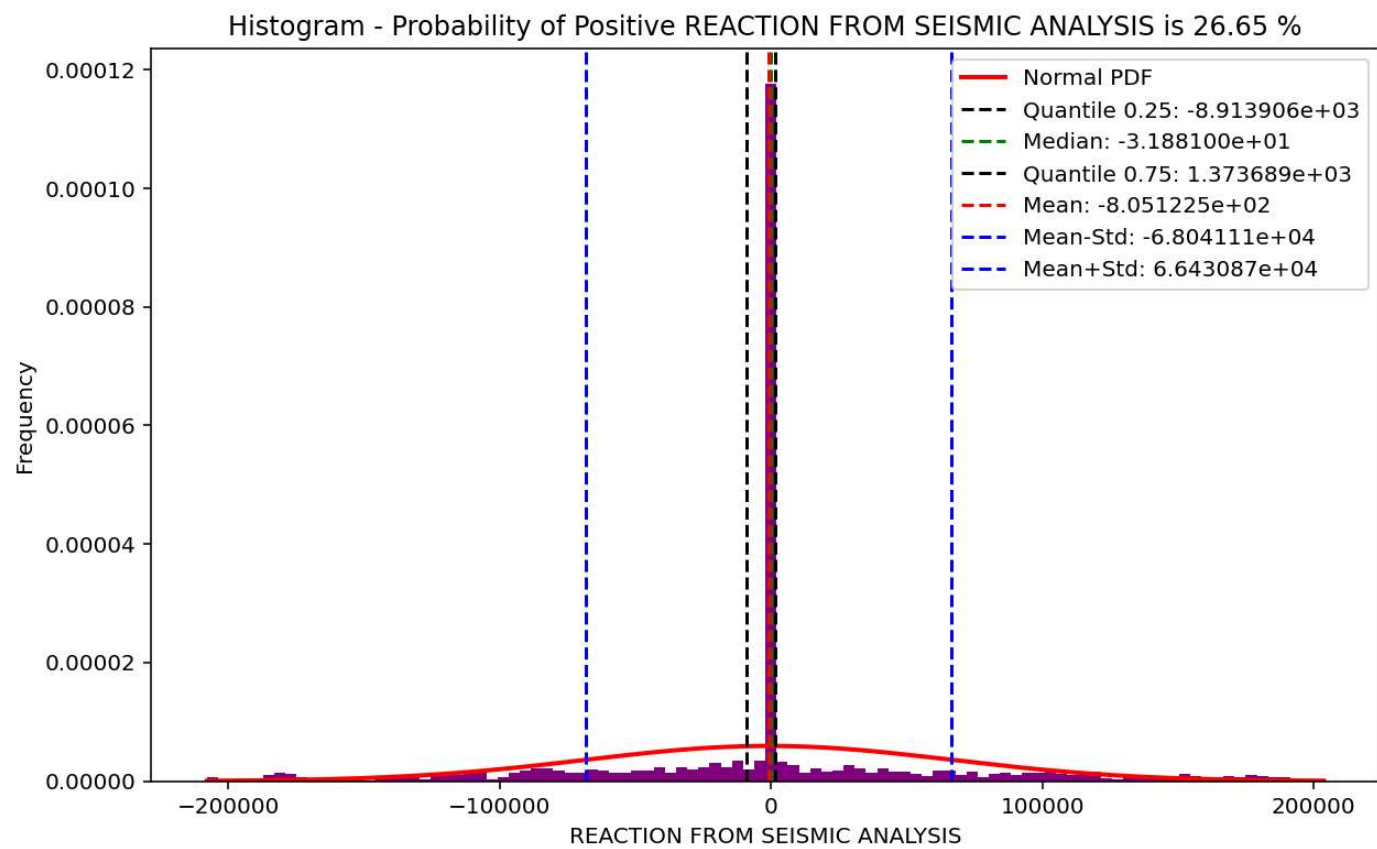
Node Displacements vs Time for During Seismic Analysis

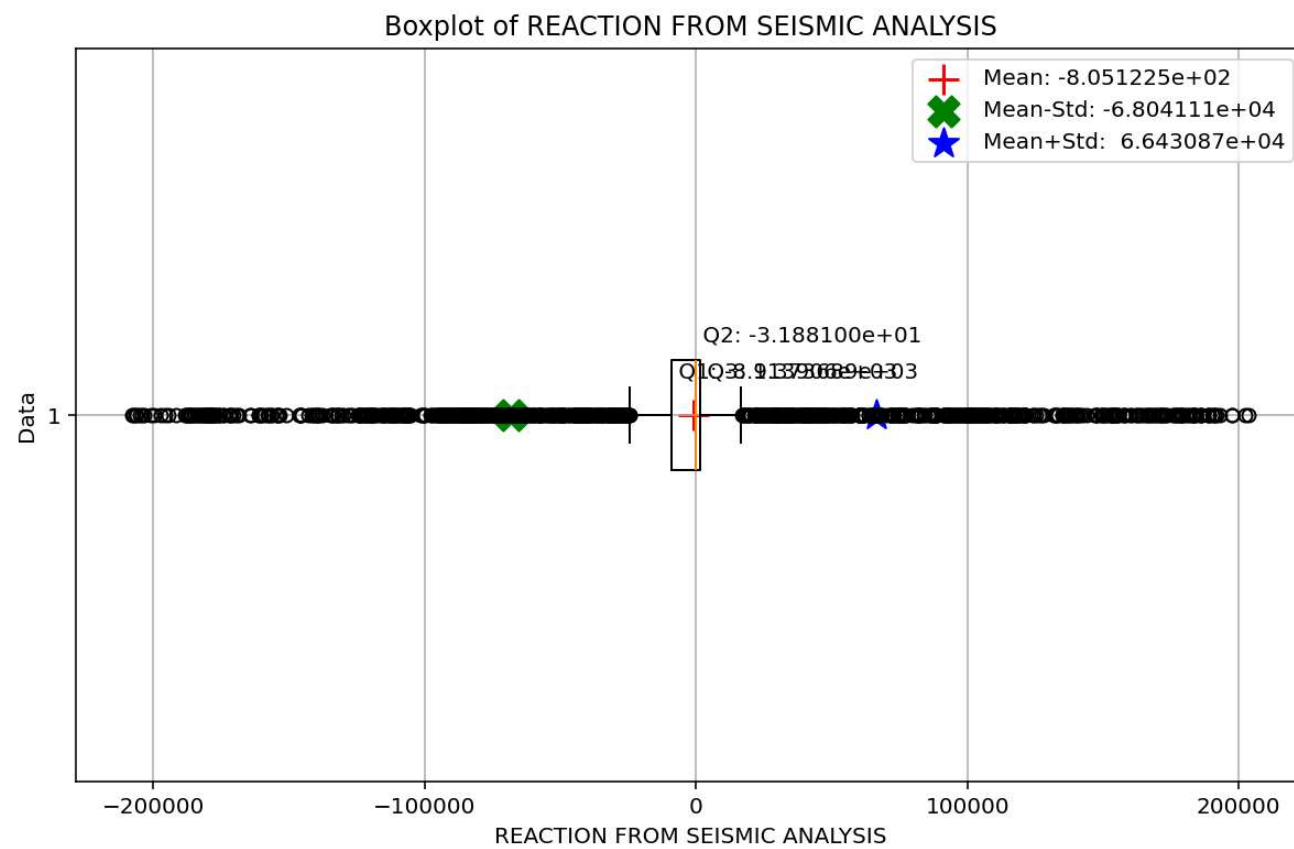


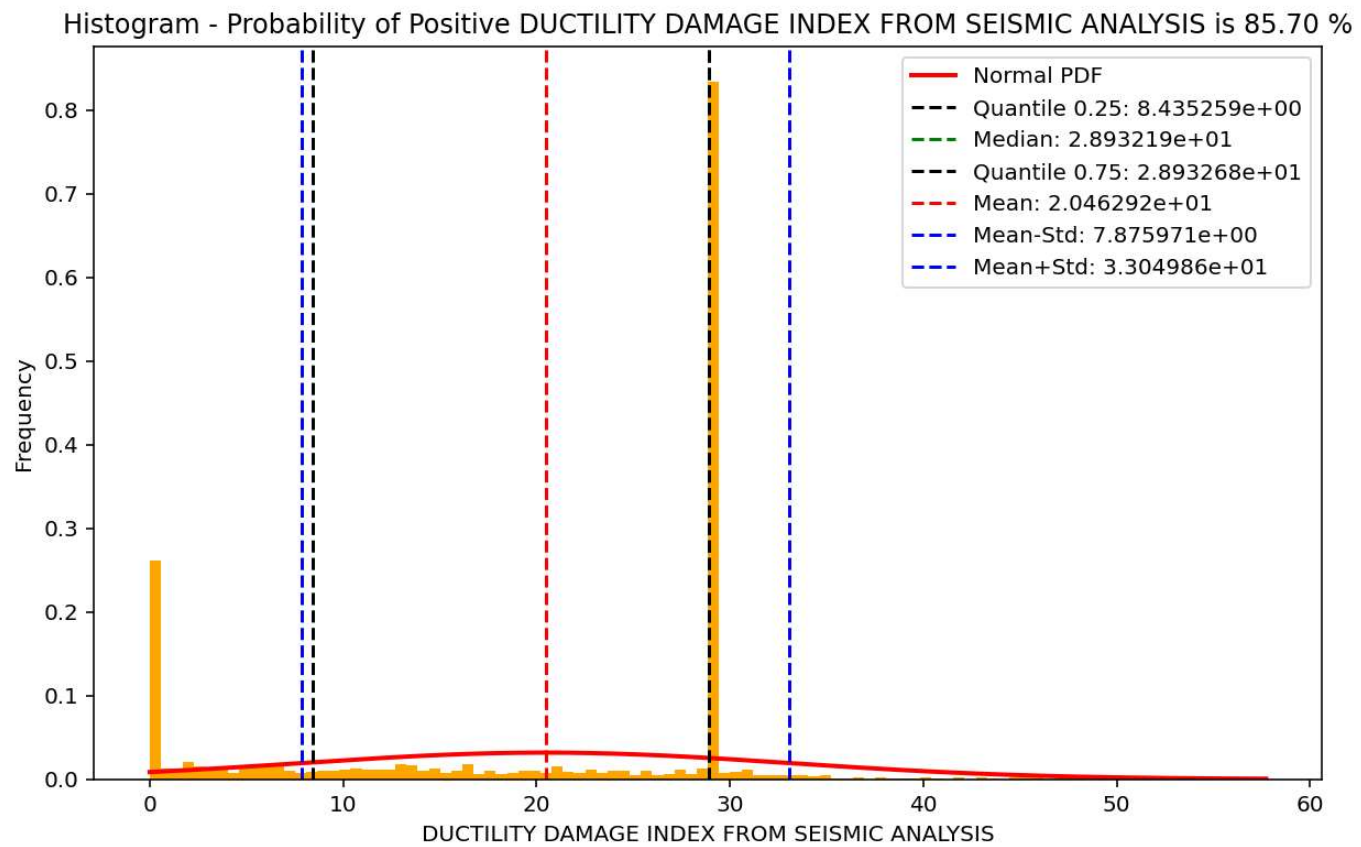


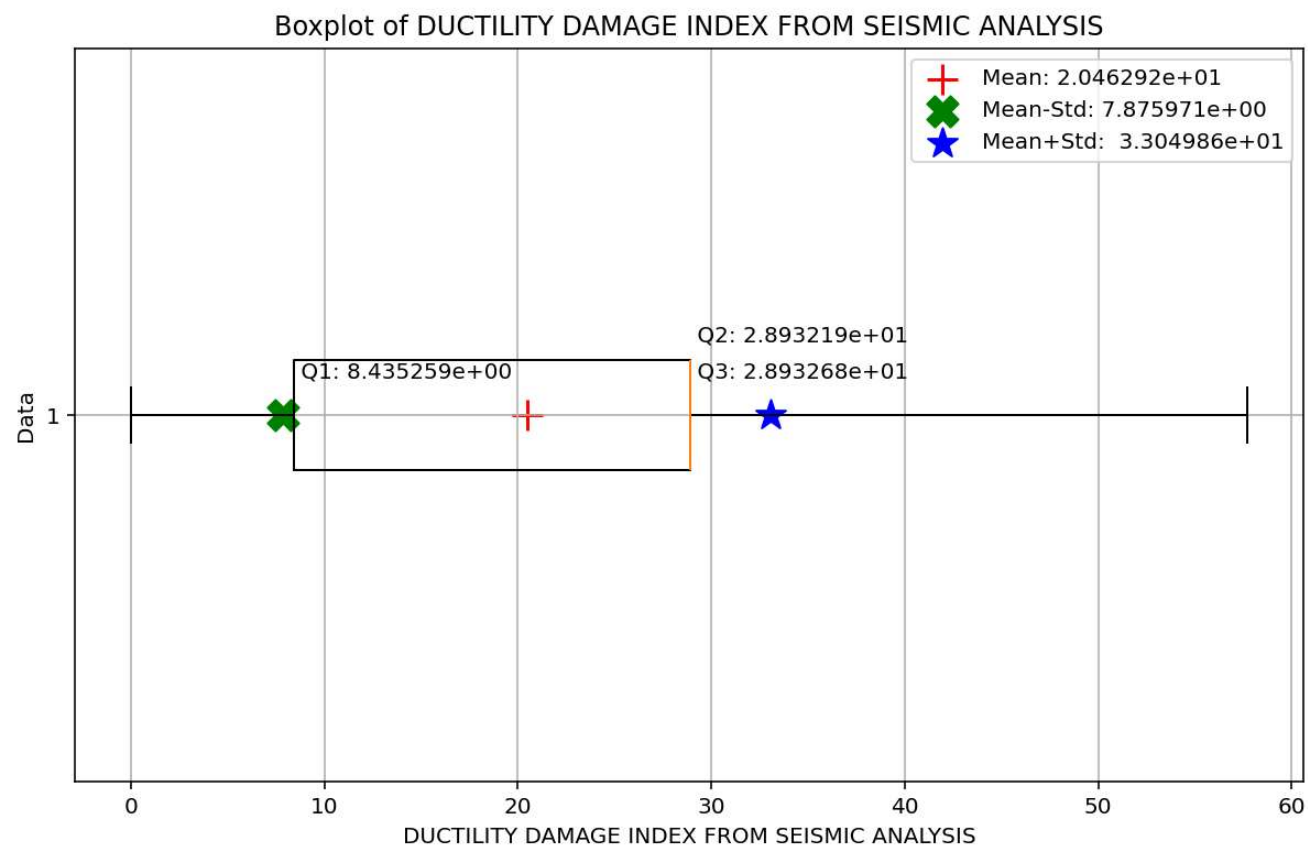




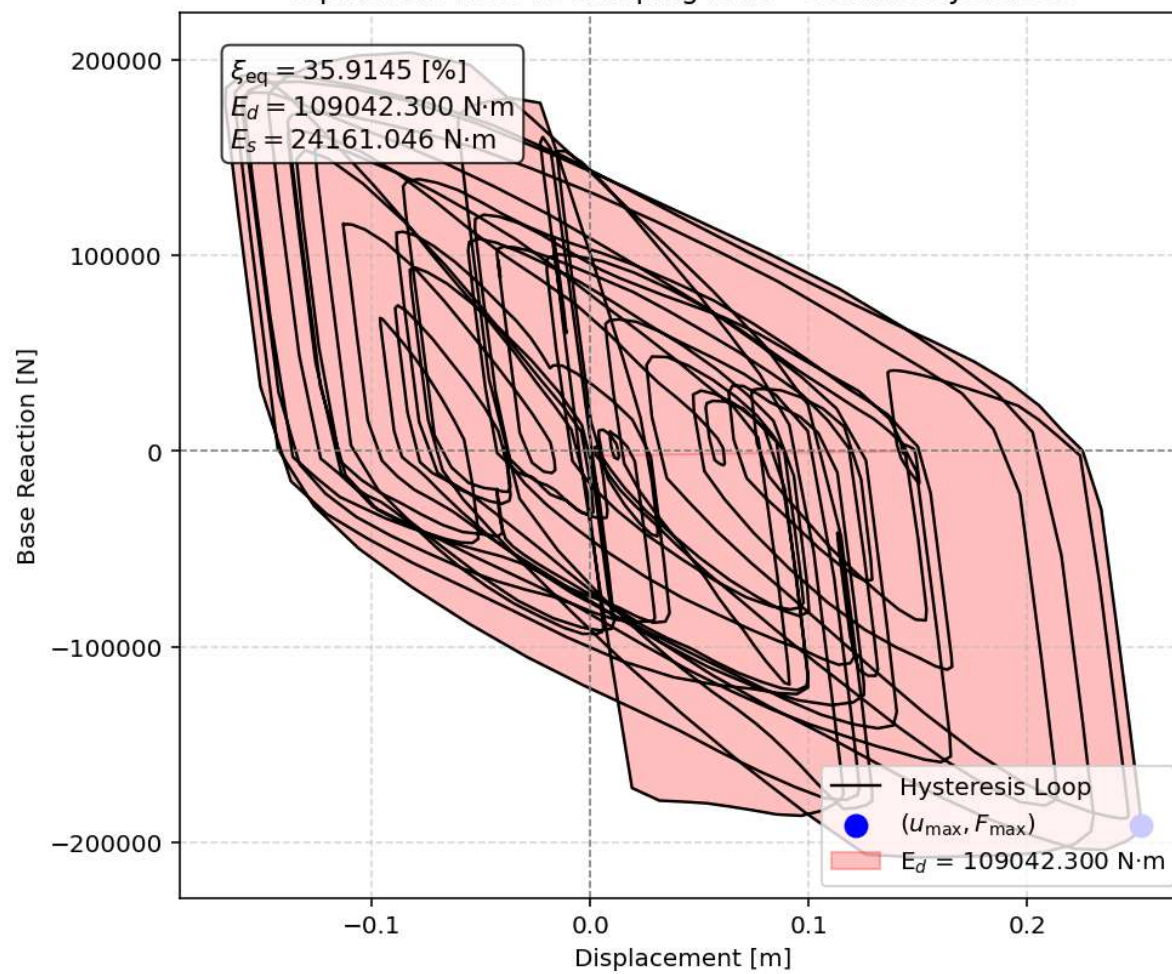




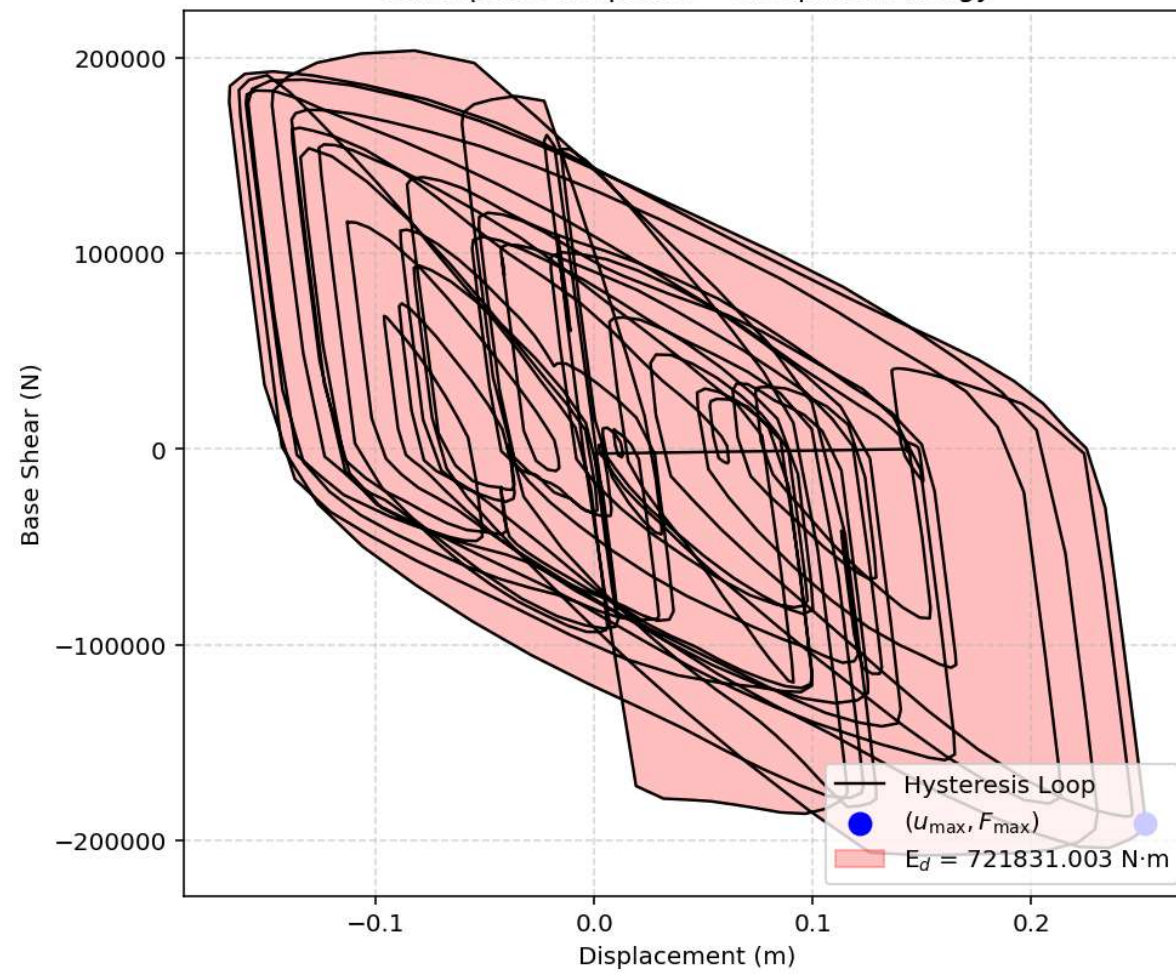


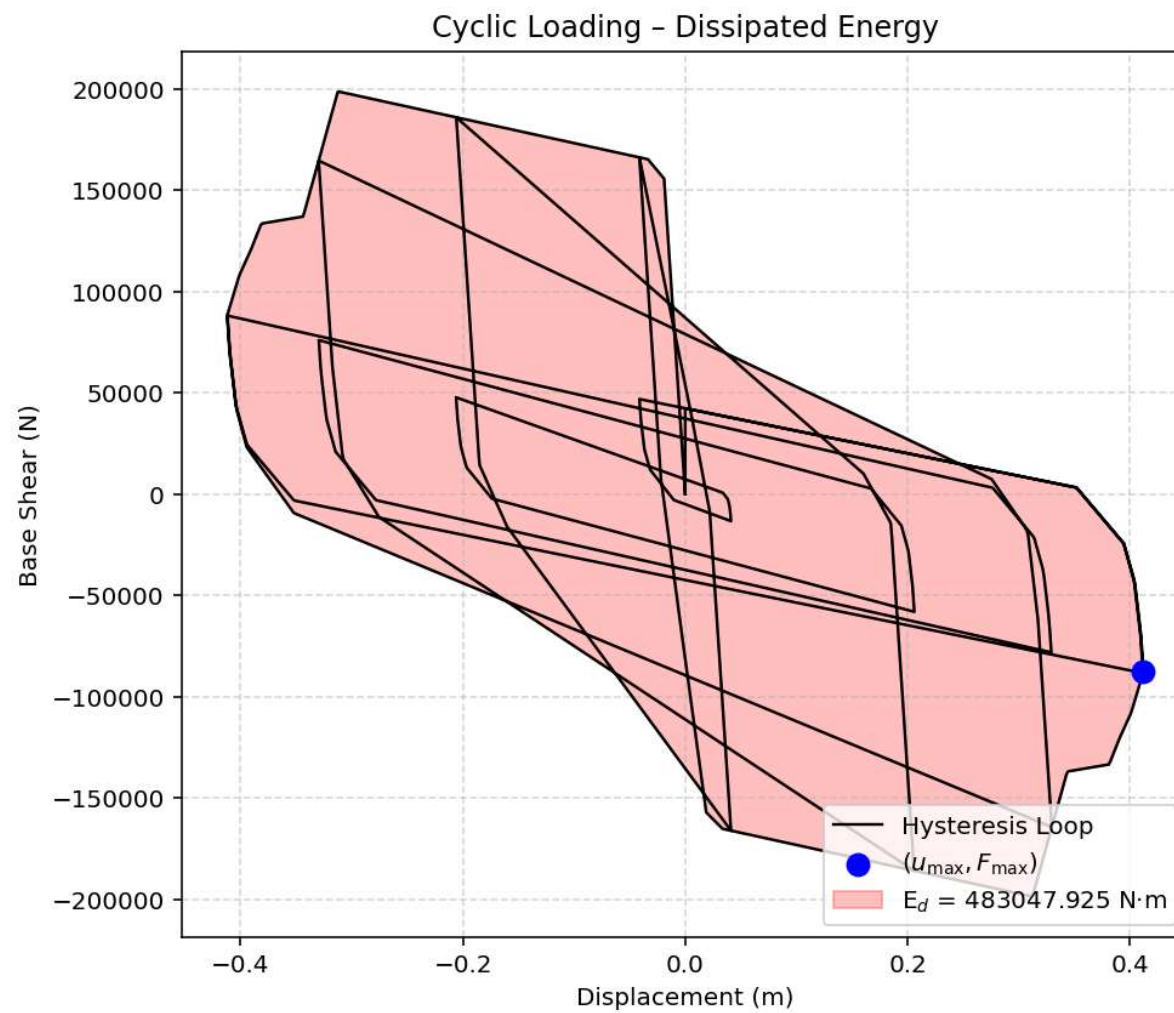


Equivalent viscous damping ratio - Seismic Hysteresis



Earthquake Response - Dissipated Energy



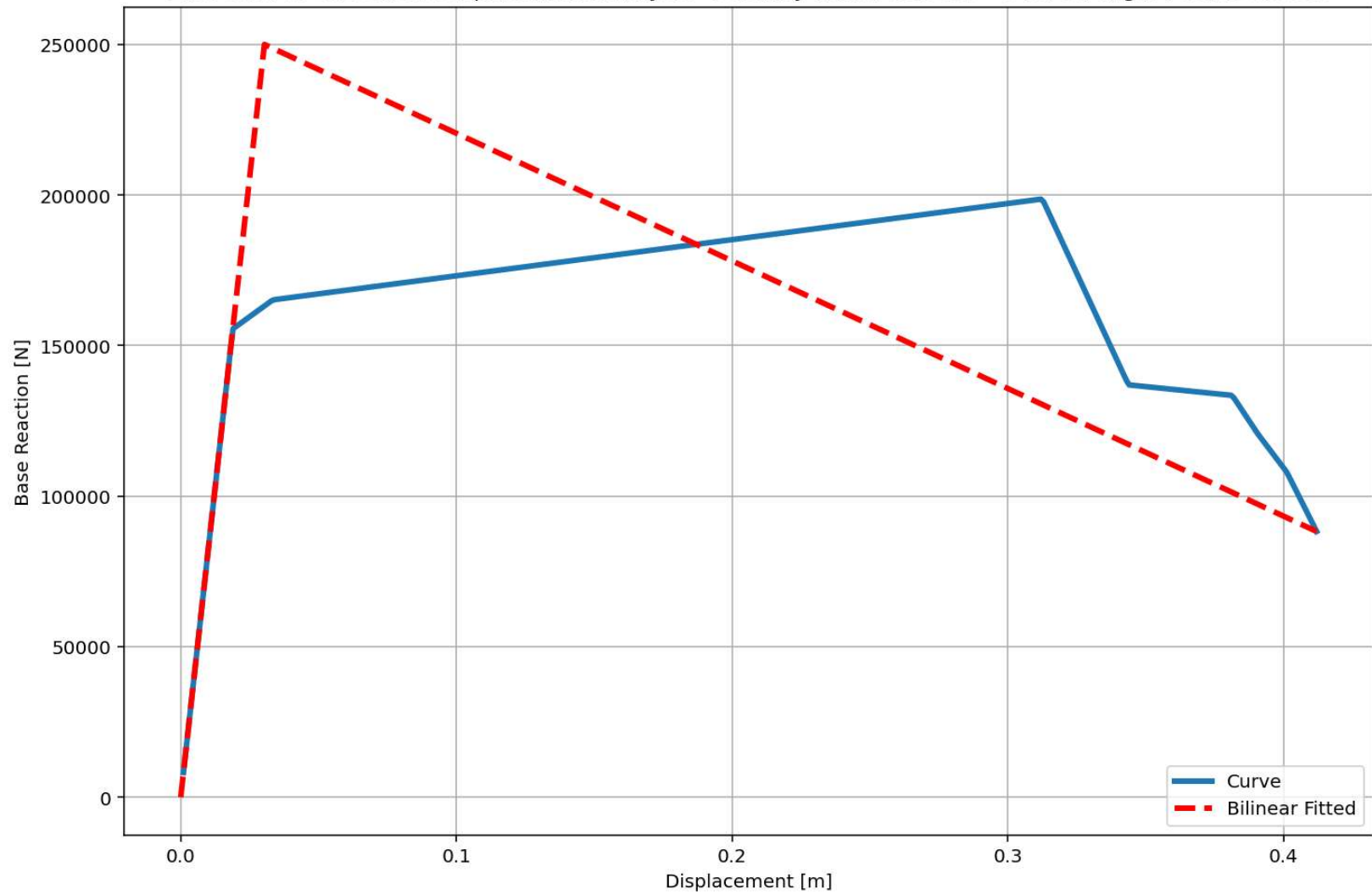


Dissipated Energy from Earthquake= 721831.00 N·m

Dissipated Energy from Cyclic Displacement= 483047.93 N·m

**FOR EVALUATION OF DISSIPATED ENERGY CAPACITY INDEX:
CHECK THE CYCLIC DISPLACEMENT ANALYSIS AND IF IT IS POSSIBLE
INCREASE THE DISPLACEMENT.**

Last Data of BaseShear-Displacement Analysis - Ductility Ratio: 13.5697 - Over Strength Factor: 0.3531



+=====+

= Analysis curve fitted =

Disp Base Shear

[[0.00000000e+00 0.00000000e+00]

[3.03616555e-02 2.50028233e+05]

[4.12000000e-01 8.82763002e+04]]

+=====+

+-----+

Structure Elastic Stiffness : 8235000.00

Structure Plastic Stiffness : 214262.86

Structure Tangent Stiffness : -423835.64

Structure Ductility Ratio : 13.57

Structure Over Strength Factor: 0.35

+-----+

Over Strength Coefficient (Ω_0): 0.3531

Displacement Ductility Ratio (μ): 13.5697

Ductility Coefficient (R_μ): 13.5697

Structural Behavior Coefficient (R): 4.7910

Structural Ductility Damage Index in X Direction: 58.1248 (%)

Seismic Fragility Curves by Damage State from DYN. TIME-HISTORY ANA. DUCTILITY DAMAGE INDEX [%]

